

BOOK 3

DYNAMICS

3 Dynamics

An understanding of forces from Cambridge IGCSE/O Level Physics or equivalent is assumed.

3.1 Momentum and Newton's laws of motion

Candidates should be able to:

- 1 understand that mass is the property of an object that resists change in motion
- 2 recall $F = ma$ and solve problems using it, understanding that acceleration and resultant force are always in the same direction
- 3 define and use linear momentum as the product of mass and velocity
- 4 define and use force as rate of change of momentum
- 5 state and apply each of Newton's laws of motion
- 6 describe and use the concept of weight as the effect of a gravitational field on a mass and recall that the weight of an object is equal to the product of its mass and the acceleration of free fall

3.2 Non-uniform motion

Candidates should be able to:

- 1 show a qualitative understanding of frictional forces and viscous/drag forces including air resistance (no treatment of the coefficients of friction and viscosity is required, and a simple model of drag force increasing as speed increases is sufficient)
- 2 describe and explain qualitatively the motion of objects in a uniform gravitational field with air resistance
- 3 understand that objects moving against a resistive force may reach a terminal (constant) velocity

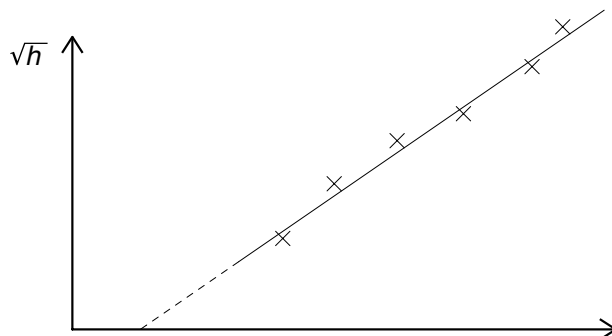
3.3 Linear momentum and its conservation

Candidates should be able to:

- 1 state the principle of conservation of momentum
- 2 apply the principle of conservation of momentum to solve simple problems, including elastic and inelastic interactions between objects in both one and two dimensions (knowledge of the concept of coefficient of restitution is not required)
- 3 recall that, for a perfectly elastic collision, the relative speed of approach is equal to the relative speed of separation
- 4 understand that, while momentum of a system is always conserved in interactions between objects, some change in kinetic energy may take place

1 9702/1/M/J/02/Q4

A student measures the time t for a ball to fall from rest through a vertical distance h . Knowing that the equation $h = \frac{1}{2}gt^2$ applies, the student plots the graph shown.



Which of the following is an explanation for the intercept on the t axis?

- A** Air resistance has not been taken into account for larger values of h .
- B** There is a constant delay between starting the timer and releasing the ball.
- C** There is an error in the timer that consistently makes it run fast.
- D** The student should have plotted h against t^2 .

2 9702/11/O/N/10/Q11*

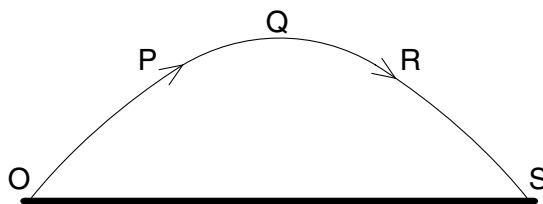
A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

- A** $\frac{M_1}{M_2}$
- B** $\frac{M_2}{M_1}$
- C** $\left(\frac{M_1}{M_2}\right)^{\frac{1}{2}}$
- D** $\left(\frac{M_2}{M_1}\right)^{\frac{1}{2}}$

3 9702/12/M/J/12/Q10

A projectile is launched at point O and follows the path OPQRS, as shown. Air resistance may be neglected.



Which statement is true for the projectile when it is at the highest point Q of its path?

- A** The horizontal component of the projectile's acceleration is zero.
- B** The horizontal component of the projectile's velocity is zero.
- C** The kinetic energy of the projectile is zero.
- D** The momentum of the projectile is zero.

4 9702/01/M/J/07/Q9*

What is meant by the weight of an object?

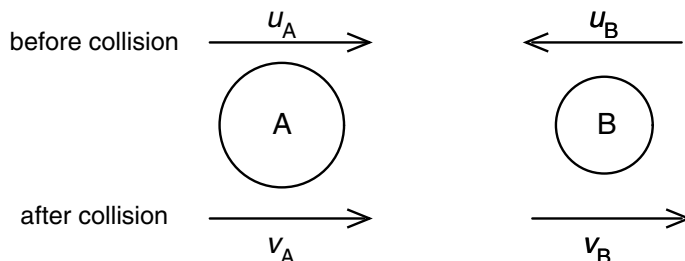
- A** the gravitational field acting on the object
- B** the gravitational force acting on the object
- C** the mass of the object multiplied by gravity
- D** the object's mass multiplied by its acceleration

5 9702/1/O/N/02/Q11

Two spheres A and B approach each other along the same straight line with speeds u_A and u_B . The spheres collide and move off with speeds v_A and v_B , both in the same direction as the initial direction of sphere A, as shown below.

Which equation applies to an elastic collision?

- A $u_A + u_B = v_B - v_A$
- B $u_A - u_B = v_B - v_A$
- C $u_A - u_B = v_B + v_A$
- D $u_A + u_B = v_B + v_A$



6 9702/1/O/N/02/Q12

Two equal masses travel towards each other on a frictionless air track at speeds of 60 cm s^{-1} and 30 cm s^{-1} . They stick together on impact.

What is the speed of the masses after impact?

- A 15 cm s^{-1}
- B 20 cm s^{-1}
- C 30 cm s^{-1}
- D 45 cm s^{-1}



7 9702/01/M/J/03/Q12

A ball of mass 2 kg travelling at 8 m s^{-1} strikes a ball of mass 4 kg travelling at 2 m s^{-1} . Both balls are moving along the same straight line as shown.

After collision, both balls move at the same velocity v .

What is the magnitude of the velocity v ?

- A 4 m s^{-1}
- B 5 m s^{-1}
- C 6 m s^{-1}
- D 8 m s^{-1}

8 9702/01/O/N/03/Q9*

Which of the following is a statement of the principle of conservation of momentum?

- A Momentum is the product of mass and velocity.
- B In an elastic collision, momentum is constant.
- C The momentum of an isolated system is constant.
- D The force acting on a body is proportional to its rate of change of momentum.

9 9702/01/O/N/08/Q9*

A ball falls vertically and bounces on the ground.

The following statements are about the forces acting while the ball is in contact with the ground.

Which statement is correct?

- A The force that the ball exerts on the ground is always equal to the weight of the ball.
- B The force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball.
- C The force that the ball exerts on the ground is always less than the weight of the ball.
- D The weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball.

10 9702/01/O/N/03/Q11

A molecule of mass m travelling horizontally with velocity u hits a vertical wall at right angles to the wall. It then rebounds horizontally with the same speed.

What is its change in momentum?

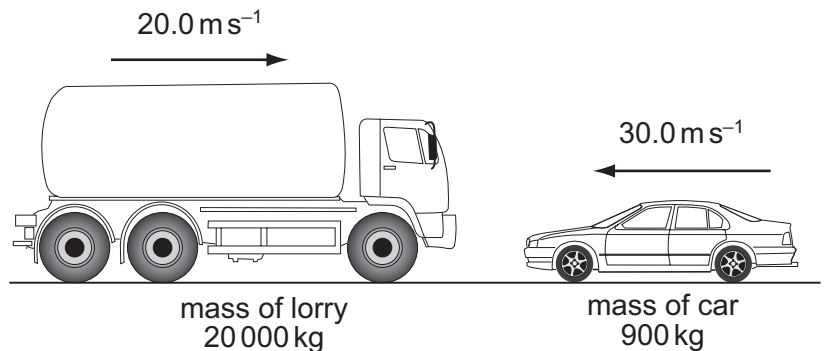
- A zero B mu C $-mu$ D $-2mu$

11 9702/01/M/J/04/Q11

The diagram shows a situation just before a head-on collision. A lorry of mass $20\,000\text{ kg}$ is travelling at 20.0 m s^{-1} towards a car of mass 900 kg travelling at 30.0 m s^{-1} towards the lorry.

What is the magnitude of the total momentum?

- A 373 kN s B 427 kN s
C 3600 kN s D 4410 kN s

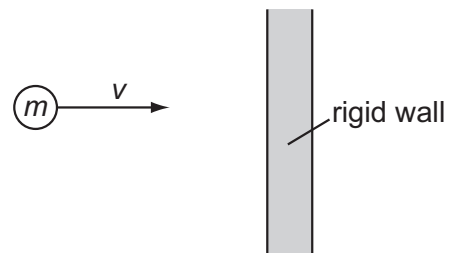


12 9702/01/O/N/04/Q11

A particle of mass m strikes a vertical rigid wall perpendicularly from the left with velocity v .

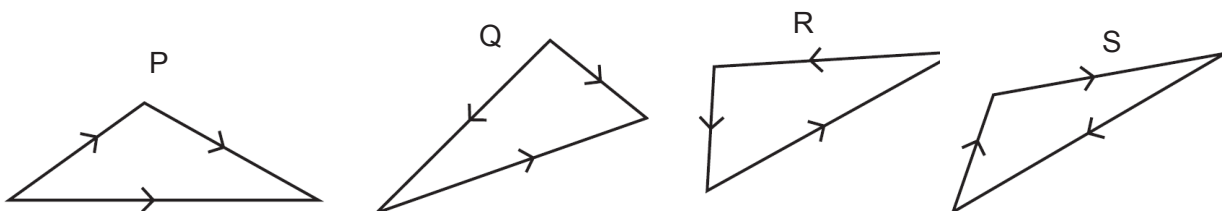
If the collision is perfectly elastic, the total change in momentum of the particle that occurs as a result of the collision is

- A $2mv$ to the right. B $2mv$ to the left.
C mv to the right. D mv to the left.



13 9702/01/O/N/04/Q12

Which two vector diagrams represent forces in equilibrium?



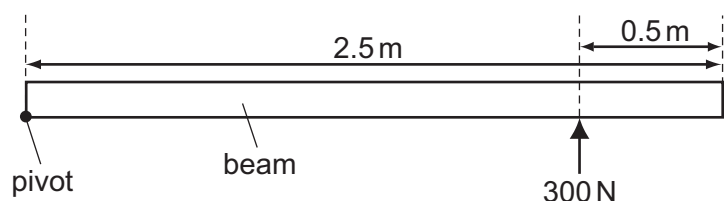
- A P and Q B Q and R C R and S D S and P

14 9702/01/O/N/04/Q13

A long uniform beam is pivoted at one end. A force of 300 N is applied to hold the beam horizontally.

What is the weight of the beam?

- A 300 N B 480 N
C 600 N D 960 N



15 9702/01/M/J/05/Q10

Which is **not** one of Newton's laws of motion?

- A The total momentum of a system of interacting bodies remains constant, providing no external force acts.
- B The rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force.
- C If body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A.
- D A body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force.

16 9702/01/M/J/05/Q11*

Two equal masses travel towards each other on a frictionless air track at speeds of 60 cm s^{-1} and 40 cm s^{-1} . They stick together on impact.

What is the speed of the masses after impact?

- A 10 cm s^{-1}
- B 20 cm s^{-1}
- C 40 cm s^{-1}
- D 50 cm s^{-1}



17 9702/01/M/J/05/Q12*

What is the centre of gravity of an object?

- A the geometrical centre of the object
- B the point about which the total torque is zero
- C the point at which the weight of the object may be considered to act
- D the point through which gravity acts

18 9702/01/O/N/05/Q9

Which is a statement of the principle of conservation of momentum?

- A A force is equal to the rate of change of momentum of the body upon which it acts.
- B In a perfectly elastic collision, the relative momentum of the bodies before impact is equal to their relative momentum after impact.
- C The momentum of a body is the product of the mass of the body and its velocity.
- D The total momentum of a system of interacting bodies remains constant, providing no external force acts.

19 9702/01/O/N/05/Q10

The gravitational field strength on the surface of planet P is one tenth of that on the surface of planet Q.

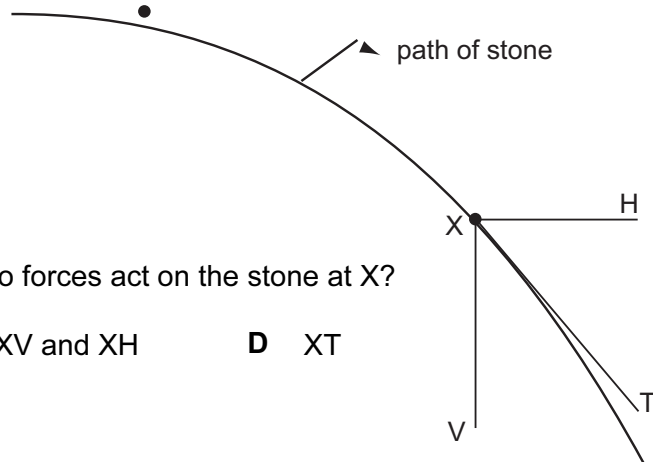
On the surface of P, a body has its mass measured to be 1.0 kg and its weight measured to be 1.0 N .

What results are obtained for measurements of the mass and weight of the same body on the surface of planet Q?

	mass on Q	weight on Q
A	1.0 kg	0.1 N
B	1.0 kg	10 N
C	10 kg	10 N
D	10 kg	100 N

20 9702/01/O/N/05/Q11

A stone is projected horizontally in a vacuum and moves along a path as shown. X is a point on this path. XV and XH are vertical and horizontal lines respectively through X. XT is the tangent to the path at X.



Along which direction or directions do forces act on the stone at X?

- A XV B XH C XV and XH D XT

21 9702/01/M/J/06/Q10

A cyclist is riding at a steady speed on a level road.

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

- A the force exerted by the cyclist on the pedals
B the forward push of the road on the back wheel
C the tension in the cycle chain
D the total air resistance and friction force

22 9702/01/M/J/06/Q11

In perfectly elastic collisions between two atoms, it is always true to say that

- A the initial speed of one atom will be the same as the final speed of the other atom.
B the relative speed of approach between the two atoms equals their relative speed of separation.
C the total momentum must be conserved, but a small amount of the total kinetic energy may be lost in the collision.
D whatever their initial states of motion, neither atom can be stationary after the collision.

23 9702/01/M/J/06/Q12*

Two railway trucks of masses m and $3m$ move towards each other in opposite directions with speeds $2v$ and v respectively. These trucks collide and stick together.

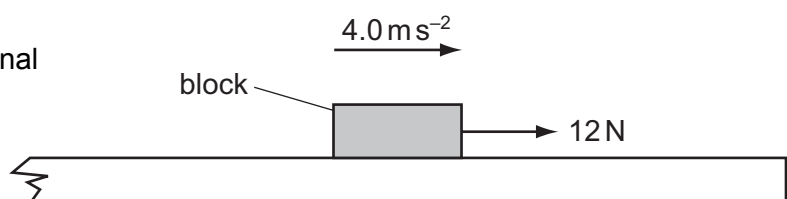
What is the speed of the trucks after the collision?

- A $\frac{v}{4}$ B $\frac{v}{2}$ C v D $\frac{5v}{4}$

24 9702/01/O/N/07/Q10

A block of mass 0.60 kg is on a rough horizontal surface. A force of 12 N is applied to the block and it accelerates at 4.0 m s^{-2} .

What is the magnitude of the frictional force acting on the block?



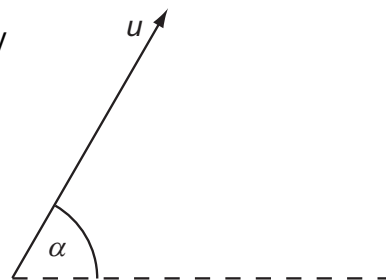
- A 2.4 N B 5.3 N
C 6.7 N D 9.6 N

25 9702/01/M/J/03/Q7

A projectile is fired at an angle α to the horizontal at a speed u , as shown.

What are the vertical and horizontal components of its velocity after a time t ?

Assume that air resistance is negligible. The acceleration of free fall is g .



	vertical component	horizontal component
A	$u \sin \alpha$	$u \cos \alpha$
B	$u \sin \alpha - gt$	$u \cos \alpha - gt$
C	$u \sin \alpha - gt$	$u \cos \alpha$
D	$u \cos \alpha$	$u \sin \alpha - gt$

26 9702/01/O/N/06/Q10*

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a .

Which quantities **must** be in the same direction?

- A** a and v only **B** a and F only **C** v and F only **D** v , F and a

27 9702/01/O/N/06/Q11

The diagram shows two identical spheres X and Y.



Initially X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically.

What happens?

	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

28 9702/01/M/J/07/Q11

A lorry of mass $20\,000\text{ kg}$ is travelling at 20.0 m s^{-1} . A car of mass 900 kg is travelling at 30.0 m s^{-1} towards the lorry.

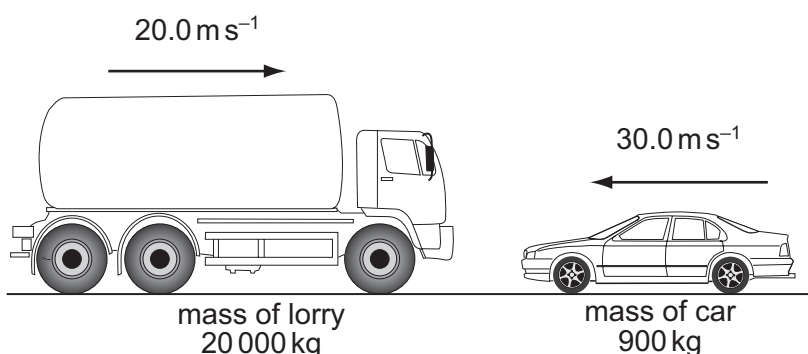
What is the magnitude of the total momentum?

A 209 kNs

B 373 kNs

C 427 kNs

D 1045 kNs



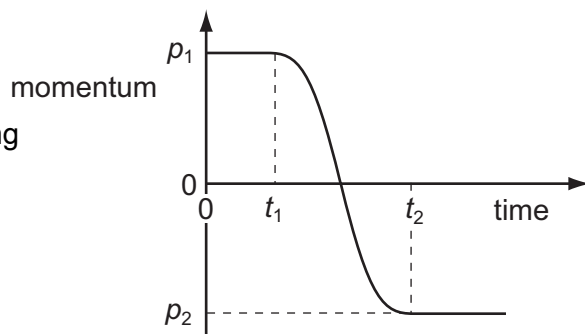
29 9702/01/M/J/07/Q10

The graph shows the variation with time of the momentum of a ball as it is kicked in a straight line.

Initially, the momentum is p_1 at time t_1 . At time t_2 the momentum is p_2 .

What is the magnitude of the average force acting on the ball between times t_1 and t_2 ?

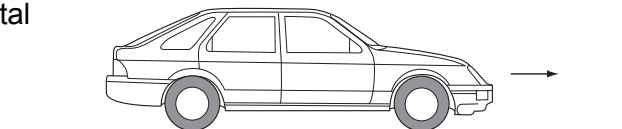
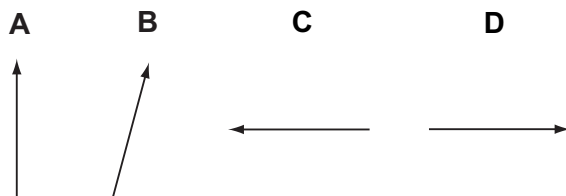
- A $\frac{p_1 - p_2}{t_2}$ B $\frac{p_1 - p_2}{t_2 - t_1}$
C $\frac{p_1 + p_2}{t_2}$ D $\frac{p_1 + p_2}{t_2 - t_1}$



30 9702/01/O/N/07/Q11

A car with front-wheel drive accelerates in the direction shown.

Which diagram best shows the direction of the total force exerted by the road on the front wheels?

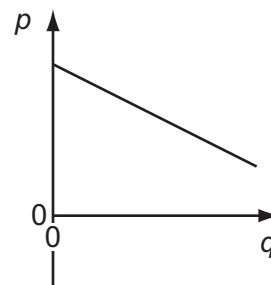


31 9702/01/O/N/07/Q12

The graph shows how a certain quantity p varies with another quantity q for a parachutist falling at terminal speed.

What are the quantities p and q , and what is represented by the magnitude of the gradient of the graph?

	quantity p	quantity q	magnitude of gradient
A	height	time	terminal speed
B	momentum	time	weight of parachutist
C	height	potential energy	mass of parachutist
D	velocity	time	acceleration of free fall



32 9702/01/M/J/08/Q9

Which is a statement of the principle of conservation of momentum?

- A Momentum is the product of mass and velocity.
B Momentum is conserved only in elastic collisions.
C Momentum is conserved by all bodies in a collision.
D Momentum is conserved providing no external forces act.

33 9702/01/M/J/08/Q12

A ball is falling at terminal speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

What is the order of increasing magnitude of these three forces?

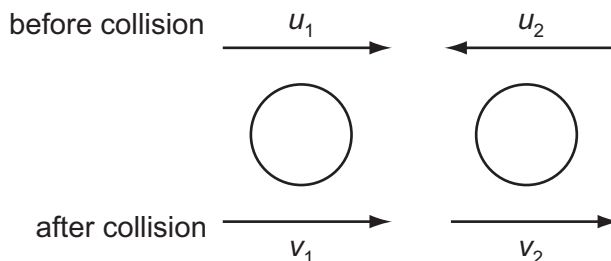
- A upthrust $\rightarrow$ viscous drag $\rightarrow$ weight C viscous drag $\rightarrow$ weight $\rightarrow$ upthrust
B viscous drag $\rightarrow$ upthrust $\rightarrow$ weight D weight $\rightarrow$ upthrust $\rightarrow$ viscous drag

34 9702/01/O/N/08/Q10

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision, and v_1 and v_2 after collision, in the directions shown below.

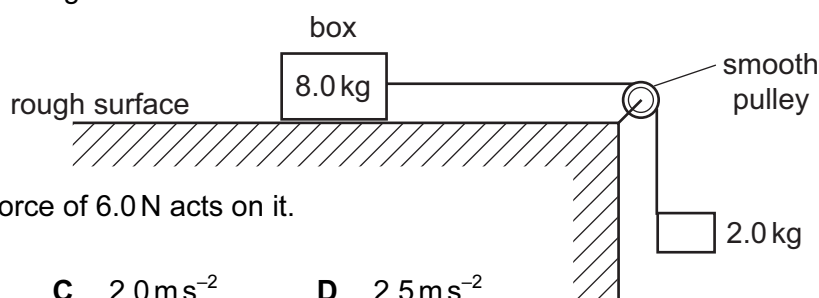
Which equation is correct if the collision is perfectly elastic?

- A $u_1 - u_2 = v_2 + v_1$ B $u_1 - u_2 = v_2 - v_1$
C $u_1 + u_2 = v_2 + v_1$ D $u_1 + u_2 = v_2 - v_1$



35 9702/01/O/N/08/Q11

A box of mass 8.0 kg rests on a horizontal, rough surface. A string attached to the box passes over a smooth pulley and supports a 2.0 kg mass at its other end.



When the box is released, a friction force of 6.0 N acts on it. What is the acceleration of the box?

- A 1.4 ms^{-2} B 1.7 ms^{-2} C 2.0 ms^{-2} D 2.5 ms^{-2}

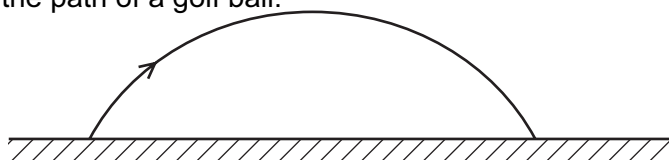
36 9702/01/M/J/09/Q7

Which statement about Newton's laws of motion is correct?

- A The first law follows from the second law.
B The third law follows from the second law.
C Conservation of energy is a consequence of the third law.
D Conservation of linear momentum is a consequence of the first law.

37 9702/01/M/J/09/Q8

The diagram shows the path of a golf ball.



Which row describes changes in the horizontal and vertical components of the golf ball's velocity, when air resistance forces are ignored?

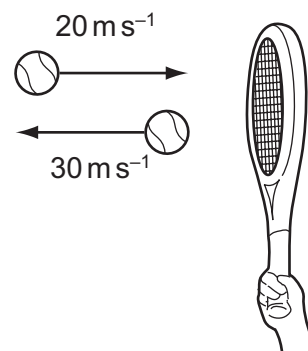
	horizontal	vertical
A	constant deceleration	constant acceleration downwards
B	constant deceleration	acceleration decreases upwards then increases downwards
C	constant velocity	constant acceleration downwards
D	constant velocity	acceleration decreases upwards then increases downwards

38 9702/01/M/J/09/Q9

A tennis ball of mass 100 g is struck by a tennis racket. The velocity of the ball is changed as shown.

What is the magnitude of the change in momentum of the ball?

- A 1 kg ms^{-1} B 5 kg ms^{-1}
C 1000 kg ms^{-1} D 5000 kg ms^{-1}



39 9702/01/M/J/09/Q12

An object, made from two equal masses joined by a light rod, falls with uniform speed through air. The rod remains horizontal.

Which statement about the equilibrium of the system is correct?

- A It is not in equilibrium because it is falling steadily.
B It is not in equilibrium because it is in motion.
C It is not in equilibrium because there is a resultant torque.
D It is in equilibrium because there is no resultant force and no resultant torque.

40 9702/11/O/N/15/Q12

Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

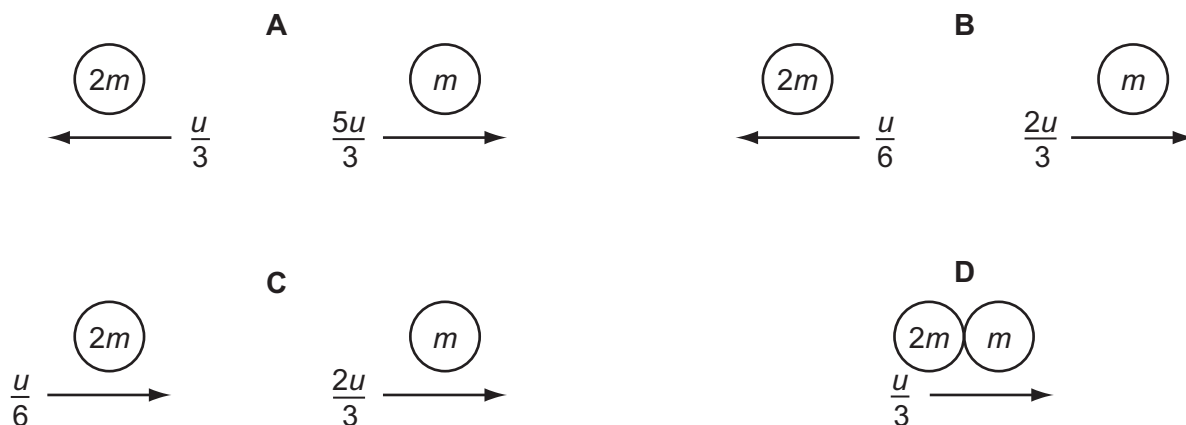
- A Total kinetic energy of the ball is conserved.
B Total kinetic energy of the system is conserved.
C Total momentum of the ball is conserved.
D Total momentum of the system is conserved.

41 9702/13/O/N/12/Q11*

The diagram shows two spherical masses approaching each other head-on at an equal speed u . One has mass $2m$ and the other has mass m .



Which diagram, showing the situation after the collision, shows the result of an elastic collision?



the spheres stick together

42 9702/12/O/N/09/Q9

A supermarket trolley, total mass 30 kg, is moving at 3.0 m s^{-1} . A retarding force of 60 N is applied to the trolley for 0.50 s in the opposite direction to the trolley's initial velocity.

What is the trolley's new velocity after the application of the force?

- A 1.0 m s^{-1} B 1.5 m s^{-1} C 2.0 m s^{-1} D 2.8 m s^{-1}

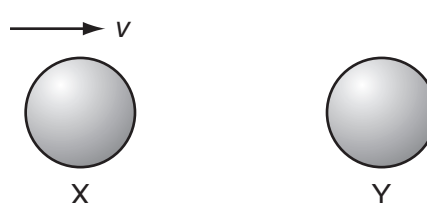
43 9702/12/O/N/09/Q10

What is the centre of gravity of an object?

- A the geometrical centre of the object
B the point about which the total torque is zero
C the point at which the weight of the object may be considered to act
D the point through which gravity acts

44 9702/12/M/J/10/Q10

The diagram shows two identical spheres X and Y.



Initially, X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically. What happens?

	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

45 9702/12/M/J/10/Q11

Two equal masses travel towards each other on a frictionless air track at speeds of 60 cm s^{-1} and 40 cm s^{-1} . They stick together on impact.

What is the speed of the masses after impact?

- A 10 cm s^{-1} B 20 cm s^{-1}
C 40 cm s^{-1} D 50 cm s^{-1}



46 9702/11/O/N/10/Q10

The gravitational field strength on the surface of planet P is one tenth of that on the surface of planet Q.

On the surface of P, a body has a mass of 1.0 kg and a weight of 1.0 N.

What are the mass and weight of the same body on the surface of planet Q?

	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

47 9702/11/O/N/10/Q11

A body, initially at rest, explodes into two masses M_1 and M_2 that move apart with speeds v_1 and v_2 respectively.

What is the ratio $\frac{v_1}{v_2}$?

- A $\frac{M_1}{M_2}$ B $\frac{M_2}{M_1}$ C $\sqrt{\frac{M_1}{M_2}}$ D $\sqrt{\frac{M_2}{M_1}}$

48 9702/11/O/N/10/Q12

Two experiments are carried out using two trolleys of equal mass. All moving parts of the trolleys are frictionless, as is the surface that the trolleys move over. In both experiments, trolley X moves towards trolley Y, which is initially stationary.



After the collision in experiment 1, X is stationary and Y moves off to the right.
After the collision in experiment 2, the trolleys join and move off together.
What types of collision occur in these experiments?

	experiment 1	experiment 2
A	elastic	elastic
B	elastic	inelastic
C	inelastic	elastic
D	inelastic	inelastic

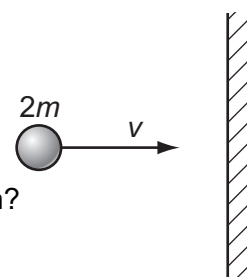
49 9702/12/O/N/10/Q9

A particle of mass $2m$ and velocity v strikes a wall.

The particle rebounds along the same path after colliding with the wall.
The collision is inelastic.

What is a possible change in the momentum of the ball during the collision?

- A mv B $2mv$ C $3mv$ D $4mv$



50 9702/12/O/N/10/Q10

Which defines the weight of a body?

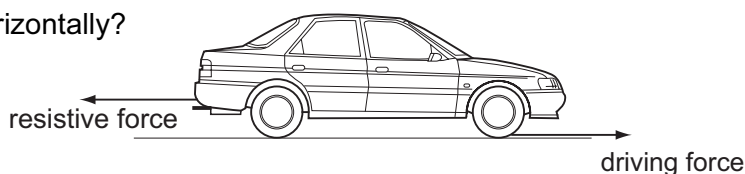
- A the amount of matter in the body C the number of particles in the body
B the force of gravity on the body D the product of the body's volume and density

51 9702/01/M/J/08/Q11

A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of 2.0 ms^{-2} .

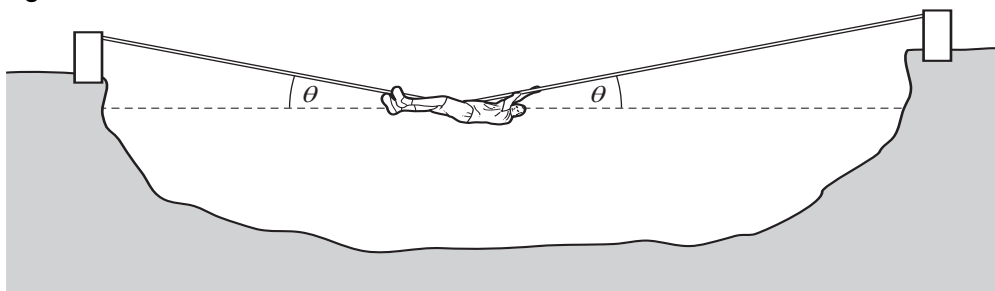
What is the resistive force acting horizontally?

- A 0.5 kN B 1.5 kN
C 2.0 kN D 3.5 kN



52 9702/12/O/N/10/Q11

The diagram shows a rope bridge that a student makes on an adventure training course. The student has a weight W .



Which formula gives the tension T in the rope?

- A $T = \frac{W}{2\cos\theta}$ B $T = \frac{W}{2\sin\theta}$ C $T = \frac{W}{\cos\theta}$ D $T = \frac{W}{\sin\theta}$

53 9702/11/M/J/11/Q8

A body has a weight of 58.9 N when on the Earth. On the Moon, the acceleration of free fall is 1.64 ms^{-2} .

What are the weight and the mass of the body when it is on the Moon?

	weight / N	mass / kg
A	9.85	1.00
B	9.85	6.00
C	58.9	1.00
D	58.9	6.00

54 9702/11/M/J/11/Q9

A body of mass m , moving at velocity v , collides with a stationary body of the same mass and sticks to it.

Which row describes the momentum and kinetic energy of the two bodies after the collision?

	momentum	kinetic energy
A	mv	$\frac{1}{4}mv^2$
B	mv	$\frac{1}{8}mv^2$
C	$2mv$	$\frac{1}{2}mv^2$
D	$2mv$	mv^2

55 9702/11/M/J/11/Q10

A molecule of mass m travelling horizontally with velocity u hits a vertical wall at right-angles to its velocity. It then rebounds horizontally with the same speed.

What is its change in momentum?

- A zero B mu C $-mu$ D $-2mu$

56 9702/12/M/J/11/Q11

The momentum of an object changes from 160 kg ms^{-1} to 240 kg ms^{-1} in 2 s.

What is the mean resultant force on the object during the change?

- A 40 N B 80 N C 200 N D 400 N

57 9702/12/M/J/11/Q10

A force F is applied to a freely moving object. At one instant of time, the object has velocity v and acceleration a .

Which quantities **must** be in the same direction?

- A a and v only B a and F only C v and F only D v , F and a

58 9702/12/M/J/11/Q12

A car accelerates in a straight line.

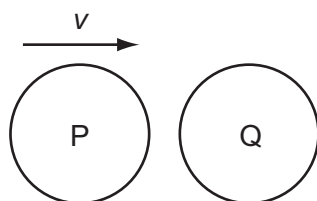
A graph of the momentum of the car is plotted against time.

What is evaluated by finding the gradient of the graph at a particular time?

- A the acceleration of the car C the kinetic energy of the car
B the resultant force on the car D the power supplied to the car

59 9702/12/M/J/11/Q13

The diagram shows a particle P, travelling at speed v , about to collide with a stationary particle Q of the same mass. The collision is perfectly elastic.



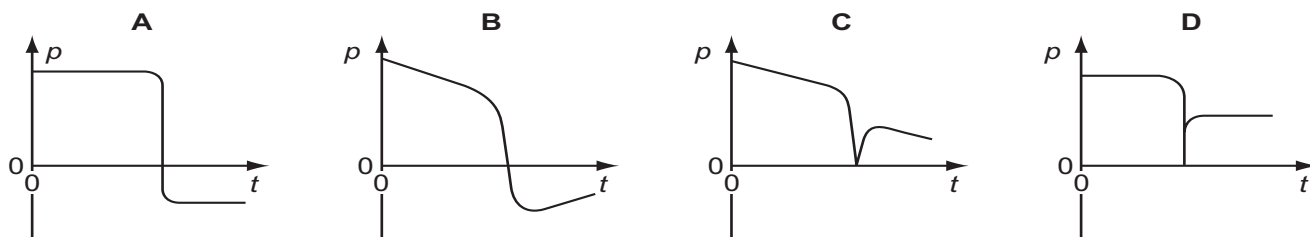
Which statement describes the motion of P and of Q immediately after the collision?

- A P rebounds with speed $\frac{1}{2}v$ and Q acquires speed $\frac{1}{2}v$.
B P rebounds with speed v and Q remains stationary.
C P and Q both travel in the same direction with speed $\frac{1}{2}v$.
D P comes to a standstill and Q acquires speed v .

60 9702/11/O/N/11/Q12

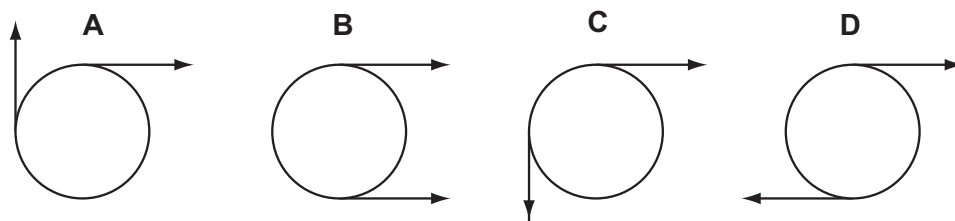
An ice-hockey puck slides along a horizontal, frictionless ice-rink surface. It collides inelastically with a wall at right angles to its path, and then rebounds along its original path.

Which graph shows the variation with time t of the momentum p of the puck?



61 9702/12/O/N/11/Q13

Two co-planar forces act on the rim of a wheel. The forces are equal in magnitude. Which arrangement of forces provides only a couple?



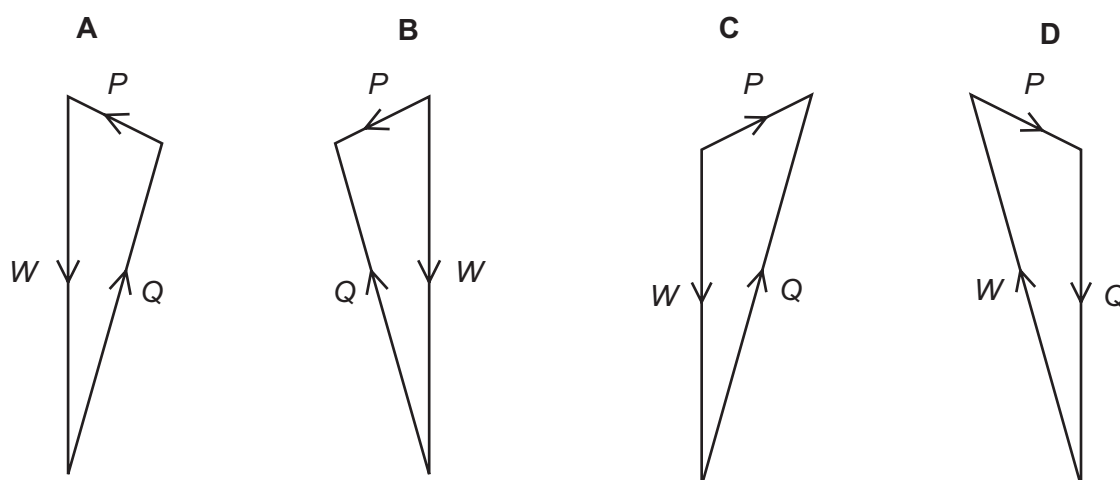
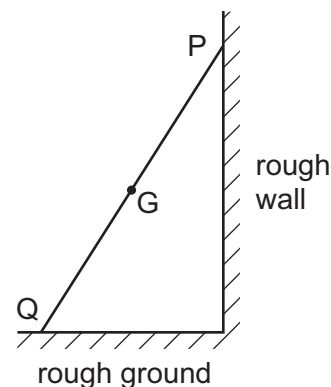
62 9702/11/O/N/11/Q13

A ladder rests in equilibrium on rough ground against a rough wall.

Its weight W acts through the centre of gravity G . Forces also act on the ladder at P and at Q .

These forces are P and Q respectively.

Which vector triangle represents the forces on the ladder?

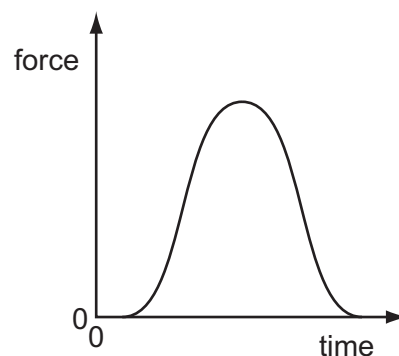


63 9702/11/O/N/15/Q10

A golf ball is hit by a club. The graph shows the variation with time of the force exerted on the ball by the club.

Which quantity, for the time of contact, **cannot** be found from the graph?

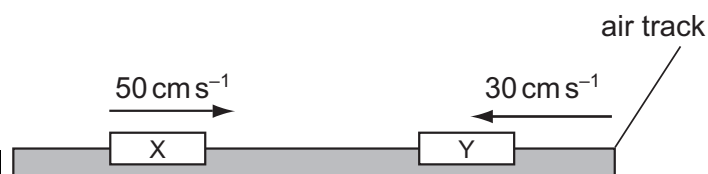
- A the average force on the ball
- B the change in momentum of the ball
- C the contact time between the ball and the club
- D the maximum acceleration of the ball



64 9702/13/O/N/15/Q12

Two equal masses X and Y are moving towards each other on a frictionless air track as shown. The masses make an elastic collision.

Which row gives possible velocities for the two masses after the collision?



	velocity of X	velocity of Y
A	zero	20 cm s^{-1} to the right
B	10 cm s^{-1} to the right	10 cm s^{-1} to the right
C	20 cm s^{-1} to the left	zero
D	30 cm s^{-1} to the left	50 cm s^{-1} to the right

65 9702/12/O/N/11/Q10

A group of students investigating the principle of conservation of momentum use a small truck travelling over a frictionless surface.

Sand is dropped into the truck as it passes X. At Y, a trapdoor in the bottom of the truck opens and the sand falls out.



How does the velocity of the truck change when the sand is added to the truck at X and then leaves the truck at Y?

	at X	at Y
A	decreases	increases
B	decreases	stays the same
C	stays the same	increases
D	stays the same	stays the same

66 9702/12/O/N/11/Q11

An object of mass 20 kg is travelling at a constant speed of 6.0 m s^{-1} .

It collides with an object of mass 12 kg travelling at a constant speed of 15 m s^{-1} in the opposite direction. The objects stick together.

What is the speed of the objects immediately after the collision?

- A** 1.9 m s^{-1} **B** 9.0 m s^{-1} **C** 9.4 m s^{-1} **D** 21 m s^{-1}

67 9702/01/O/N/08/Q9

A ball falls vertically and bounces on the ground.

The following statements are about the forces acting while the ball is in contact with the ground.

Which statement is correct?

- A** The force that the ball exerts on the ground is always equal to the weight of the ball.
B The force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball.
C The force that the ball exerts on the ground is always less than the weight of the ball.
D The weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball.

68 9702/11/M/J/16/Q10

Two spheres approach each other along the same straight line. Their speeds are u_1 and u_2 before collision. After the collision, the spheres separate with speeds v_1 and v_2 in the directions shown below.



Which equation must be correct if the collision is perfectly elastic?

- A** $u_1 - u_2 = v_2 + v_1$ **C** $u_1 + u_2 = v_2 + v_1$
B $u_1 - u_2 = v_2 - v_1$ **D** $u_1 + u_2 = v_2 - v_1$

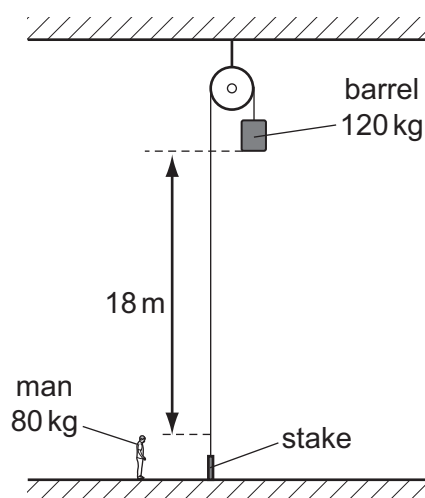
69 9702/12/M/J/12/Q11

Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	momentum	kinetic energy
A	conserved	conserved
B	conserved	not conserved
C	not conserved	conserved
D	not conserved	not conserved

70 9702/13/M/J/12/Q10

The diagram shows a barrel suspended from a frictionless pulley on a building. The rope supporting the barrel goes over the pulley and is secured to a stake at the bottom of the building.



A man stands close to the stake. The bottom of the barrel is 18 m above the man's head. The mass of the barrel is 120 kg and the mass of the man is 80 kg.

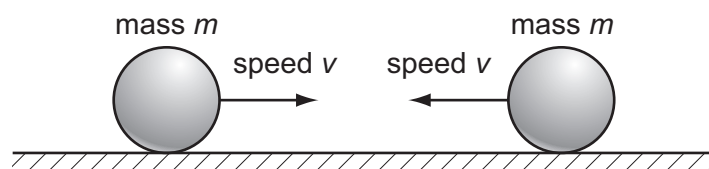
The man keeps hold of the rope after untying it from the stake and is lifted upwards as the barrel falls.

What is the man's upward speed when his head is level with the bottom of the barrel? (Use $g = 10 \text{ ms}^{-2}$.)

- A** 6 ms^{-1} **B** 8 ms^{-1} **C** 13 ms^{-1} **D** 19 ms^{-1}

71 9702/12/O/N/12/Q13

Two identical, perfectly elastic spheres have the same mass m . They travel towards each other with the same speed v along a horizontal frictionless surface.



Which statement about the sum of the kinetic energies of the spheres is correct?

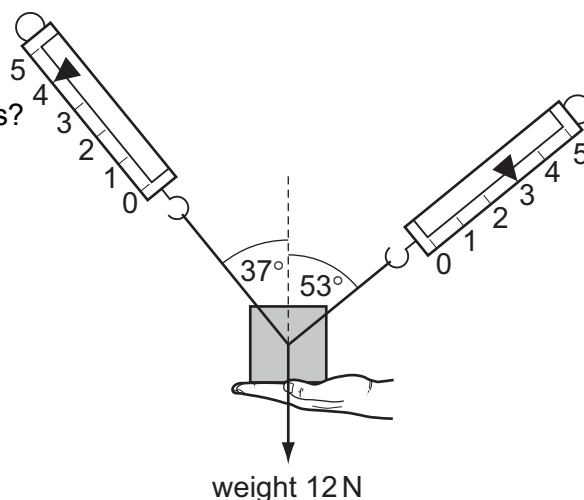
- A** The sum of their kinetic energies before impact is zero.
B The sum of their kinetic energies before impact is $\frac{1}{2}mv^2$.
C The sum of their kinetic energies after impact is zero.
D The sum of their kinetic energies after impact is mv^2 .

72 9702/12/M/J/12/Q14

A 1.2 kg mass is supported by a person's hand and two newton-meters as shown.

When the person's hand is removed,
What is the initial vertical acceleration of the mass?

- A 0.6 ms^{-2}
- B 2 ms^{-2}
- C 4 ms^{-2}
- D 6 ms^{-2}



73 9702/12/O/N/12/Q15

A lorry of mass 20 000 kg has a constant resultant force F acting on it.

It accelerates from 6.0 ms^{-1} to 30.0 ms^{-1} in a time of 300 s.

What is the change in momentum of the lorry and the value of F ?

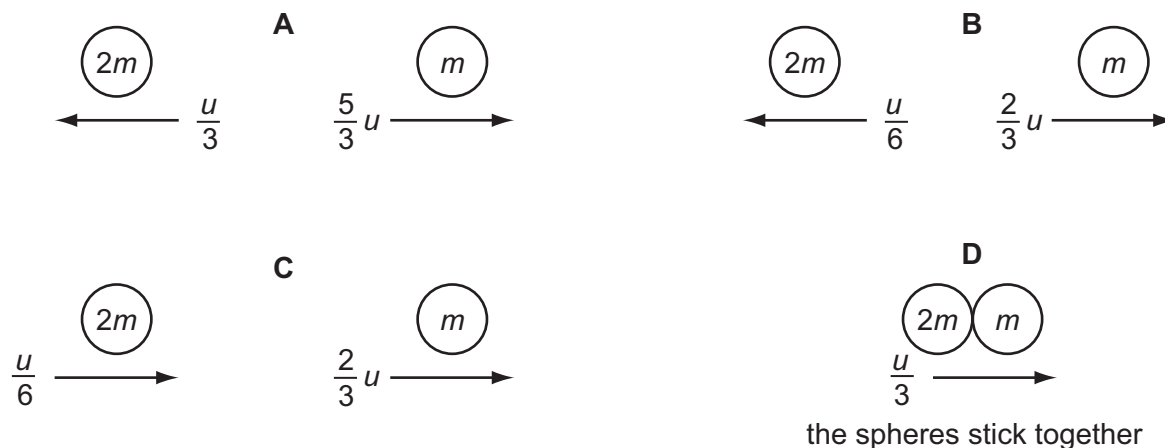
	change in momentum / N s	force F / N
A	48 000	160
B	480 000	1600
C	600 000	2000
D	600 000	20 000

74 9702/13/O/N/12/Q11

The diagram shows two spherical masses approaching each other head-on at an equal speed u . One is of mass m and the other of mass $2m$.



Which diagram, showing the situation after the collision, is **not** consistent with the principle of conservation of momentum?



75 9702/13/O/N/12/Q11*

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the kinetic energy and in the momentum of the molecule caused by the collision?

	change in momentum	change in kinetic energy
A	0	0
B	0	mv^2
C	$2mv$	0
D	mv^2	0

76 9702/13/O/N/12/Q13

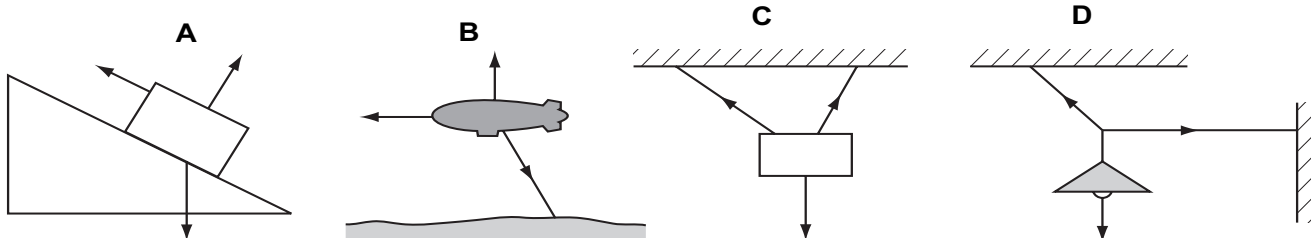
The IKAROS satellite has mass 320 kg and moves through space using a solar sail of area 20 m^2 . The average solar wind pressure is $1.0 \times 10^{-5} \text{ N m}^{-2}$.

What is the acceleration of the satellite caused by the solar wind?

- A $3.1 \times 10^{-8} \text{ m s}^{-2}$ C $3.2 \times 10^{-3} \text{ m s}^{-2}$
 B $6.3 \times 10^{-7} \text{ m s}^{-2}$ D $6.4 \times 10^{-2} \text{ m s}^{-2}$

77 9702/13/O/N/12/Q17

The diagrams show the forces acting on different bodies. Which body **cannot** be in equilibrium?

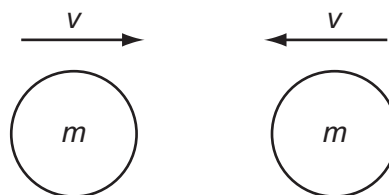


78 9702/1/M/J/02/Q9

Two similar spheres, each of mass m and travelling with speed v , are moving towards each other. The spheres have a head-on elastic collision.

Which statement is correct?

- A The spheres stick together on impact.
 B The total kinetic energy after impact is mv^2 .
 C The total kinetic energy before impact is zero.
 D The total momentum before impact is $2mv$.

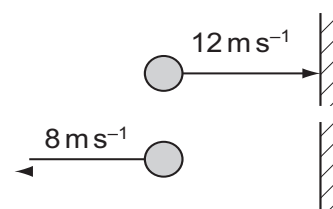


79 9702/12/O/N/12/Q12

A ball of mass 0.5 kg is thrown against a wall at a speed of 12 m s^{-1} . It bounces back with a speed of 8 m s^{-1} . The collision lasts for 0.10 s.

What is the average force on the ball due to the collision?

- A 0.2 N B 1 N
 C 20 N D 100 N



80 9702/11/M/J/13/Q10

Which of the following is a statement of the principle of conservation of momentum?

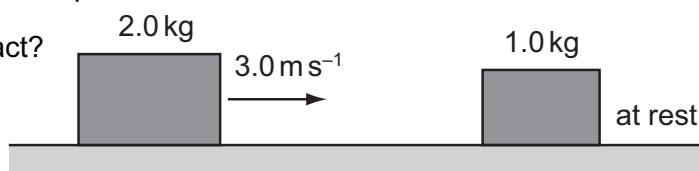
- A In an elastic collision momentum is constant.
- B Momentum is the product of mass and velocity.
- C The force acting on a body is proportional to its rate of change of momentum.
- D The momentum of an isolated system is constant.

81 9702/11/M/J/13/Q11

A 2.0 kg mass travelling at 3.0 ms^{-1} on a frictionless surface collides head-on with a stationary 1.0 kg mass. The masses stick together on impact.

How much kinetic energy is lost on impact?

- A zero
- B 2.0 J
- C 2.4 J
- D 3.0 J

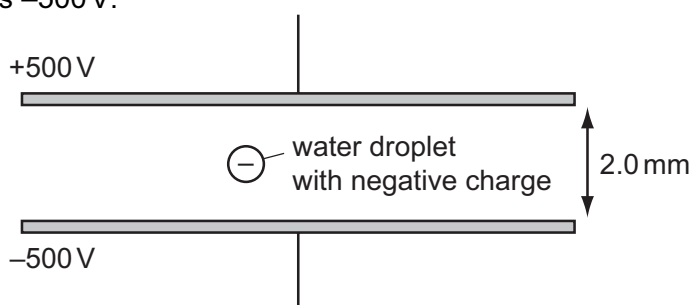


82 9702/11/M/J/13/Q13

A small water droplet of mass $3.0 \mu\text{g}$ carries a charge of $-6.0 \times 10^{-11} \text{ C}$. The droplet is situated in the Earth's gravitational field between two horizontal metal plates. The potential of the upper plate is +500 V and the potential of the lower plate is -500 V.

What is the motion of the droplet?

- A It accelerates downwards.
- B It remains stationary.
- C It accelerates upwards.
- D It moves upwards at a constant velocity.

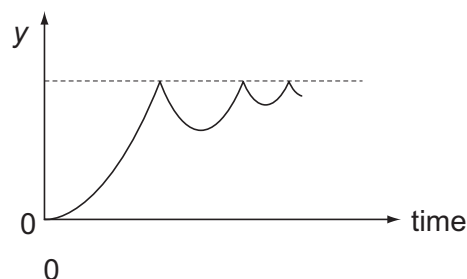


83 9702/12/M/J/13/Q8

A ball is released from rest above a horizontal surface and bounces several times. The graph shows how, for this ball, a quantity y varies with time.

What is the quantity y ?

- A acceleration
- B displacement
- C kinetic energy
- D velocity



84 9702/12/M/J/13/Q9

A strong wind of speed 33 ms^{-1} blows against a wall. The density of the air is 1.2 kg m^{-3} . The wall has an area of 12 m^2 at right angles to the wind velocity. The air has its speed reduced to zero when it hits the wall.

What is the approximate force exerted by the air on the wall?

- A 330 N
- B 400 N
- C 480 N
- D 16 000 N

85 9702/12/O/N/13/Q10

An astronaut of mass m in a spacecraft experiences a gravitational force $F = mg$ when stationary on the launchpad.

What is the gravitational force on the astronaut when the spacecraft is launched vertically upwards with an acceleration of $0.2g$?

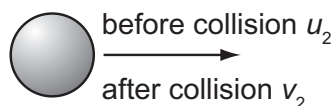
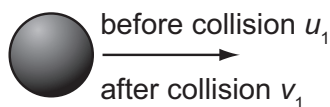
- A $1.2mg$
- B mg
- C $0.8mg$
- D 0

86 9702/11/O/N/16/Q13

Two spheres travel along the same line with velocities u_1 and u_2 . They collide and after collision their velocities are v_1 and v_2 .

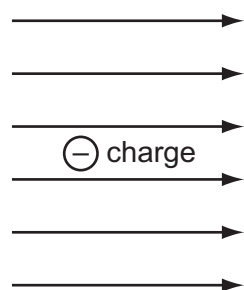
Which collision is **not** elastic?

	$u_1 / \text{m s}^{-1}$	$u_2 / \text{m s}^{-1}$	$v_1 / \text{m s}^{-1}$	$v_2 / \text{m s}^{-1}$
A	2	-5	-5	-2
B	3	-3	0	6
C	3	-2	1	6
D	5	2	3	6

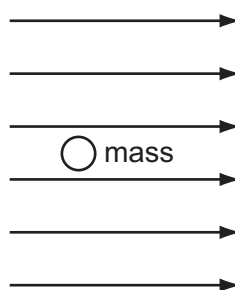


87 9702/13/M/J/13/Q12

The diagrams show a negative electric charge situated in a uniform electric field and a mass situated in a uniform gravitational field.



uniform electric field



uniform gravitational field

Which row shows the directions of the forces acting on the charge and on the mass?

	charge	mass
A		
B		
C		
D		

88 9702/12/M/J/13/Q10

Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements **must** be correct?

- A The initial speed of one body will be the same as the final speed of the other body.
- B The relative speed of approach between the two bodies equals their relative speed of separation.
- C The total momentum is conserved but the total kinetic energy will be reduced.
- D One of the bodies will be stationary at one instant.

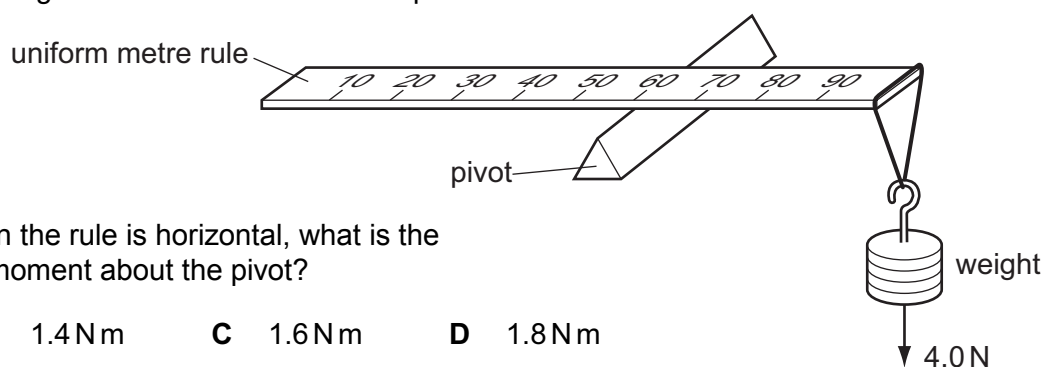
89 9702/12/O/N/13/Q9

What is meant by the mass and by the weight of an object on the Earth?

	mass	weight
A	its momentum divided by its velocity	the work done in lifting it one metre
B	the gravitational force on it	the property that resists its acceleration
C	the pull of the Earth on it	its mass divided by the acceleration of free fall
D	the property that resists its acceleration	the pull of the Earth on it

90 9702/12/O/N/13/Q15

A uniform metre rule of weight 2.0 N is pivoted at the 60 cm mark. A 4.0 N weight is suspended from one end, causing the rule to rotate about the pivot.



At the instant when the rule is horizontal, what is the resultant turning moment about the pivot?

- A** zero **B** 1.4 N m **C** 1.6 N m **D** 1.8 N m

91 9702/11/M/J/14/Q9

An object of mass 4.0 kg moving with a speed of 3.0 ms^{-1} strikes a stationary object in an inelastic collision.

Which statement is correct?

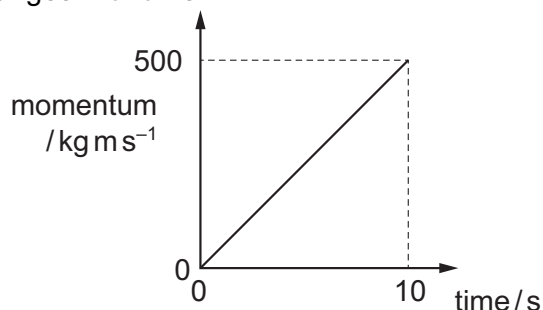
- A** After collision, the total kinetic energy is 18 J.
B After collision, the total kinetic energy is less than 18 J.
C Before collision, the total kinetic energy is 12 J.
D Before collision, the total kinetic energy is less than 12 J.

92 9702/11/M/J/14/Q10

The graph shows how the momentum of a motorcycle changes with time.

What is the resultant force on the motorcycle?

- A** 50 N **B** 500 N
C 2500 N **D** 5000 N



93 9702/11/O/N/14/Q9

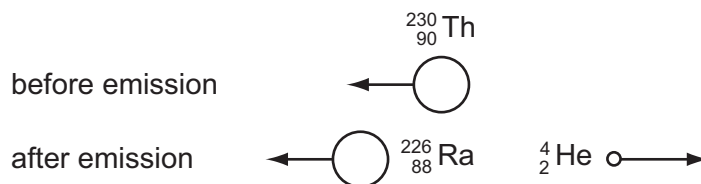
Two railway trucks of masses m and $3m$ move towards each other in opposite directions with speeds $2v$ and v respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

- A** $\frac{v}{4}$ **B** $\frac{v}{2}$ **C** v **D** $\frac{5v}{4}$

94 9702/13/O/N/13/Q10

A moving thorium nucleus ${}^{230}_{90}\text{Th}$ spontaneously emits an α -particle. The nucleus formed is a radium nucleus ${}^{226}_{88}\text{Ra}$, as shown.



Which statement is correct?

- A The kinetic energy of the α -particle equals the kinetic energy of the radium nucleus.
- B The momentum of the α -particle equals the momentum of the radium nucleus.
- C The total momentum before the emission equals the total momentum after the emission.
- D The velocity of the α -particle equals the velocity of the radium nucleus.

95 9702/13/O/N/13/Q11

An isolated system consists of two bodies on which no external forces act. The two bodies collide with each other and stick together on impact.

Which row correctly compares the total kinetic energy and the total momentum of the bodies before and after the collision?

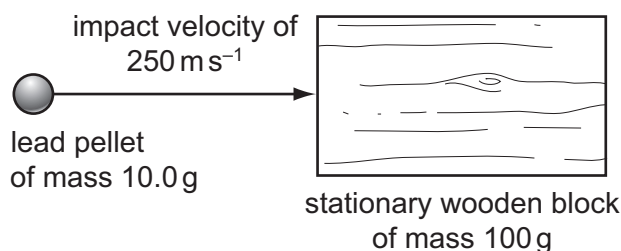
	total kinetic energy before and after the collision	total momentum before and after the collision
A	different	different
B	different	the same
C	the same	different
D	the same	the same

96 9702/13/O/N/13/Q13

A lead pellet of mass 10.0 g is shot horizontally into a stationary wooden block of mass 100 g. The pellet hits the block with an impact velocity of 250 m s^{-1} . It embeds itself in the block and it does not emerge.

What will be the speed of the block immediately after the pellet is embedded?

- A 23 m s^{-1}
- B 25 m s^{-1}
- C 75 m s^{-1}
- D 79 m s^{-1}



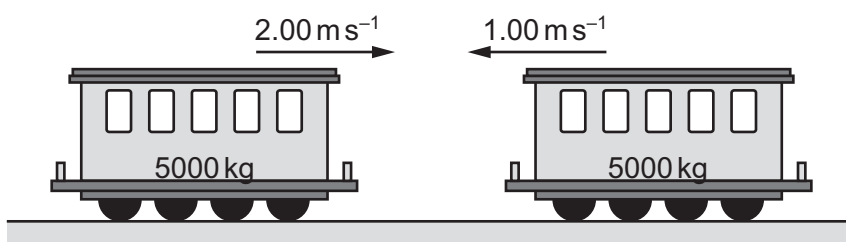
97 9702/12/M/J/14/Q7

Two train carriages each of mass 5000 kg roll toward one another on a level track. One is travelling at 2.00 m s^{-1} and the other at 1.00 m s^{-1} , as shown.

They collide and join together.

What is the kinetic energy lost during the collision?

- A 1250 J
- B 7500 J
- C 11 250 J
- D 12 500 J



98 9702/12/M/J/14/Q8

A resultant force causes a body to accelerate.

What is equal to the resultant force?

- A** the acceleration of the body per unit mass
- B** the change in kinetic energy of the body per unit time
- C** the change in momentum of the body per unit time
- D** the change in velocity of the body per unit time

99 9702/12/M/J/14/Q9

A ship of mass 8.4×10^4 kg is approaching a harbour with speed 16.4 m s^{-1} . By using reverse thrust it can maintain a constant total stopping force of 920 000 N.

How long will it take to stop?

- A** 15 seconds **B** 150 seconds **C** 25 minutes **D** 250 minutes

100 9702/11/O/N/14/Q7

What is the principle of conservation of momentum?

- A** Force is equal to the rate of change of momentum.
- B** Momentum is the product of mass and velocity.
- C** The total momentum of a system remains constant provided no external force acts on it.
- D** The total momentum of two bodies after collision is equal to their total momentum before collision.

101 9702/11/O/N/14/Q8

Water is pumped through a hose-pipe at a rate of 90 kg per minute. It emerges from the hose-pipe horizontally with a speed of 20 m s^{-1} .

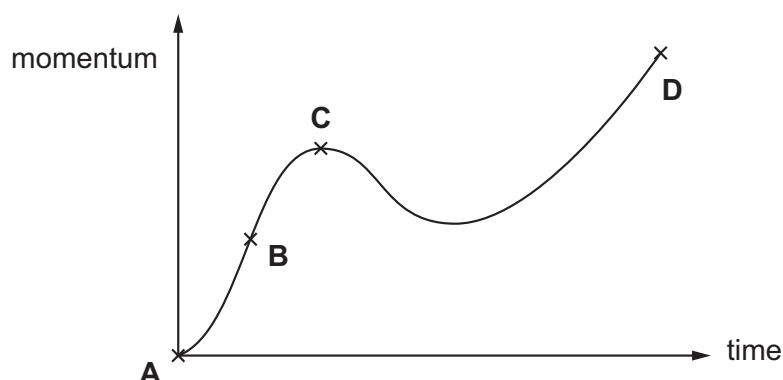
Which force is required from a person holding the hose-pipe to prevent it moving backwards?

- A** 30 N **B** 270 N **C** 1800 N **D** 10 800 N

102 9702/13/O/N/14/Q10

A body experiences a varying resultant force that causes its momentum to vary, as shown in the graph.

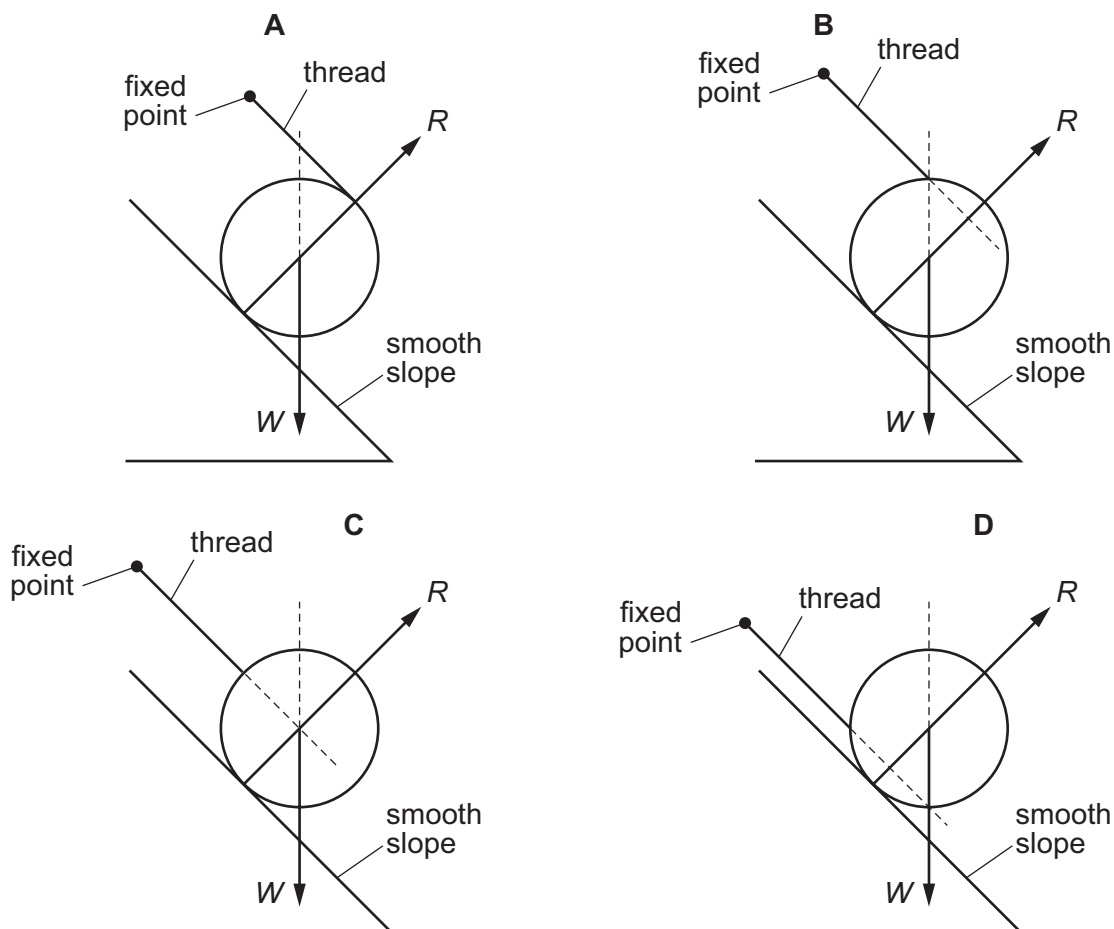
At which point does the resultant force have the largest value?



103 9702/13/O/N/14/Q13

A cylinder of weight W is placed on a smooth slope. The contact force of the slope on the cylinder is R . A thread is attached to the surface of the cylinder. The other end of the thread is fixed.

Which diagram shows the cylinder in equilibrium?



104 9702/12/M/J/15/Q11

Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

- A Some of the kinetic energy of trolley X is changed to momentum in the collision.
- B Some of the momentum of trolley X is changed to kinetic energy in the collision.
- C Trolley X loses some of its momentum as heat in the collision.
- D Trolley X shares its momentum with trolley Y but some of its kinetic energy is lost.

105 9702/12/M/J/15/Q12

An astronaut throws a stone with a horizontal velocity near to the Moon's surface.

Which row describes the horizontal and vertical forces acting on the stone after release?

	horizontal force	vertical force
A	constant	constant
B	constant	decreasing
C	zero	constant
D	zero	decreasing

106 9702/11/M/J/15/Q10

What is a reasonable estimate of the average gravitational force acting on a fully grown woman standing on the Earth?

- A** 60 N **B** 250 N **C** 350 N **D** 650 N

107 9702/13/O/N/12/Q12

A molecule of mass m travelling at speed v hits a wall in a direction perpendicular to the wall. The collision is elastic.

What are the changes in the momentum and in the kinetic energy of the molecule caused by the collision?

	change in momentum	change in kinetic energy
A	0	0
B	0	mv^2
C	$2mv$	0
D	mv^2	0

108 9702/11/M/J/15/Q12

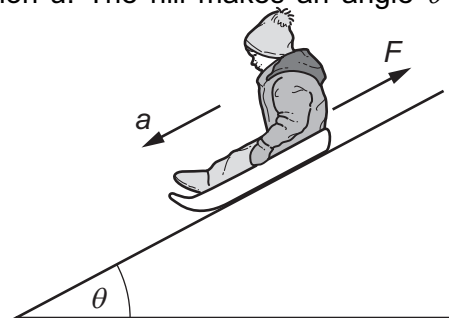
A child on a sledge slides down a hill with acceleration a . The hill makes an angle θ with the horizontal.

The total mass of the child and the sledge is m

The acceleration of free fall is g .

What is the friction force F ?

- A** $m(g \cos \theta - a)$
- B** $m(g \cos \theta + a)$
- C** $m(g \sin \theta - a)$
- D** $m(g \sin \theta + a)$



109 9702/12/M/J/17/Q8

A golf ball of mass m is dropped onto a hard surface from a height h_1 and rebounds to a height h_2 .

The momentum of the golf ball just as it reaches the surface is different from its momentum just as it leaves the surface.

What is the total change in the momentum of the golf ball between these two instants? (Ignore air resistance.)

- A** $m\sqrt{2gh_1} - m\sqrt{2gh_2}$ **C** $m\sqrt{2g(h_1 - h_2)}$
B $m\sqrt{2gh^1} + m\sqrt{2gh^2}$ **D** $m\sqrt{2g(h^1 + h^2)}$

110 9702/12/M/J/15/Q13

Newton's third law of motion is often summarised as 'Every action (force) has an equal and opposite reaction.'

A book rests on a table.

If the weight of the book is the 'action' force, what is the 'reaction' force?

- A** the pull of the book on the Earth **C** the push of the book on the table
B the pull of the Earth on the book **D** the push of the table on the book

111 9702/13/M/J/15/Q11

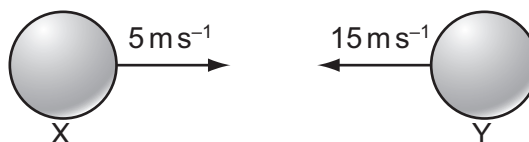
A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

	total momentum before collision / kg m s^{-1}	total momentum after collision / kg m s^{-1}	total kinetic energy before collision / J	total kinetic energy after collision / J
A	6	2	90	30
B	6	6	30	90
C	6	6	90	30
D	6	6	90	90

112 9702/13/M/J/15/Q12

Two balls X and Y are moving towards each other with speeds of 5 m s^{-1} and 15 m s^{-1} respectively.



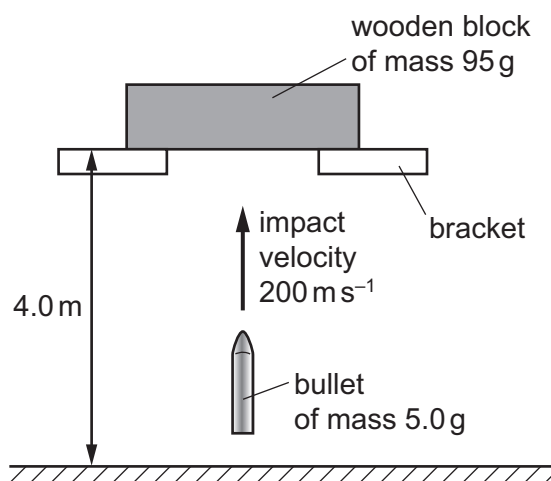
They make a perfectly elastic head-on collision and ball Y moves to the right with a speed of 7 m s^{-1} .

What is the speed and direction of ball X after the collision?

- A 3 m s^{-1} to the left C 3 m s^{-1} to the right
B 13 m s^{-1} to the left D 13 m s^{-1} to the right

113 9702/13/M/J/15/Q13

A wooden block is freely supported on brackets at a height of 4.0 m above the ground, as shown.



A bullet of mass 5.0 g is shot vertically upwards into the wooden block of mass 95 g . It embeds itself in the block. The impact causes the block to rise above its supporting brackets.

The bullet hits the block with a velocity of 200 m s^{-1} . How far above the ground will the block be at the maximum height of its path?

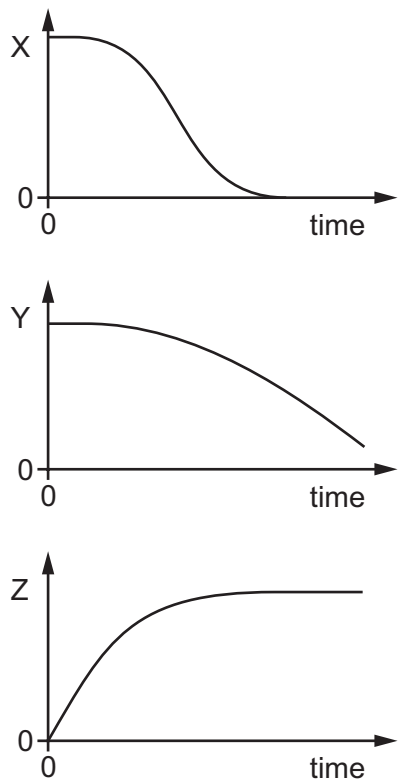
- A 5.1 m B 5.6 m C 9.1 m D 9.6 m

114 9702/11/M/J/17/Q8

An object is dropped at time $t = 0$ from a high building. Air resistance is significant.

Three graphs are plotted against time.

- the height of the object above the ground
- the speed of the object
- the magnitude of the resultant force on the object



What are the quantities X, Y and Z?

	height of the object above the ground	speed of the object	magnitude of the resultant force on the object
A	X	Y	Z
B	X	Z	Y
C	Y	Z	X
D	Z	Y	X

115 9702/11/M/J/17/Q7

The mass of a rocket-propelled truck is approximately equal to the mass of the fuel in its tank. The fuel is ignited and the truck is propelled along horizontal tracks by a constant force. The effect of air resistance is negligible.

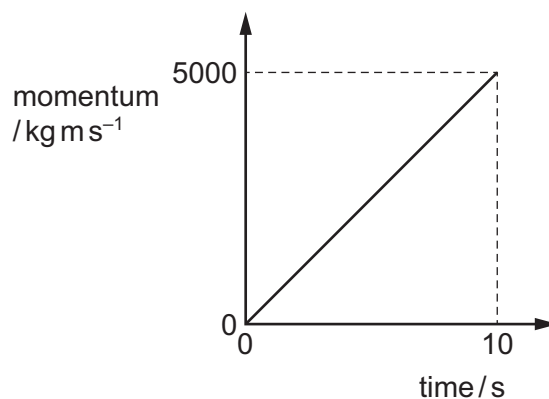
During a test run the fuel is consumed at a constant rate.

Which statement describes the acceleration of the truck during the test run?

- A** The acceleration of the truck decreases as the fuel is consumed.
- B** The acceleration of the truck increases as the fuel is consumed.
- C** The acceleration of the truck remains constant.
- D** The acceleration of the truck is zero and the truck moves at a constant velocity.

116 9702/11/M/J/17/Q10

The graph shows how the momentum of a motorcycle changes with time.



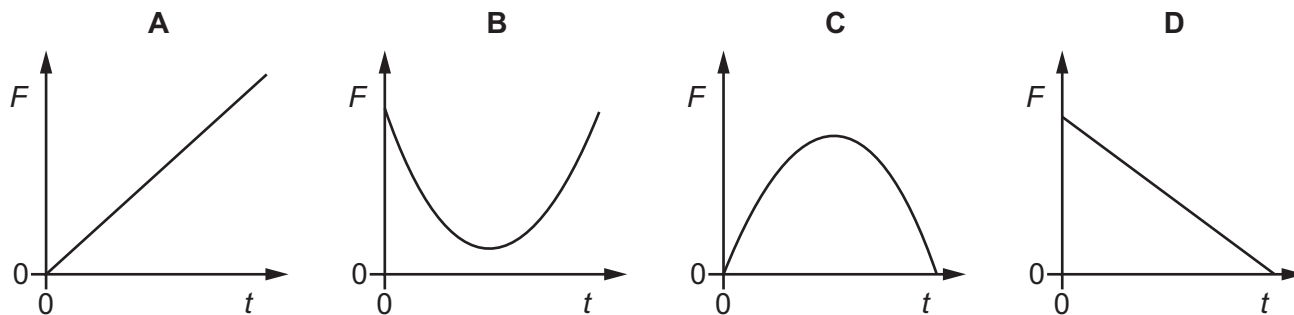
What is the resultant force on the motorcycle?

- A** 500 N
- B** 5000 N
- C** 25000 N
- D** 50000 N

117 9702/12/M/J/17/Q7

A rubber ball is dropped onto a table and bounces back up. The table exerts a force F on the ball.

Which graph best shows the variation with time t of the force F for the short time that the ball is in contact with the table?



118 9702/12/M/J/17/Q8

A golf ball of mass m is dropped onto a hard surface from a height h_1 and rebounds to a height h_2 .

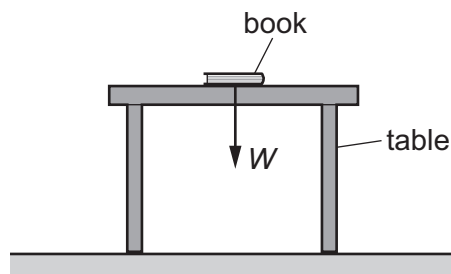
The momentum of the golf ball just as it reaches the surface is different from its momentum just as it leaves the surface.

What is the total change in the momentum of the golf ball between these two instants? (Ignore air resistance.)

- A** $m\sqrt{2gh_1} - m\sqrt{2gh_2}$ **C** $m\sqrt{2g(h_1 - h_2)}$
B $m\sqrt{2gh_1} + m\sqrt{2gh_2}$ **D** $m\sqrt{2g(h_1 + h_2)}$

119 9702/12/M/J/17/Q9

A book of weight W is at rest on a table. A student attempts to state Newton's third law of motion by saying that 'action equals reaction'.



If the weight of the book is the 'action' force, what is the 'reaction' force?

- A** the force W acting downwards on the Earth from the table
B the force W acting upwards on the book from the table
C the force W acting upwards on the Earth from the book
D the force W acting upwards on the table from the floor

120 9702/12/M/J/17/Q14

Which quantities are conserved in an inelastic collision?

	kinetic energy	total energy	linear momentum
A	conserved	not conserved	conserved
B	conserved	not conserved	not conserved
C	not conserved	conserved	conserved
D	not conserved	conserved	not conserved

121 9702/12/M/J/17/Q16

A stone of mass m falls from rest at the top of a cliff of height h into the sea below. Just before hitting the sea the stone has speed v .

What is the average force of air resistance acting on the stone during its fall?

- A** mg **B** $\frac{m(v^2 - 2gh)}{h}$ **C** $m\left(g - \frac{v^2}{2h}\right)$ **D** $m\left(gh - \frac{v^2}{2}\right)$

122 9702/11/O/N/17/Q8

The three forces acting on a hot-air balloon that is moving vertically are its weight, the force due to air resistance and the upthrust force.

The hot-air balloon descends vertically at constant speed. The force of air resistance on the balloon is F .

Which weight of material must be released from the balloon so that it ascends vertically at the same constant speed?

- A** F **B** $2F$ **C** $3F$ **D** $4F$

123 9702/11/O/N/17/Q9

A car is moving at constant speed in a straight line with the engine providing a driving force equal to the resistive force F .

When the engine is switched off, the car is brought to rest in a distance of 100 m by the resistive force.

It may be assumed that F is constant during the deceleration.

The process is then repeated for the same car with the same initial speed but with a constant resistive force of $0.800 F$.

How far will the car travel while decelerating?

- A** 120 m **B** 125 m **C** 156 m **D** 250 m

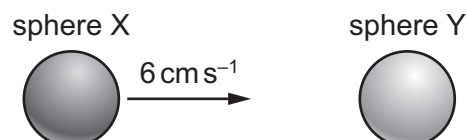
124 9702/11/O/N/17/Q10

What is a statement of the principle of conservation of momentum?

- A** In an elastic collision momentum is constant.
B Momentum is the product of mass and velocity.
C The force acting on a body is proportional to its rate of change of momentum.
D The momentum of an isolated system is constant.

125 9702/11/O/N/17/Q11

Two solid spheres form an isolated system. Sphere X moves with speed 6 cm s^{-1} in a straight line directly towards a stationary sphere Y, as shown.



The spheres have a perfectly elastic collision. After the collision, sphere X moves with speed 2 cm s^{-1} in the same direction as before the collision.

What is the speed of sphere Y?

- A** 2 cm s^{-1} **B** 4 cm s^{-1} **C** 6 cm s^{-1} **D** 8 cm s^{-1}

126 9702/12/O/N/17/Q9

A slow vehicle and a fast vehicle travel towards each other in a straight line and then collide. Which outcome is **never** possible, regardless of the masses of the vehicles?

- A Both vehicles stop.
- B Only one vehicle stops.
- C The fast vehicle's speed increases.
- D The slow vehicle's speed increases.

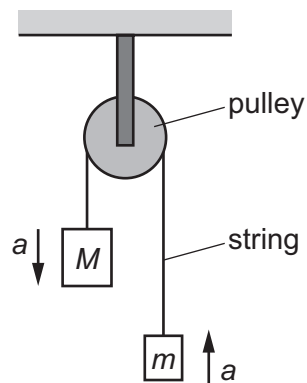
127 9702/12/O/N/17/Q10

Two blocks of masses M and m are joined by a thin string which passes over a frictionless pulley, as shown.

The acceleration of free fall is g .

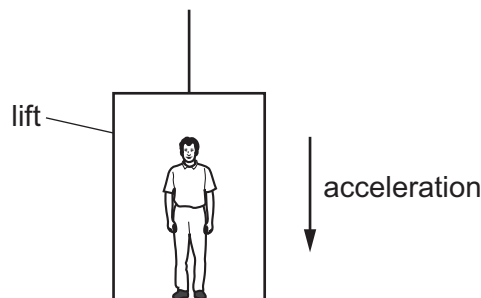
What is the acceleration a of the two blocks?

- A $\frac{(M+m)}{(M-m)}g$
- B $\frac{(M-m)}{(M+m)}g$
- C $\frac{M}{m}g$
- D $\frac{m}{M}g$



128 9702/13/O/N/17/Q8

A man stands in a lift that is accelerating vertically downwards, as shown.



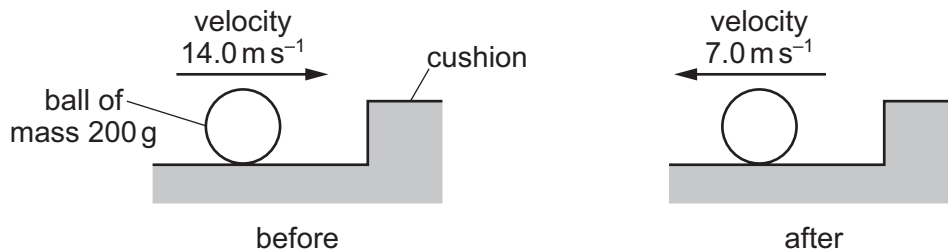
Which statement describes the force exerted by the man on the floor?

- A It is equal to the weight of the man.
- B It is greater than the force exerted by the floor on the man.
- C It is less than the force exerted by the floor on the man.
- D It is less than the weight of the man.

129 9702/13/O/N/17/Q9

A snooker ball of mass 200 g hits the cushion of a snooker table at right-angles with a speed of 14.0 m s^{-1} .

The ball rebounds with half of its initial speed. The ball is in contact with the cushion for 0.60 s .



What is the average force exerted on the ball by the cushion?

- A** 2.3 N **B** 7.0 N **C** 2300 N **D** 7000 N

130 9702/13/O/N/17/Q10

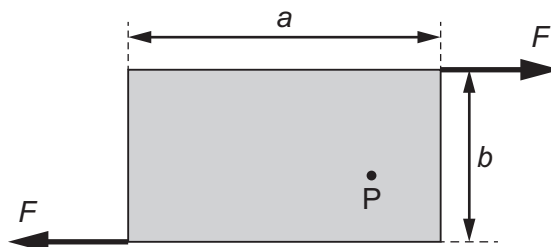
Two railway trucks of masses m and $3m$ move towards each other in opposite directions with speeds $2v$ and v respectively. These trucks collide and stick together.

What is the speed of the trucks after the collision?

- A** $\frac{v}{4}$ **B** $\frac{v}{2}$ **C** v **D** $\frac{5v}{4}$

131 9702/13/O/N/17/Q12

Two forces, each of magnitude F , act along the edges of a rectangular metal plate, as shown.



The plate has length a and width b . What is the torque about point P ?

- A** Fa **B** Fb **C** $2Fa$ **D** $2Fb$

132 9702/13/O/N/17/Q15

A ball is thrown vertically upwards. Air resistance is negligible.

Which statement is correct?

- A** By the principle of conservation of energy, the total energy of the ball is constant throughout its motion.
B By the principle of conservation of momentum, the momentum of the ball is constant throughout its motion.
C The kinetic energy of the ball is greatest at the greatest height attained.
D The potential energy of the ball increases at a constant rate during its ascent.

133 9702/13/O/N/17/Q16

A car of total mass 1560 kg is travelling with a constant speed of 32 ms^{-1} . The driving force provided by the car is 680 N. The kinetic energy of the car is 800 kJ and its momentum is 50 000 N s.

Which two items of data could be used to calculate the useful power output of the car?

- | | |
|-------------------------------------|----------------------------------|
| A driving force and momentum | C mass and momentum |
| B kinetic energy and mass | D speed and driving force |

134 9702/11/M/J/18/Q8

The momentum of a car of mass m increases from p_1 to p_2 .

What is the increase in the kinetic energy of the car?

- | | | | |
|---------------------------------------|-------------------------------------|---------------------------------|---------------------------------|
| A $\frac{(p_2^2 - p_1^2)}{2m}$ | B $\frac{(p_2 - p_1)^2}{2m}$ | C $\frac{p_2 - p_1}{2m}$ | D $\frac{p_1 - p_2}{2m}$ |
|---------------------------------------|-------------------------------------|---------------------------------|---------------------------------|

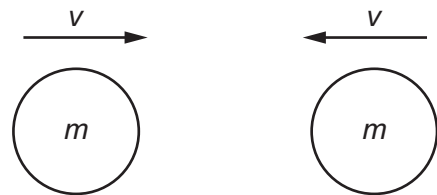
135 9702/11/M/J/18/Q9

Two similar spheres, each of mass m and travelling with speed v , are moving towards each other.

The spheres have a head-on elastic collision.

Which statement is correct?

- A** The spheres stick together on impact.
B The total kinetic energy after impact is mv^2 .
C The total kinetic energy before impact is zero.
D The total momentum before impact is $2mv$.



136 9702/12/M/J/18/Q8

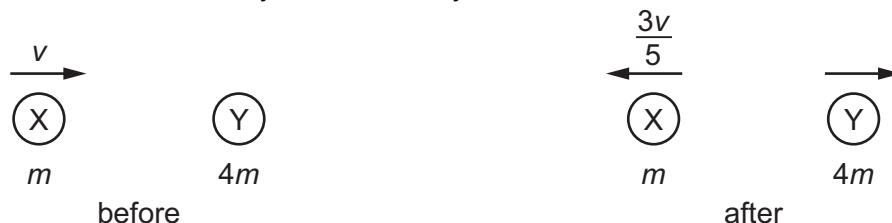
A tennis ball of mass 55 g is travelling horizontally with a speed of 30 ms^{-1} . The ball makes contact with a wall before rebounding in the horizontal direction with a speed of 20 ms^{-1} . The ball is in contact with the wall for a time of $5.0 \times 10^{-3} \text{ s}$.

What is the average force exerted on the wall by the ball?

- | | | | |
|----------------|----------------|----------------|----------------|
| A 110 N | B 220 N | C 330 N | D 550 N |
|----------------|----------------|----------------|----------------|

137 9702/12/M/J/18/Q9

An elastic collision occurs between two bodies X and Y. The mass of body X is m and the mass of body Y is $4m$. Body X travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Body Y is stationary before the collision.

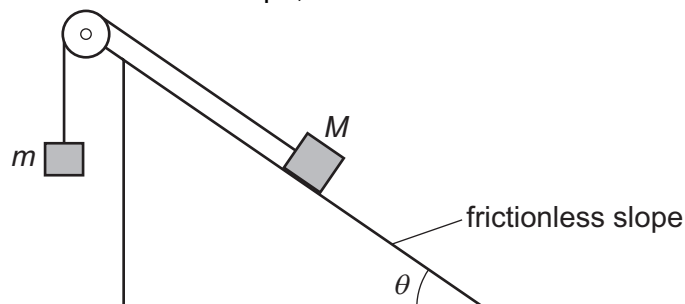


What is the kinetic energy of body Y after the collision?

- A $\frac{8}{10}mv^2$ B $\frac{34}{50}mv^2$ C $\frac{16}{50}mv^2$ D $\frac{1}{5}mv^2$

138 9702/13/M/J/18/Q7

Two masses, M and m , are connected by an inextensible string which passes over a frictionless pulley. Mass M rests on a frictionless slope, as shown.



The slope is at an angle θ to the horizontal.

The two masses are initially held stationary and then released. Mass M moves down the slope.

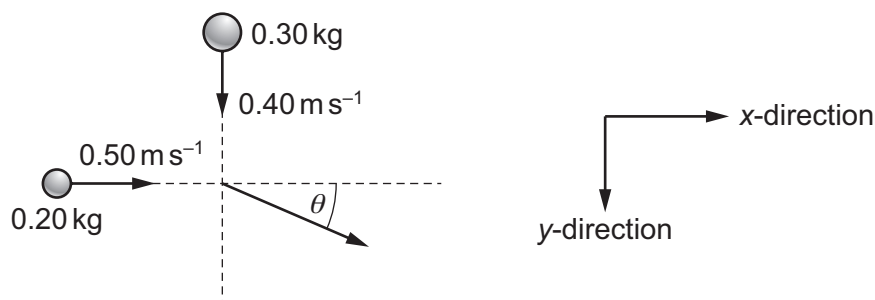
Which expression **must** be correct?

- A $\sin \theta < \frac{m}{M}$ B $\cos \theta < \frac{m}{M}$ C $\sin \theta > \frac{m}{M}$ D $\cos \theta > \frac{m}{M}$

139 9702/13/M/J/18/Q9

A ball of mass 0.20 kg , travelling in the x -direction at a speed of 0.50 ms^{-1} , collides with a ball of mass 0.30 kg travelling in the y -direction at a speed of 0.40 ms^{-1} .

The two balls stick together after the collision, travelling at an angle θ to the x -direction.



What is the value of θ ?

- A 39° B 40° C 50° D 51°

140 9702/12/F/M/18/Q10

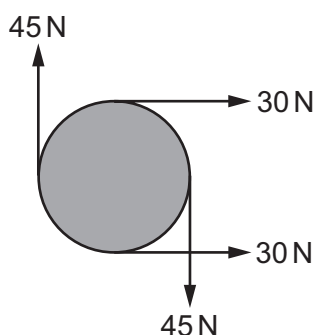
Steel pellets, each with a mass of 0.60 g, fall vertically onto a horizontal plate at a rate of 100 pellets per minute. They strike the plate with a velocity of 5.0 m s^{-1} and rebound with a velocity of 4.0 m s^{-1} .

What is the average force exerted on the plate by the pellets?

- A** 0.0010 N **B** 0.0054 N **C** 0.0090 N **D** 0.54 N

141 9702/12/F/M/18/Q11

The diagram shows four forces applied to a circular object.



Which row describes the resultant force and resultant torque on the object?

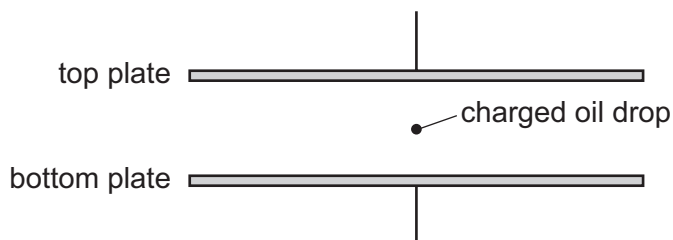
	resultant force	resultant torque
A	non-zero	non-zero
B	non-zero	zero
C	zero	non-zero
D	zero	zero

142 9702/12/F/M/18/Q12

A charged oil drop is held stationary between two charged parallel plates.

Which forces act on the oil drop?

- A** both electric and gravitational
B electric only
C gravitational only
D neither electric nor gravitational



143 9702/11/O/N/18/Q7

Water is pumped through a hose-pipe at a rate of 90 kg per minute. Water emerges horizontally from the hose-pipe with a speed of 20 m s^{-1} .

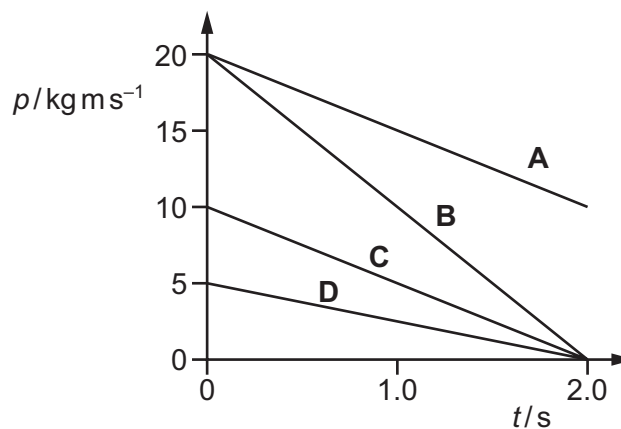
What is the minimum force required from a person holding the hose-pipe to prevent it moving backwards?

- A 30 N B 270 N C 1800 N D 108 000 N

144 9702/12/O/N/18/Q7

A resultant force of 10 N acts on a body for a time of 2.0 s.

Which graph could show the variation with time t of the momentum p of the body?



145 9702/12/O/N/18/Q9

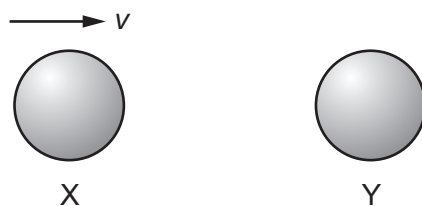
Two bodies travelling along the same straight line collide in a perfectly elastic collision.

Which statement **must** be correct?

- A The initial speed of one body will be the same as the final speed of the other body.
- B The relative speed of approach between the two bodies equals their relative speed of separation.
- C The total momentum is conserved but the total kinetic energy will be reduced.
- D One of the bodies will be stationary at one instant.

146 9702/12/O/N/18/Q10

The diagram shows two identical spheres X and Y.



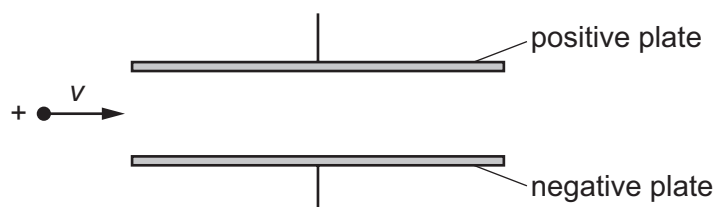
Initially, X moves with speed v directly towards Y. Y is stationary. The spheres collide elastically.

What happens?

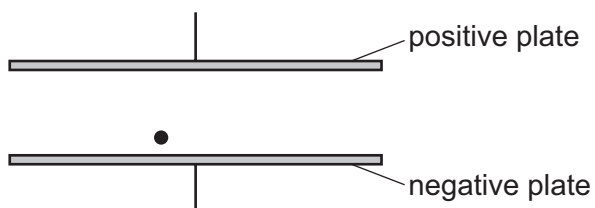
	X	Y
A	moves with speed $\frac{1}{2}v$ to the right	moves with speed $\frac{1}{2}v$ to the right
B	moves with speed v to the left	remains stationary
C	moves with speed $\frac{1}{2}v$ to the left	moves with speed $\frac{1}{2}v$ to the right
D	stops	moves with speed v to the right

147 9702/12/O/N/18/Q11

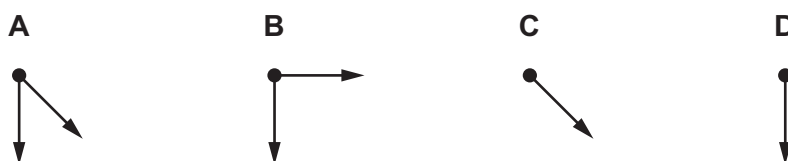
A positively-charged particle of negligible mass, moving at constant velocity v in a vacuum, enters a uniform electric field between two parallel plates, as shown.



A short time later, the particle is at the position shown.



Which diagram represents the force or forces acting on the particle?



148 9702/13/O/N/18/Q7

Two isolated spheres have masses 2.0 kg and 4.0 kg. The spheres collide and then move apart.

During the collision, the 2.0 kg mass has an average acceleration of 8.0 ms^{-2} .

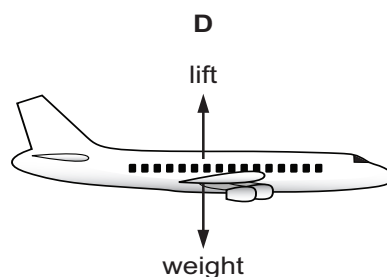
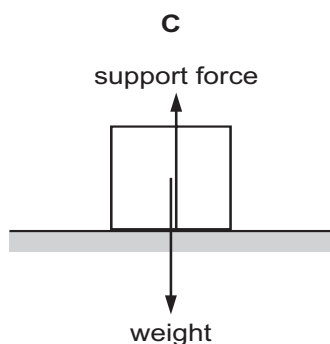
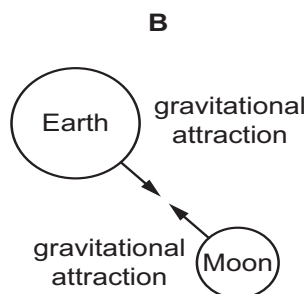
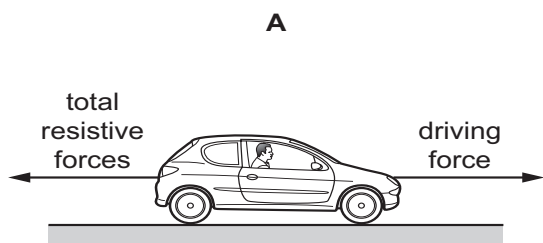
What is the average acceleration of the 4.0 kg mass?

- A** 2.0 ms^{-2} **B** 4.0 ms^{-2} **C** 8.0 ms^{-2} **D** 16 ms^{-2}

149 9702/11/M/J/19/Q9

Each diagram illustrates a pair of forces of equal magnitude.

Which diagram gives an example of a pair of forces that is described by Newton's third law of motion?

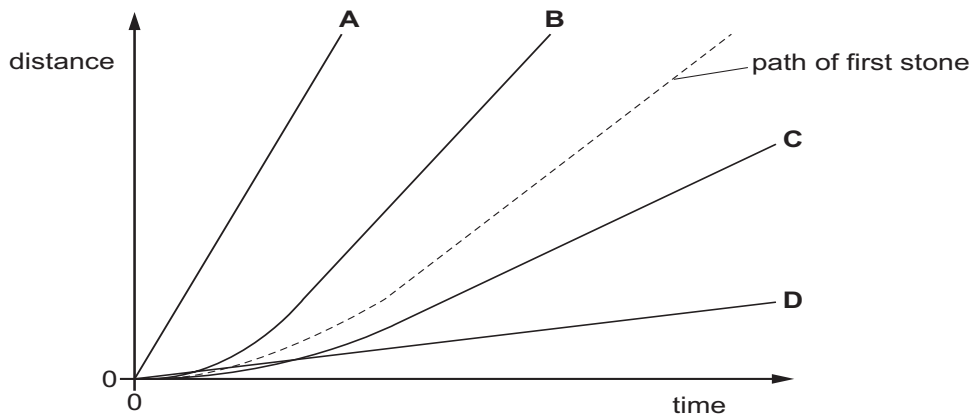


150 9702/11/M/J/19/Q10

A stone is dropped from a tall building. Air resistance is significant. The variation of distance fallen with time is shown by the dashed line.

A second stone with the same dimensions but a smaller mass is dropped from the same building.

Which line represents the motion of the second stone?



151 9702/11/M/J/19/Q11

A helium atom of mass m collides normally with a wall. The atom arrives at the wall with speed v and then rebounds along its original path. Assume that the collision is perfectly elastic.

What is the change in the momentum of the atom during its collision?

- A zero B $0.5mv$ C mv D $2mv$

152 9702/12/M/J/19/Q9

What describes the mass of an object?

- A the force the object experiences due to gravity
 B the momentum of the object before a collision
 C the resistance of the object to changes in motion
 D the weight of the object as measured by a balance

153 9702/12/M/J/19/Q10

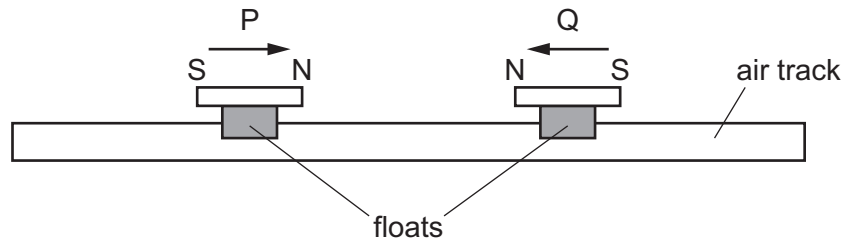
A car has mass m . A person needs to push the car with force F in order to give the car acceleration a . The person needs to push the car with force $2F$ in order to give the car acceleration $3a$.

Which expression gives the constant resistive force opposing the motion of the car?

- A ma B $2ma$ C $3ma$ D $4ma$

154 9702/12/M/J/19/Q11

Two bar magnets P and Q are mounted on floats which can slide without friction along an air track.



The two magnets slide towards each other along the air track and interact, without making contact.

The relative speed of approach of the magnets is equal to their relative speed of separation.

Which statement about P and Q must be correct?

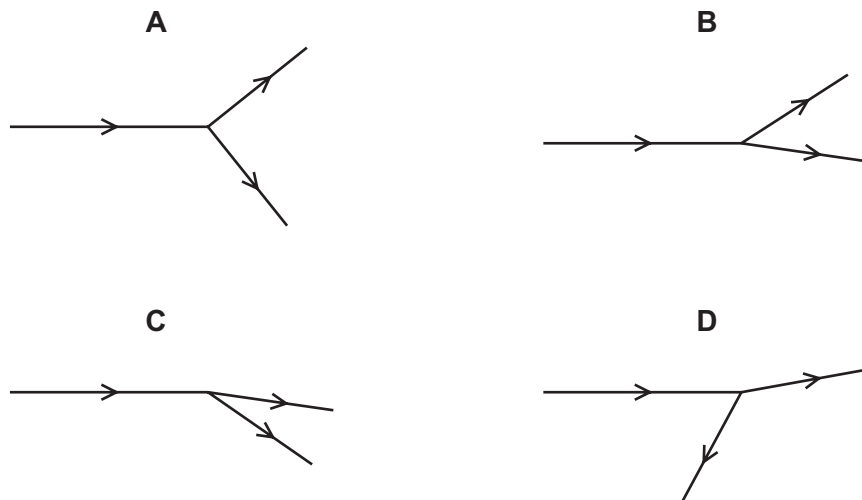
- A During the interaction between P and Q some of the total kinetic energy is lost.
- B During the interaction between P and Q some of the total momentum is lost.
- C The momentum of Q after the interaction is equal to the momentum of P before the interaction.
- D The values of (kinetic energy of P + kinetic energy of Q) before and after the interaction are equal.

155 9702/13/M/J/19/Q9

A nucleus collides with a stationary nucleus in a vacuum. The diagrams show the paths of the nuclei before and after the collision.

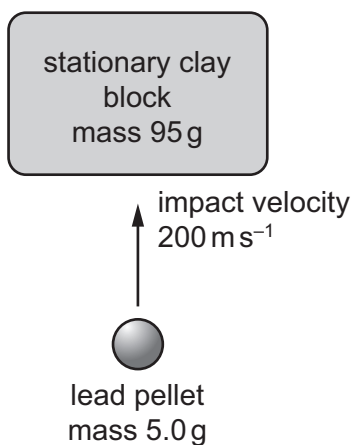
No other particles are involved in the collision.

Which diagram is **not** possible?



156 9702/13/M/J/19/Q17

A lead pellet is shot vertically upwards into a clay block that is stationary at the moment of impact but is able to rise freely after impact.



The mass of the pellet is 5.0 g and the mass of the clay block is 95 g.

The pellet hits the block with an initial vertical velocity of 200 ms^{-1} . It embeds itself in the block and does not emerge.

How high above its initial position will the block rise?

- A** 5.1 m **B** 5.6 m **C** 10 m **D** 100 m

157 9702/11/O/N/19/Q7

Which statement follows directly from Newton's first law?

- A** A body remains at constant velocity unless acted upon by a resultant force.
- B** A satellite in circular motion about the Earth has a constant velocity.
- C** A water drop leaving a spinning umbrella travels at a constant velocity.
- D** The force acting on an object is equal to its change in momentum.

158 9702/11/O/N/19/Q8

A resultant force causes an object to accelerate.

What is equal to the resultant force?

- A** the acceleration of the object per unit mass
- B** the change in kinetic energy of the object per unit time
- C** the change in momentum of the object per unit time
- D** the change in velocity of the object per unit time

159 9702/11/O/N/19/Q9

A skydiver falls from an aircraft that is moving horizontally.

The vertical component of the velocity of the skydiver is v .

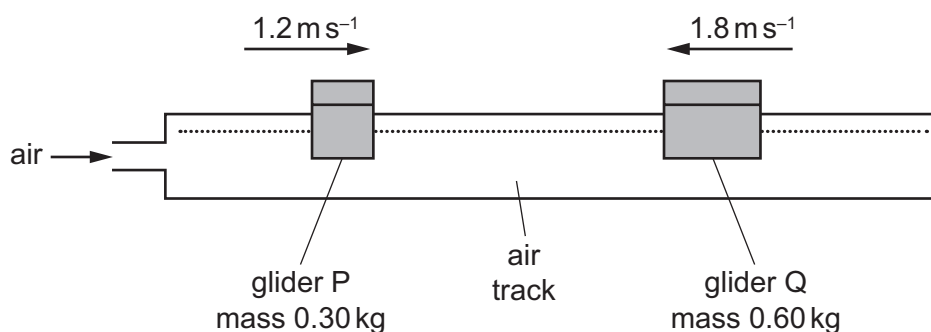
The vertical component of the acceleration of the skydiver is a .

Which row describes v and a during the first few seconds after the skydiver leaves the aircraft?

	v	a
A	constant	constant
B	constant	decreasing
C	increasing	constant
D	increasing	decreasing

160 9702/11/O/N/19/Q10

Two gliders are travelling towards each other on a horizontal air track. Glider P has mass 0.30 kg and is moving with a constant speed of 1.2 m s^{-1} . Glider Q has mass 0.60 kg and is moving with a constant speed of 1.8 m s^{-1} .



The gliders have a perfectly elastic collision.

What are the speeds of the two gliders after the collision?

	speed of P $/\text{m s}^{-1}$	speed of Q $/\text{m s}^{-1}$
A	1.2	0.6
B	2.0	1.4
C	2.8	0.2
D	3.6	0.6

161 9702/12/O/N/19/Q7

A snooker ball has a mass of 200 g. It hits the cushion of a snooker table and rebounds along its original path.

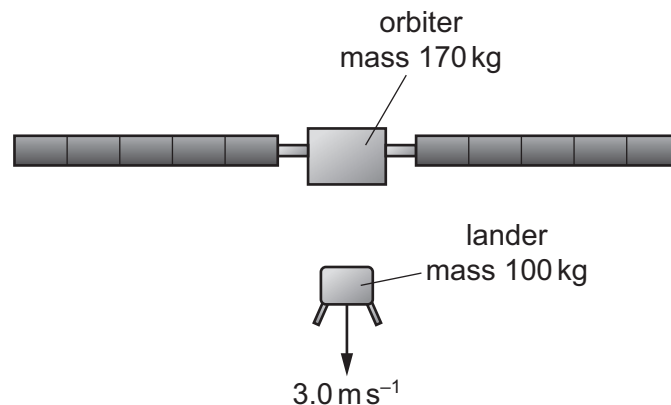
The ball arrives at the cushion with a speed of 14.0 m s^{-1} and then leaves it with a speed of 7.0 m s^{-1} . The ball and the cushion are in contact for a time of 0.60 s.

What is the average force exerted on the ball by the cushion?

- A** 1.4 N **B** 2.3 N **C** 4.2 N **D** 7.0 N

162 9702/12/O/N/19/Q9

The space probe Rosetta was designed to investigate a comet. The probe consisted of an orbiter and a lander. The orbiter had a mass of 170 kg and the lander had a mass of 100 kg. When the two parts separated, the lander was pushed towards the surface of the comet so that its change in velocity towards the comet was 3.0 m s^{-1} .



Assume that the orbiter and lander were an isolated system.

The orbiter moved away from the comet during the separation.

What was the change in the speed of the orbiter?

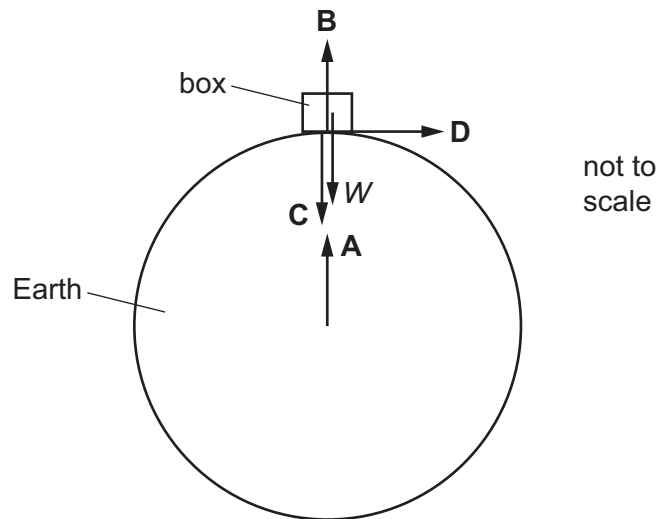
- A** 1.8 m s^{-1} **B** 2.3 m s^{-1} **C** 3.0 m s^{-1} **D** 5.1 m s^{-1}

163 9702/13/O/N/19/Q7

A box rests on the Earth, as shown.

Newton's third law describes how forces of the same type act in pairs. One of the forces of a pair is the weight W of the box.

Which arrow represents the other force of this pair?



164 9702/13/O/N/19/Q8

A snowflake is falling from the sky on a still day. Its weight acts vertically downwards and air resistance acts vertically upwards. As the snowflake falls, air resistance increases until it is equal to the weight and there is no resultant force acting on the snowflake.



When the forces become equal, which statement is correct?

- A** The snowflake accelerates.
- B** The snowflake decelerates.
- C** The snowflake is stationary.
- D** The snowflake moves at a constant velocity.

165 9702/13/O/N/19/Q9

Two objects X and Y in an isolated system undergo a perfectly elastic collision. The velocities of the objects before and after the collision are shown.

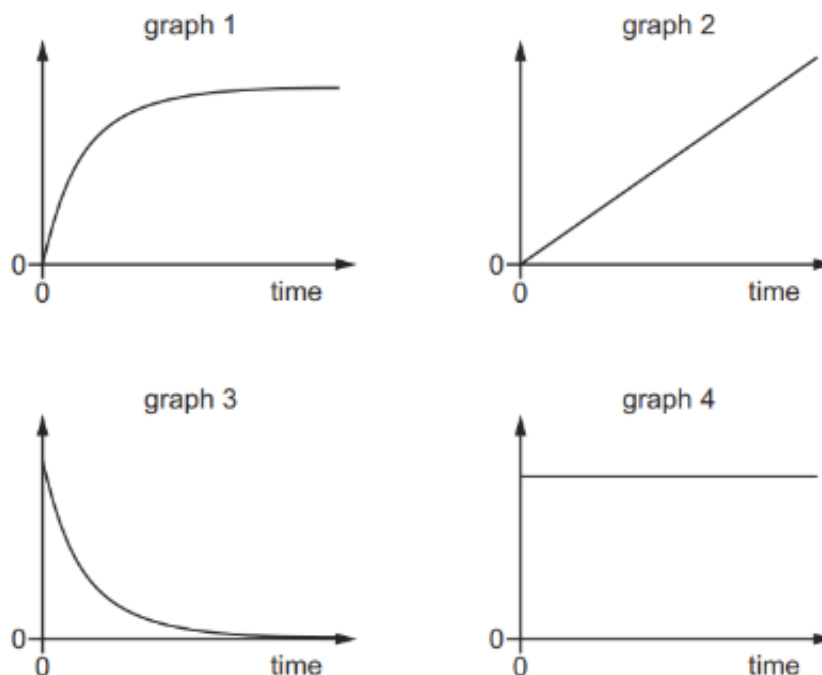


What is the speed v of Y after the collision?

- A 2.0 ms^{-1} B 18 ms^{-1} C 22 ms^{-1} D 24 ms^{-1}

166 9702/12/F/M/20/Q9

The diagram shows graphs of various quantities plotted against time for an object dropped from a stationary balloon high in the atmosphere.



Which statement could be correct?

- A Graph 1 is acceleration against time and graph 3 is resultant force against time.
 B Graph 1 is acceleration against time and graph 4 is resultant force against time.
 C Graph 3 is acceleration against time and graph 1 is velocity against time.
 D Graph 3 is acceleration against time and graph 2 is velocity against time.

167. 9702/12/F/M/20/Q8

A person of mass 60 kg stands on accurate bathroom scales, placed on the floor of an elevator (lift) which operates in a tall building.

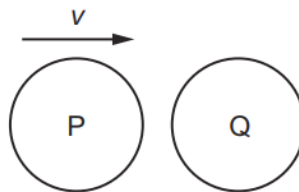
At a certain instant the bathroom scales read 58 kg.

Which row could give the person's direction of movement and type of motion?

	direction	motion
A	downwards	constant speed
B	downwards	slowing down
C	upwards	constant speed
D	upwards	slowing down

168. 9702/12/F/M/20/Q10

The diagram shows a particle P, travelling at speed v , about to collide with a stationary particle Q of the same mass. The collision is perfectly elastic.



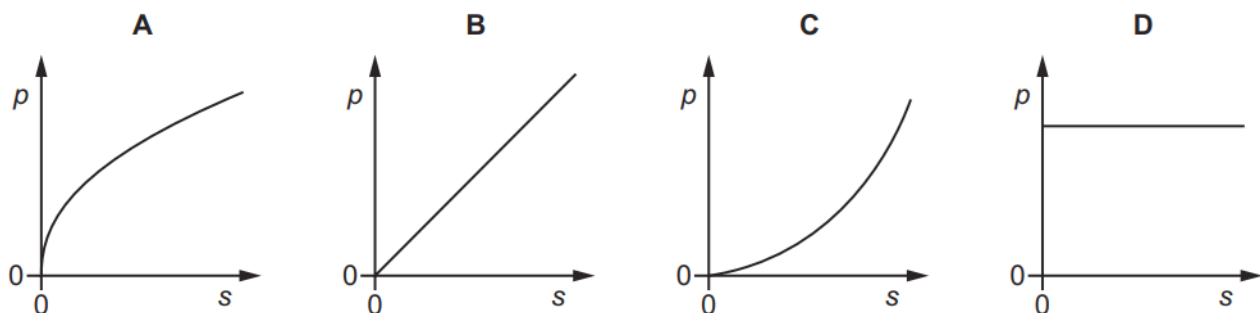
Which statement describes the motion of P and of Q immediately after the collision?

- A** P and Q both travel in the same direction with speed $\frac{1}{2}v$.
- B** P comes to rest and Q acquires speed v .
- C** P rebounds with speed $\frac{1}{2}v$ and Q acquires speed $\frac{1}{2}v$.
- D** P rebounds with speed v and Q remains stationary.

169. 9702/11/M/J/20/Q8

A car accelerates from rest in a straight line with constant acceleration.

Which graph best represents the variation of the momentum p of the car with the distance s travelled by the car?



170. 9702/11/M/J/20/Q10

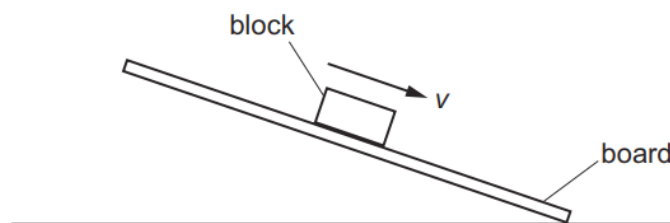
A stationary toy gun fires a bullet.

Which statement about the bullet and the gun, immediately after firing, is **not** correct?

- A** The force exerted on the bullet by the gun has the same magnitude as the force exerted on the gun by the bullet.
- B** The force exerted on the bullet by the gun is in the opposite direction to the force exerted on the gun by the bullet.
- C** The gun and the bullet have the same magnitude of momentum.
- D** The kinetic energy of the gun must equal the kinetic energy of the bullet.

171. 9702/11/M/J/20/Q11

A wooden block rests on the rough surface of a board. One end of the board is then raised until the block slides down the board at constant velocity v .



What describes the forces acting on the block when it is sliding with constant velocity?

	frictional force on block	resultant force on block
A	down the board	down the board
B	down the board	zero
C	up the board	down the board
D	up the board	zero

172. 9702/12/M/J/20/Q8

An astronaut has a weight of 660 N when she is standing on the Earth's surface.

The acceleration of free fall on the surface of Mars is 3.71 ms^{-2} .

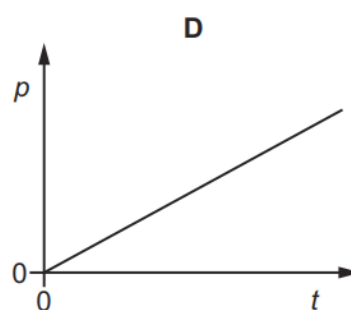
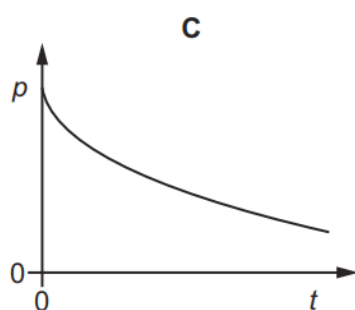
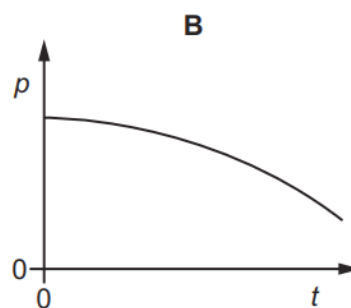
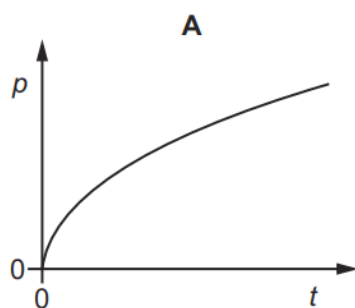
What would be the weight of the astronaut if she stood on the surface of Mars?

- A** 67.3 N **B** 178 N **C** 250 N **D** 660 N

173. 9702/12/M/J/20/Q7

The resultant force acting on an object is slowly increased.

Which graph could show the variation with time t of the momentum p of the object?



174. 9702/12/M/J/20/Q9

A mass m_1 travelling with speed u_1 collides with a mass m_2 travelling with speed u_2 in the same direction. After the collision, mass m_1 has speed v_1 and mass m_2 has speed v_2 in the same direction. The collision is perfectly elastic.



before the collision



after the collision

Which equation is **not** correct?

A $m_1 u_1^2 - m_1 v_1^2 = m_2 v_2^2 - m_2 u_2^2$

B $v_2 + u_2 = v_1 + u_1$

C $m_1(u_1 - v_1) = m_2(v_2 - u_2)$

D $m_1(u_1 - v_1)^2 = m_2(u_2 - v_2)^2$

175. 9702/13/M/J/20/Q8

A ball of mass m travels vertically downwards and then hits a horizontal floor at speed u .

It rebounds vertically upwards with speed v .

The collision lasts a time Δt .

What is the average resultant force exerted on the ball during the collision?

A $\frac{mv - mu}{\Delta t}$ downwards

B $\frac{mv - mu}{\Delta t}$ upwards

C $\frac{mv + mu}{\Delta t}$ downwards

D $\frac{mv + mu}{\Delta t}$ upwards

176. 9702/13/M/J/20/Q10

A ball of mass m , moving at a velocity v , collides with a stationary ball of mass $2m$.

The two balls stick together.

Which fraction of the initial kinetic energy is lost on impact?

A $\frac{1}{9}$

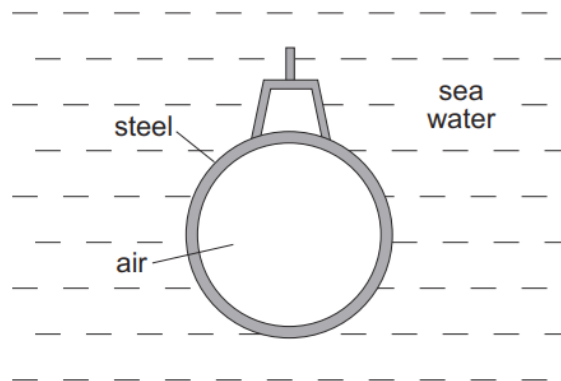
B $\frac{1}{3}$

C $\frac{2}{3}$

D $\frac{8}{9}$

177. 9702/13/M/J/20/Q11

A submarine is in equilibrium in a fully submerged position.



What causes the upthrust on the submarine?

A The air in the submarine is less dense than sea water.

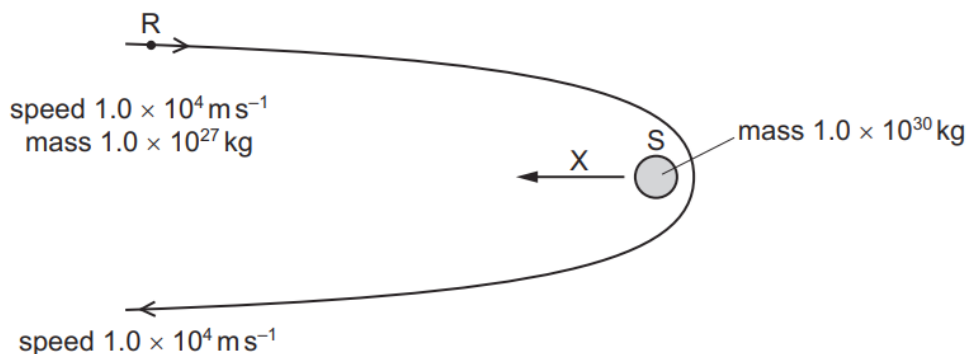
B There is a difference in water pressure acting on the top and on the bottom of the submarine.

C The sea water exerts a greater upward force on the submarine than the weight of the steel.

D The submarine displaces its own volume of sea water.

178. 9702/11/O/N/20/Q9

A rock R of mass 1.0×10^{27} kg is a large distance from a star S and is travelling at a speed of $1.0 \times 10^4 \text{ ms}^{-1}$. The star has mass 1.0×10^{30} kg. The rock travels around the star on the path shown so that it reverses its direction of motion and, when finally again a large distance from the star, has the same speed as initially.



Which statement is correct?

- A** The change in the momentum of S is in the direction of arrow X.
- B** The change in the velocity of S is approximately 20 ms^{-1} .
- C** The magnitude of the change of momentum of R is 10^3 times greater than the magnitude of the change of momentum of S.
- D** The momentum of R does not change.

179. 9702/11/O/N/20/Q10

The diagram shows the masses and velocities of two trolleys that are about to collide.



After the impact they move off together.

What is the kinetic energy lost in the collision?

- A** 4 J
- B** 6 J
- C** 12 J
- D** 14 J

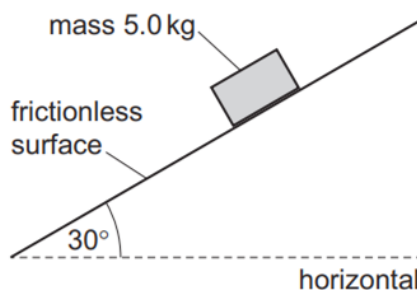
180. 9702/12/O/N/20/Q8

What is **not** a statement of one of Newton's laws of motion?

- A** If body X exerts a force on body Y, body Y exerts an equal and opposite force on body X.
- B** If no resultant force acts on a body it has constant velocity.
- C** The rate of change of momentum of a body is proportional to the resultant force acting on it and takes place in the direction of the force.
- D** The total momentum of a system of interacting bodies is constant if there is no external force.

181. 9702/12/O/N/20/Q7

A mass of 5.0 kg is released from rest on a frictionless surface inclined at 30° to the horizontal. Air resistance is negligible.



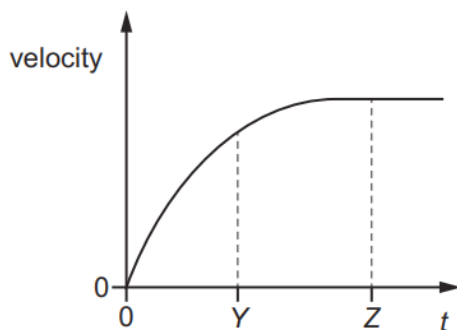
How far does the mass travel in a time of 0.80 s?

- A** 1.6 m **B** 2.0 m **C** 2.7 m **D** 3.1 m

182. 9702/12/O/N/20/Q9

An object falls from a tall building.

The graph shows how the velocity of the object changes with time t .



The acceleration of free fall is g .

What describes the acceleration of the object at times $t = Y$ and $t = Z$?

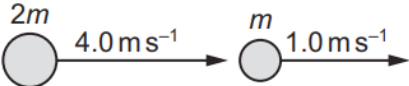
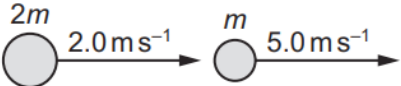
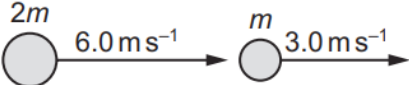
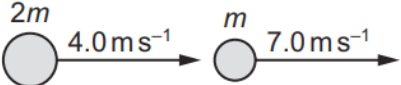
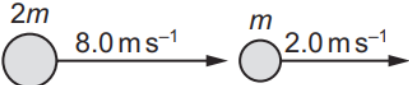
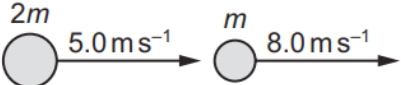
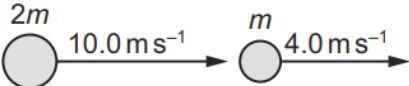
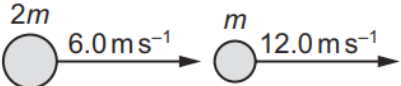
	acceleration at $t = Y$	acceleration at $t = Z$
A	decreasing	g
B	decreasing	0
C	constant	g
D	constant	0

183. 9702/12/O/N/20/Q10

Two balls, one of mass $2m$ and one of mass m , collide.

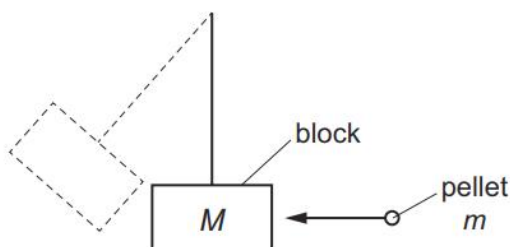
The diagrams show the initial and final velocities of the balls.

Which collision is **not** elastic?

	before collision	after collision
A		
B		
C		
D		

184. 9702/12/O/N/20/Q11

The diagram shows a 'ballistic pendulum'.



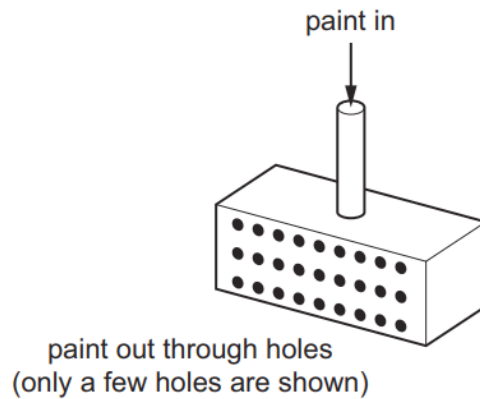
A pellet of mass m travelling at a speed u hits a stationary block of mass M . The pellet becomes embedded in the block and causes the block to move at a speed v immediately after the impact.

When a pellet of mass $2m$, travelling at a speed $2u$, hits a block of mass $2M$, what is the speed of the block immediately after the impact? (Neglect the small increase in the mass of the block as the pellet's mass is added during the collision.)

- A** v **B** $v\sqrt{2}$ **C** $2v$ **D** $4v$

185. 9702/13/O/N/20/Q8

A device for spraying paint consists of a box with its axes horizontal and vertical. One of its vertical faces contains small holes. Paint is fed into the box under pressure via a vertical tube and exits through the holes as fine streams moving horizontally.



The paint is ejected at a speed of 2.5 m s^{-1} through 400 holes, each of area 0.4 mm^2 . The density of the paint is 900 kg m^{-3} .

What is the horizontal force required to hold the device stationary as it ejects the paint?

- A** 0.36 N **B** 0.90 N **C** 2.3 N **D** 900 N

186. 9702/13/O/N/20/Q9

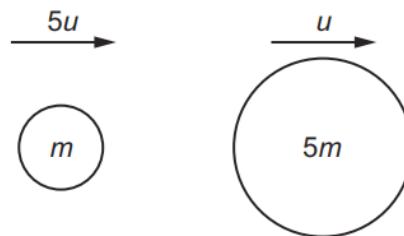
A party balloon is filled with air and held stationary at a height of several metres above the ground. The balloon is then dropped in still air.

Which statement describes the motion of the balloon from the moment of release until just before it hits the floor?

- A** The balloon decelerates continuously.
B The balloon falls at a constant speed and then decelerates.
C The balloon falls at a constant speed.
D The balloon initially accelerates and then reaches a constant speed.

187. 9702/13/O/N/20/Q10

An object of mass m travelling with speed $5u$ collides with, and sticks to, an object of mass $5m$ travelling in the same direction with speed u .



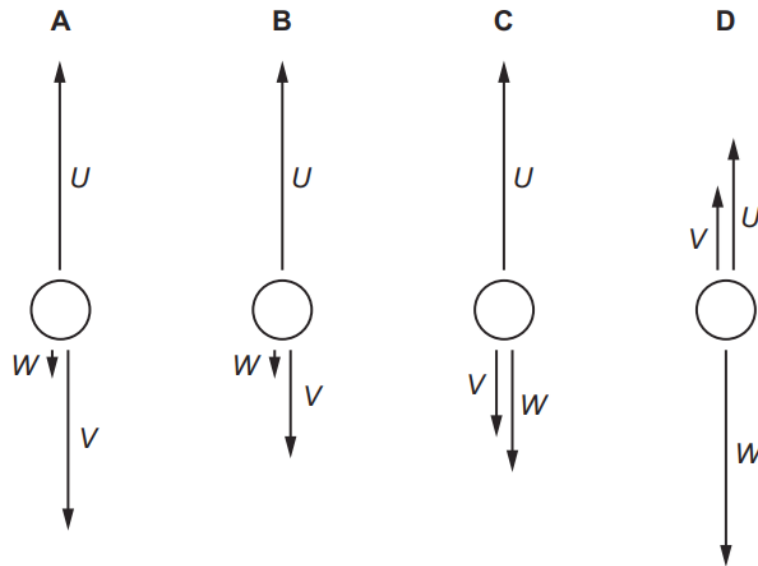
What is the speed with which the two objects travel together in the original direction?

- A** $\frac{3}{10}u$ **B** u **C** $\frac{6}{5}u$ **D** $\frac{10}{6}u$

188. 9702/13/O/N/20/Q11

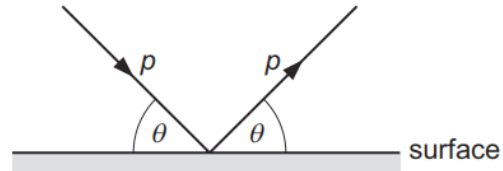
An air bubble is rising through a liquid at a constant speed. The forces on it are the upthrust U , the viscous drag V and its weight W .

Which diagram shows the directions and relative sizes of the forces?



189. 9702/12/F/M/21/Q8

A ball strikes a horizontal surface with momentum p at an angle θ to the surface, as shown.



The ball rebounds with the same magnitude of momentum at an angle θ to the surface.

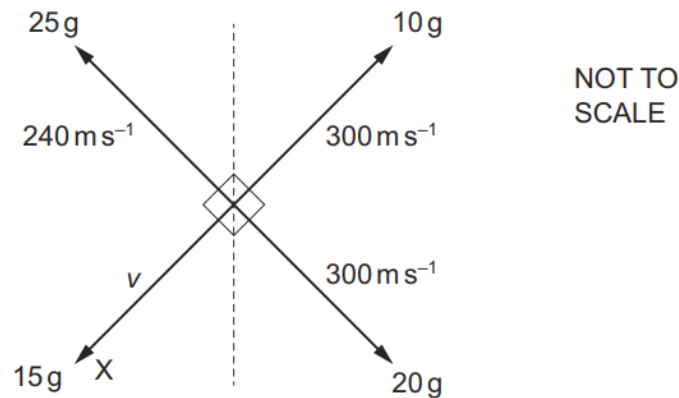
The ball is in contact with the surface for time t .

What is the magnitude of the average resultant force acting on the ball during the collision?

- A** zero **B** $\frac{2p}{t}$ **C** $\frac{2p \cos \theta}{t}$ **D** $\frac{2p \sin \theta}{t}$

190. 9702/12/F/M/21/Q10

A stationary firework explodes into four fragments which travel in different directions in a horizontal plane. The diagram shows the velocity and mass of each fragment.



What is the speed v of fragment X?

- A** 200 ms⁻¹ **B** 240 ms⁻¹ **C** 300 ms⁻¹ **D** 360 ms⁻¹

191. 9702/11/M/J/21/Q8

A rocket is fired from the Earth into space.

Newton's third law of motion describes how forces act in pairs. One of the forces of a pair is the weight of the rocket.

What is the other force of this pair?

- A** air resistance
B force of the exhaust gases on the rocket
C force of the rocket on the exhaust gases
D gravitational force of the rocket on the Earth

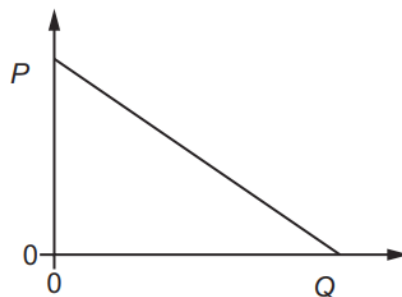
192. 9702/11/M/J/21/Q10

Which quantities are conserved in an inelastic collision?

	kinetic energy	total energy	linear momentum
A	conserved	not conserved	conserved
B	conserved	not conserved	not conserved
C	not conserved	conserved	conserved
D	not conserved	conserved	not conserved

193. 9702/11/M/J/21/Q9

The graph shows how quantity P varies with quantity Q for a body falling vertically downwards in a uniform gravitational field with air resistance.

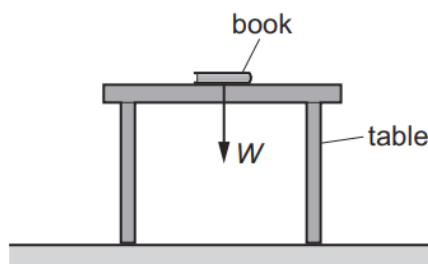


Which pair of quantities could be represented by P and Q ?

	P	Q
A	acceleration	force of air resistance
B	acceleration	time
C	velocity	force of air resistance
D	velocity	time

194. 9702/12/M/J/21/Q8

A book of weight W is at rest on a table. A student attempts to state Newton's third law of motion by saying that 'action equals reaction'.



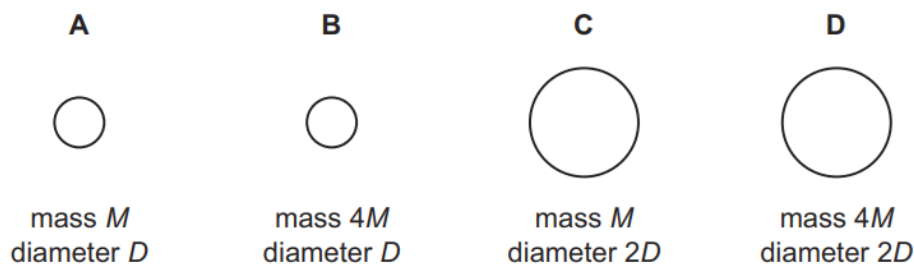
If the weight of the book is the 'action' force, what is the 'reaction' force?

- A** the force W acting downwards on the Earth from the table
- B** the force W acting upwards on the book from the table
- C** the force W acting upwards on the Earth from the book
- D** the force W acting upwards on the table from the floor

195. 9702/12/M/J/21/Q9

Four balls are dropped at the same time from the top of a very tall tower. There is no wind blowing.

Which ball hits the ground first?

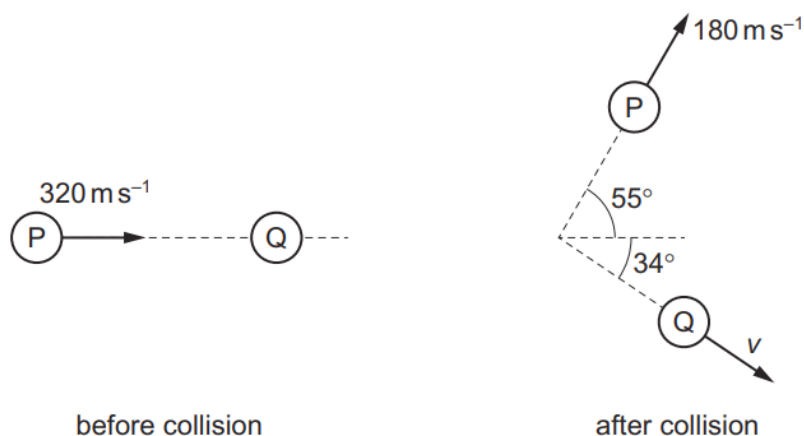


196. 9702/12/M/J/21/Q10

A nitrogen molecule P travelling at a speed of 320 ms^{-1} in a vacuum collides with a stationary nitrogen molecule Q.

After the collision, P travels at a velocity of 180 ms^{-1} at an angle of 55° to its original path.

Q travels in a direction at an angle of 34° to the initial path of P.



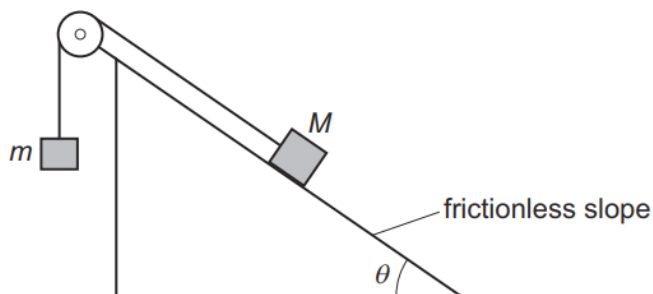
Assume that there are no external forces acting on the molecules.

What is the magnitude v of the velocity of Q after the collision?

- A** 120 ms^{-1} **B** 140 ms^{-1} **C** 180 ms^{-1} **D** 260 ms^{-1}

197. 9702/13/M/J/21/Q8

Two masses, M and m , are connected by an inextensible string which passes over a frictionless pulley. Mass M rests on a frictionless slope, as shown.



The slope is at an angle θ to the horizontal.

The two masses are initially held stationary and then released. Mass M accelerates down the slope.

Which expression **must** be correct?

- A** $\sin \theta < \frac{m}{M}$ **B** $\cos \theta < \frac{m}{M}$ **C** $\sin \theta > \frac{m}{M}$ **D** $\cos \theta > \frac{m}{M}$

198. 9702/13/M/J/21/Q9

The weights and masses of four different objects on the surfaces of four different planets are shown.

Which planet has the lowest value of acceleration of free fall at its surface?

	weight	mass
A	40 mN	6.0 g
B	3.0 N	500 g
C	10 N	1 kg
D	2.6 kN	750 kg

199. 9702/13/M/J/21/Q10

A rock in deep space is travelling towards a distant star and collides with a stationary spacecraft.

What is **not** a possible outcome of the collision?

- A** The rock becomes stationary and the spacecraft moves towards the star.
B The rock moves away from the star and so does the spacecraft.
C The rock moves away from the star and the spacecraft moves towards the star.
D The rock moves towards the star and so does the spacecraft.

200. 9702/13/M/J/21/Q11

A steel ball is falling at a constant (terminal) speed in still air. The forces acting on the ball are upthrust, viscous drag and weight.

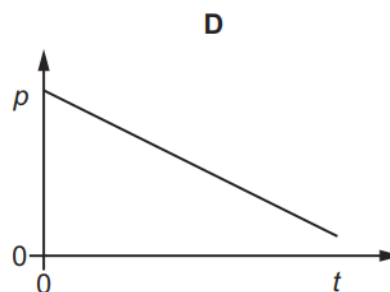
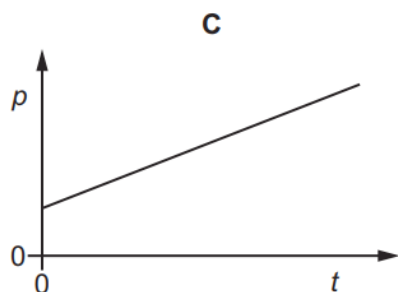
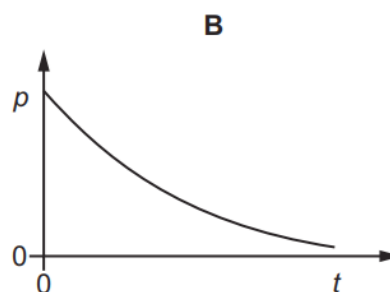
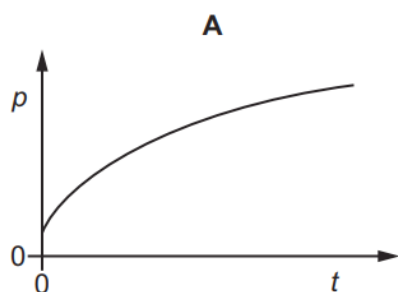
What is the order of increasing magnitude of these three forces?

- A upthrust $\rightarrow$ viscous drag $\rightarrow$ weight
- B viscous drag $\rightarrow$ upthrust $\rightarrow$ weight
- C viscous drag $\rightarrow$ weight $\rightarrow$ upthrust
- D weight $\rightarrow$ upthrust $\rightarrow$ viscous drag

201. 9702/11/O/N/21/Q8

A constant resultant force acts on an object in the direction of the object's velocity.

Which graph could show the variation with time t of the momentum p of the object?



202. 9702/11/O/N/21/Q9

Which statement **must** be true for an object in a gravitational field?

- A If the object has mass then the field causes it to accelerate.
- B If the object has mass then the field causes it to have weight.
- C If the object has weight then the field causes it to accelerate.
- D If the object has weight then the field causes it to have mass.

203. 9702/11/O/N/21/Q10

A ball of mass 0.16 kg is travelling horizontally at a speed of 20 ms^{-1} .

It collides with a wall and rebounds with a speed of 15 ms^{-1} along its original path. The ball is in contact with the wall for a time of 1.0 ms .

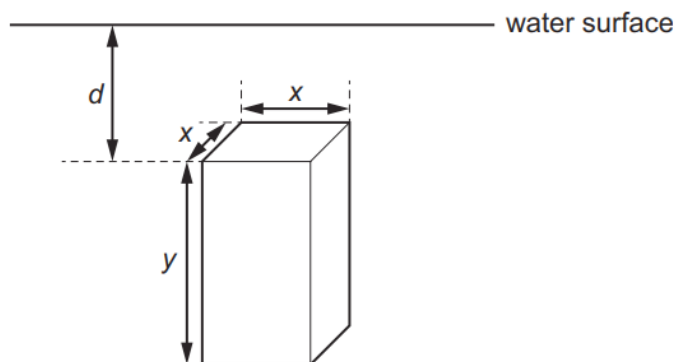
What is the average force exerted by the wall on the ball?

- A** 800 N **B** 2400 N **C** 3200 N **D** 5600 N

204. 9702/11/O/N/21/Q11

A uniform solid block is fully submerged in a tank of water.

The dimensions of the block are x and y , as shown.



The block is held vertically in the position shown. The density of the block is the same as the density of the water.

If the block is always held at the same depth d below the surface of the water, which single change would **increase** the magnitude of the upthrust force on the block?

- A** decrease the density of the block
B hold the block horizontally
C increase dimension y
D increase the density of the block

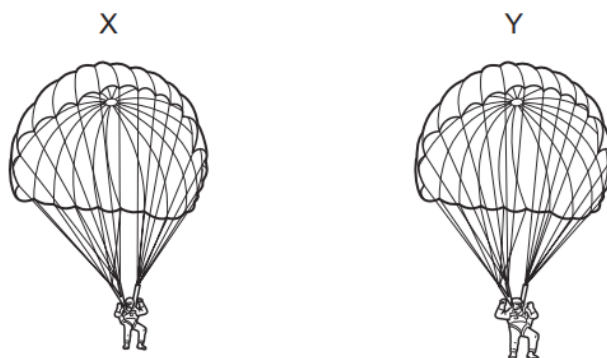
205. 9702/12/O/N/21/Q8

What is meant by the mass of an object?

- A** the property of the object that resists a change in motion
B the pull of the Earth on the object
C the total number of atoms in the object
D the weight of the object

206. 9702/12/O/N/21/Q9

The diagram shows two parachutists, X and Y, moving vertically downwards.



The total mass of parachutist Y and his parachute is twice the total mass of parachutist X and his parachute. At this moment, the air resistance on parachute Y is twice the air resistance on parachute X. Neither parachutist has reached his constant (terminal) velocity.

Which statement describes the acceleration of Y compared with the acceleration of X?

- A** The acceleration of Y is half the acceleration of X.
- B** The acceleration of Y is the same as the acceleration of X.
- C** The acceleration of Y is more than the acceleration of X, but less than twice the value.
- D** The acceleration of Y is twice the acceleration of X.

207. 9702/12/O/N/21/Q10

The table shows four different collisions between two blocks, each of mass 0.50 kg.

Which collision is perfectly elastic?

	before collision		after collision	
A	$4.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	0.0 m s^{-1} 0.50 kg	$2.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	0.50 kg
B	$2.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	$\leftarrow 2.0 \text{ m s}^{-1}$ 0.50 kg	0.0 m s^{-1} 0.50 kg	0.50 kg
C	$2.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	$\leftarrow 1.0 \text{ m s}^{-1}$ 0.50 kg	$\leftarrow 2.0 \text{ m s}^{-1}$ 0.50 kg	$3.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg
D	$4.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	$1.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	$1.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg	$4.0 \text{ m s}^{-1} \rightarrow$ 0.50 kg

208. 9702/12/O/N/21/Q11

A cylindrical block of wood has cross-sectional area A and weight W . It is totally immersed in water with its axis vertical. The block experiences pressures p_t and p_b at its top and bottom surfaces respectively.

What is the upthrust on the block?

- A $(p_b - p_t)$
- B $(p_b - p_t)A$
- C $(p_b - p_t)A - W$
- D $(p_b - p_t)A + W$

209. 9702/13/O/N/21/Q8

Water is pumped through a hose-pipe at a rate of 90 kg per minute. Water emerges horizontally from the hose-pipe with a speed of 20 m s^{-1} .

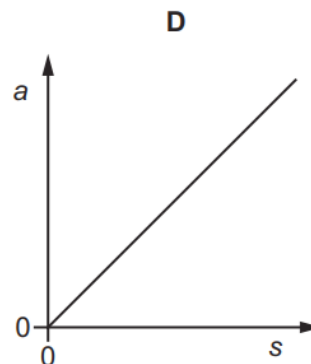
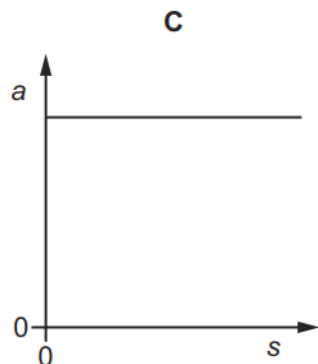
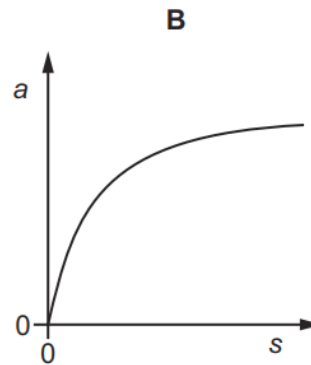
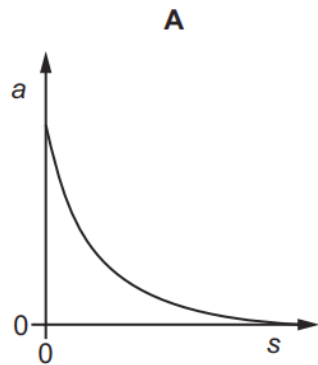
What is the minimum force required from a person holding the hose-pipe to prevent it moving backwards?

- A 30 N B 270 N C 1800 N D 110 000 N

210. 9702/13/O/N/21/Q9

A skydiver leaves a stationary balloon and falls vertically through a long distance.

Which graph best represents the variation of the acceleration a of the skydiver with the distance s travelled as she falls through the air?

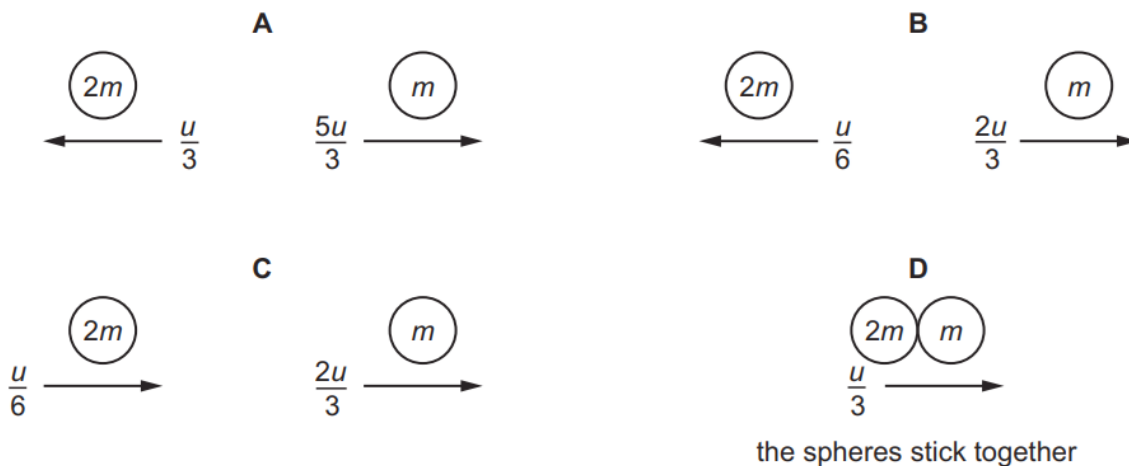


211. 9702/13/O/N/21/Q10

The diagram shows two spheres approaching each other head-on. Each sphere has speed u . One sphere has mass $2m$ and the other has mass m .



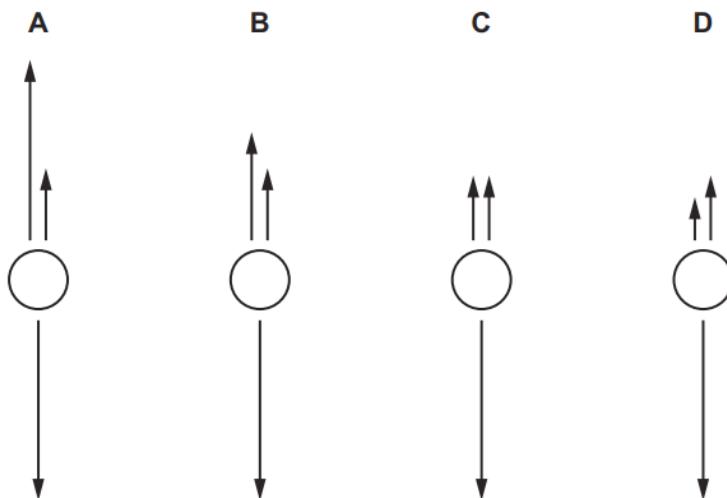
Which diagram shows the result of a perfectly elastic collision?



212. 9702/13/O/N/21/Q11

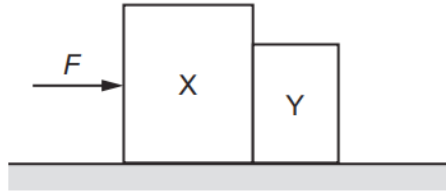
A spherical object falls through water at a constant speed. Three forces act on the object.

Which diagram, showing these three forces to scale, is correct?



213. 9702/12/F/M/22/Q7

A single horizontal force F is applied to a block X which is in contact with a separate block Y, as shown.



The blocks remain in contact as they accelerate along a horizontal frictionless surface. Air resistance is negligible. X has a greater mass than Y.

Which statement is correct?

- A** The acceleration of X is equal to force F divided by the mass of X.
- B** The force that X exerts on Y is equal to F .
- C** The force that X exerts on Y is less than F .
- D** The force that X exerts on Y is less than the force that Y exerts on X.

214. 9702/12/F/M/22/Q8

A car of mass 750 kg has a horizontal driving force of 2.0 kN acting on it. It has a forward horizontal acceleration of 2.0 ms^{-2} .



What is the resistive force acting horizontally?

- A** 0.50 kN **B** 1.5 kN **C** 2.0 kN **D** 3.5 kN

215. 9702/12/F/M/22/Q10

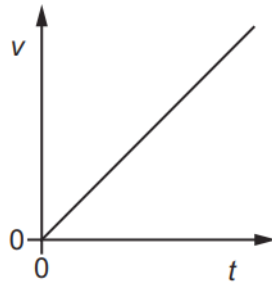
A rock of mass $2m$, travelling in deep space at velocity v , explodes into two parts of equal mass, one of which is then stationary.

What is the kinetic energy of the moving part after the explosion?

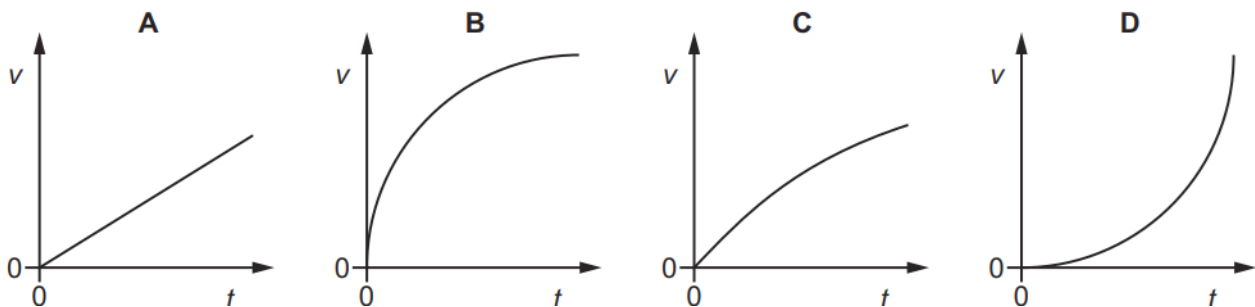
- A** $\frac{1}{2}mv^2$ **B** mv^2 **C** $\frac{3}{2}mv^2$ **D** $2mv^2$

216. 9702/12/F/M/22/Q9

An object falls freely from rest in a vacuum. The graph shows the variation with time t of the velocity v of the object.



Which graph, **using the same scales**, represents the object falling in air?



217. 9702/11/M/J/22/Q7

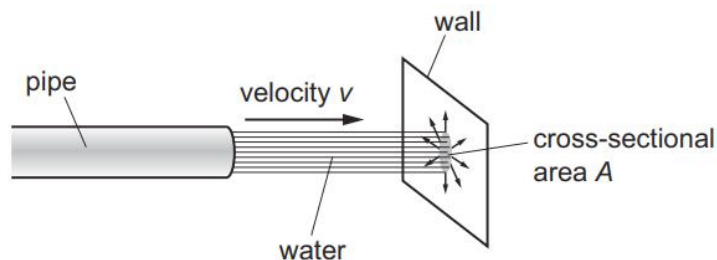
An object is moving along the ground in a straight line at a constant speed.

Which statement about the resultant force acting on the object is correct?

- A** The resultant force acting on the object is equal to its weight.
- B** The resultant force acting on the object is equal to the product of its mass and its velocity.
- C** The resultant force acting on the object is equal to the resistive force.
- D** The resultant force acting on the object is equal to zero.

218. 9702/11/M/J/22/Q8

Water flows out of a pipe and hits a wall.



When the jet of water hits the wall, it has horizontal velocity v and cross-sectional area A .

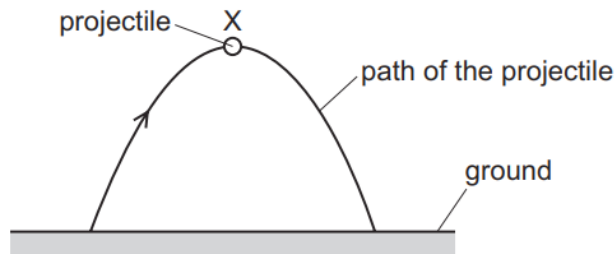
The density of the water is ρ . The water does not rebound from the wall.

What is the force exerted on the wall by the water?

- A** $\frac{\rho v}{A}$
- B** $\frac{\rho v^2}{A}$
- C** $\rho A v$
- D** $\rho A v^2$

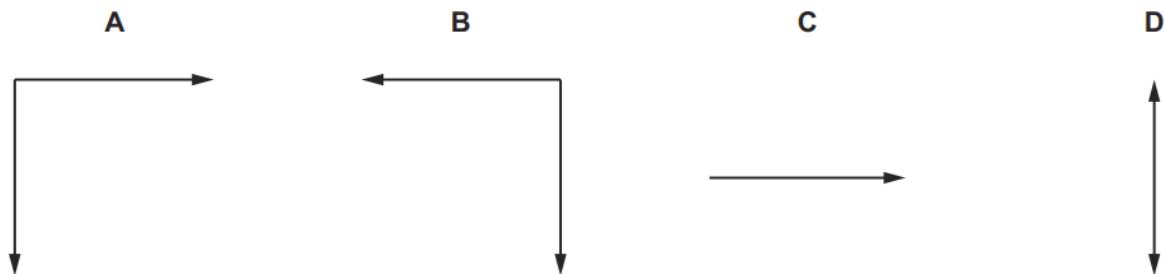
219. 9702/11/M/J/22/Q9

A projectile is launched at an angle above horizontal ground and travels through the air.



The projectile reaches its maximum height at position X. Assume that no upthrust acts on the projectile.

Which diagram shows the **directions** of the force or forces acting on the projectile at position X?



220. 9702/12/M/J/22/Q7

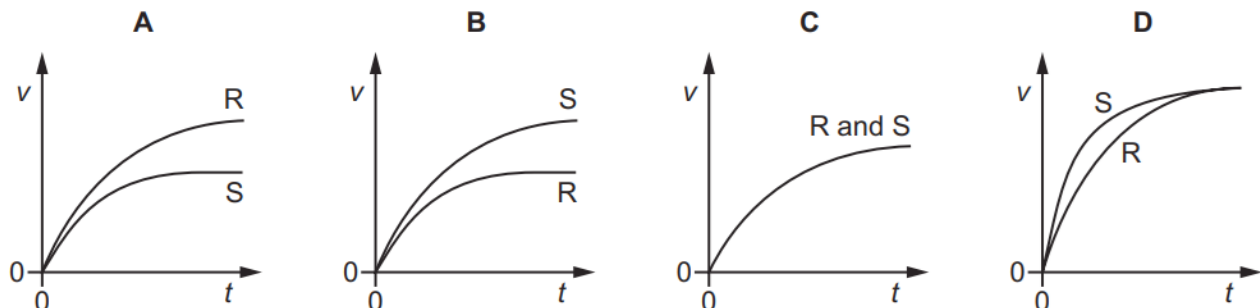
Which statement is **not** a requirement of a pair of forces that obey Newton's third law of motion?

- A** The forces act in opposite directions.
- B** The forces act on different objects.
- C** The forces act on objects in contact.
- D** The forces are of equal magnitude.

221. 9702/12/M/J/22/Q9

A stone S and a foam rubber ball R are identical spheres of equal volume. They are released from rest at time $t = 0$ and fall vertically through the air. Both reach terminal velocity.

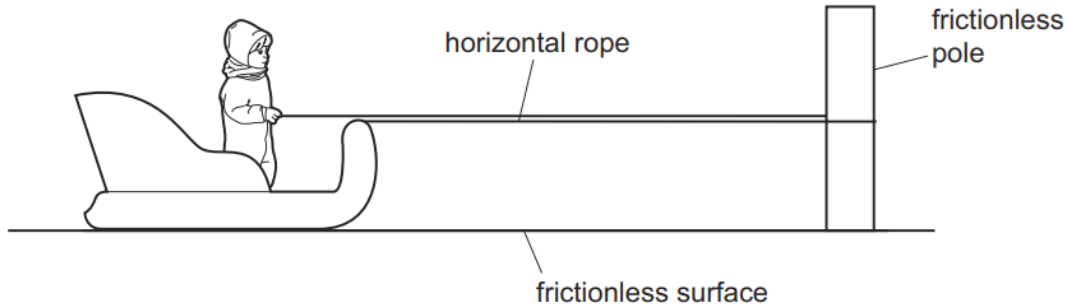
Which graph best shows the variation with time t of the speed v of the stone and of the rubber ball?



222. 9702/12/M/J/22/Q8

A child of mass 20 kg stands on the rough surface of a sledge of mass 40 kg. The sledge can slide on a horizontal frictionless surface.

One end of a rope is attached to the sledge. The rope passes around a fixed frictionless pole, and the other end of the rope is held by the child, as shown.



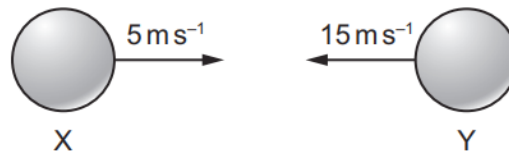
The rope is horizontal. The child pulls on the rope with a horizontal force of 12 N. This causes the child and the sledge to move with equal acceleration towards the pole.

What is the frictional force between the child and the sledge?

- A** 4.0 N **B** 6.0 N **C** 8.0 N **D** 12 N

223. 9702/12/M/J/22/Q10

Two balls X and Y are moving towards each other with speeds of 5 m s^{-1} and 15 m s^{-1} respectively.



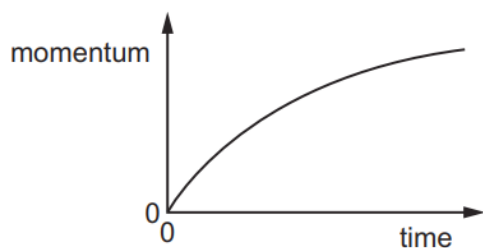
They make a perfectly elastic head-on collision and ball Y moves to the right with a speed of 7 m s^{-1} .

What is the speed and direction of ball X after the collision?

- A** 3 m s^{-1} to the left
B 13 m s^{-1} to the left
C 3 m s^{-1} to the right
D 13 m s^{-1} to the right

224. 9702/13/M/J/22/Q8

A car accelerates from rest. The graph shows the variation of the momentum of the car with time.



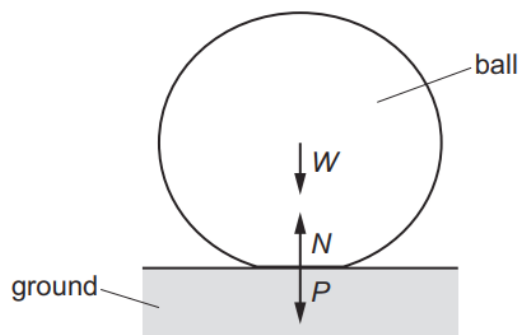
What is the meaning of the gradient of the graph at a particular time?

- A** the kinetic energy of the car
- B** the rate of change of kinetic energy of the car
- C** the resultant force on the car
- D** the velocity of the car

225. 9702/13/M/J/22/Q9

A ball is dropped onto horizontal ground and bounces vertically upwards. When the ball is in contact with the ground, the following forces act:

- the weight W of the ball
- the contact force P exerted on the ground by the ball
- the contact force N exerted on the ball by the ground.



When the ball is in contact with the ground, the ball is momentarily stationary.

At this instant, which relationship is correct?

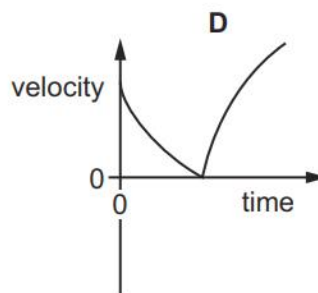
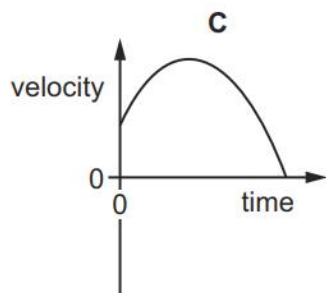
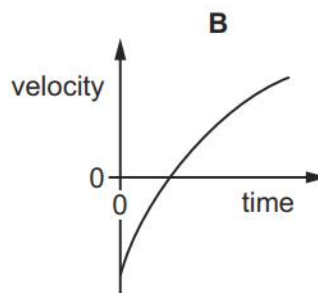
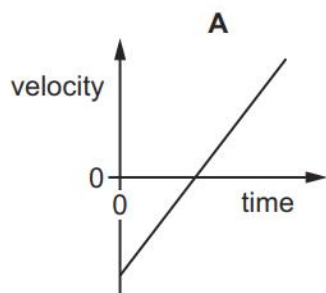
- A** $N = P + W$ **B** $N > P + W$ **C** $N = W$ **D** $N > W$

226. 9702/13/M/J/22/Q10

A person stands on the edge of a high cliff that is next to the sea. The person throws a stone vertically upwards. Air resistance acts on the stone.

The stone eventually hits the sea.

Which velocity–time graph best shows the motion of the stone from when it is released until it hits the sea?

**227. 9702/13/M/J/22/Q11**

Skaters of masses 80 kg and 40 kg move directly towards each other and collide.

Before the collision, the heavier skater is moving to the right at a speed of 2.0 m s^{-1} and the lighter skater is moving to the left at a speed of 1.0 m s^{-1} .

After the collision, the heavier skater moves to the right at a speed of 0.80 m s^{-1} .

What is the relative speed of separation of the two skaters?

- A** 0.6 m s^{-1} **B** 1.4 m s^{-1} **C** 2.2 m s^{-1} **D** 2.6 m s^{-1}

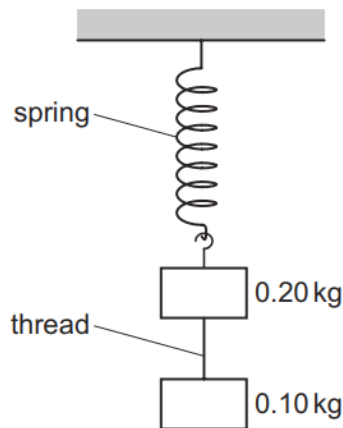
228. 9702/11/O/N/22/Q7

What is the definition of linear momentum?

- A** force per unit time
B product of force and time
C product of velocity and mass
D velocity per unit mass

229. 9702/11/O/N/22/Q8

A mass of 0.20 kg is suspended from the lower end of a light spring. A second mass of 0.10 kg is suspended from the first mass by a thread. The arrangement is allowed to come into static equilibrium and then the thread is cut.



Immediately after the thread is cut, what is the upward acceleration of the 0.20 kg mass?

- A** 4.9 ms^{-2} **B** 6.5 ms^{-2} **C** 9.8 ms^{-2} **D** 15 ms^{-2}

230. 9702/11/O/N/22/Q9

A snowflake and a raindrop are in still air. They both fall from rest at the same time and from the same height, far above the ground.

The snowflake and raindrop contain the same mass of water. Assume that there is no evaporation or melting. Also assume that, for a given speed, the drag force acting on the snowflake is greater than the drag force acting on the raindrop.

Which statement about the snowflake and raindrop is correct?

- A** The raindrop takes more time than the snowflake to reach terminal velocity.
B The raindrop takes more time than the snowflake to reach the ground.
C They reach the same terminal velocity.
D They take the same amount of time to reach the ground.

231. 9702/12/O/N/22/Q8

A constant resultant force F acts on an object of mass m for time t .

What is the change in momentum of the object?

- A** $\frac{F}{t}$ **B** $\frac{Ft}{m}$ **C** Ft **D** $\frac{F}{mt}$

232. 9702/12/O/N/22/Q9

The acceleration of free fall on the surface of planet P is one-tenth of that on the surface of planet Q.

On the surface of P, an object has a mass of 1.0 kg and a weight of 1.0 N.

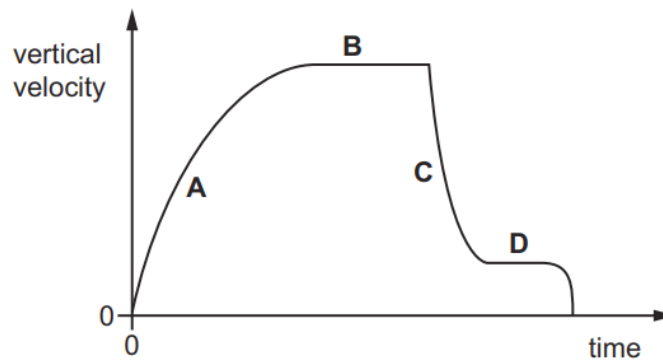
What are the mass and the weight of the same object on the surface of planet Q?

	mass on Q / kg	weight on Q / N
A	1.0	0.1
B	1.0	10
C	10	10
D	10	100

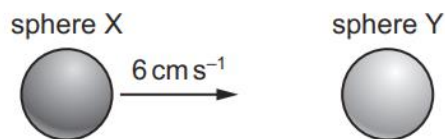
233. 9702/12/O/N/22/Q10

A parachutist falls from rest from a balloon. The variation with time of the vertical velocity of the parachutist is shown.

In which region is the force due to air resistance much greater than the weight of the parachutist?

**234. 9702/12/O/N/22/Q11**

Two solid spheres form an isolated system. Sphere X moves with speed 6 cm s^{-1} in a straight line directly towards a stationary sphere Y, as shown.



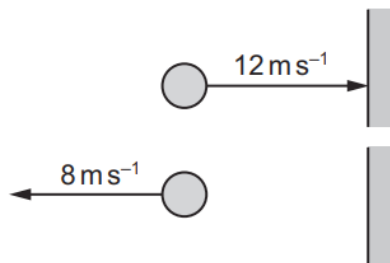
The spheres have a perfectly elastic collision. After the collision, sphere X moves with speed 2 cm s^{-1} in the same direction as before the collision.

What is the speed of sphere Y?

- A** 2 cm s^{-1} **B** 4 cm s^{-1} **C** 6 cm s^{-1} **D** 8 cm s^{-1}

235. 9702/13/O/N/22/Q8

A ball of mass 0.5 kg hits a vertical wall at a speed of 12 m s^{-1} . It bounces back along its original path with a speed of 8 m s^{-1} . The collision lasts for 0.10 s .

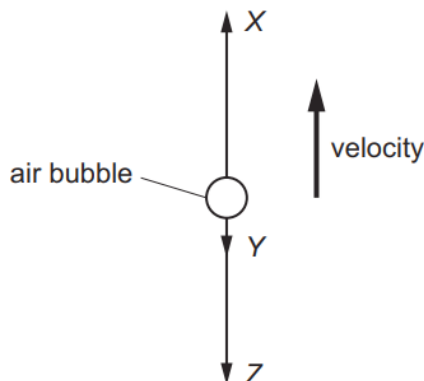


What is the average force on the ball due to the collision?

- A** 0.2 N **B** 1 N **C** 20 N **D** 100 N

236. 9702/13/O/N/22/Q9

An air bubble in a tank of water is rising with terminal (constant) velocity. The forces acting on the bubble are X , Y and Z , as shown.



The upthrust on the bubble is X .

Which statement about the forces is correct?

- A** Z is the viscous force on the bubble, Y is the weight of the bubble and $X = Y + Z$.
B Z is the viscous force on the bubble, Y is the weight of the bubble and $X > Y + Z$.
C Z is the weight of the bubble, Y is the viscous force on the bubble and $X = Y + Z$.
D Z is the weight of the bubble, Y is the viscous force on the bubble and $X > Y + Z$.

237. 9702/13/O/N/22/Q10

Two blocks are at rest on a frictionless horizontal surface. One block is made of wood and the other block is made of steel.

A steel ball is fired horizontally with a speed v at the wooden block. The ball embeds itself in the block, and the ball and block move together after impact.

A second identical steel ball is fired horizontally with speed v at the steel block. The steel ball then rebounds back along its original path with speed $\frac{v}{2}$.



The wooden block and the steel block have equal mass.

Which statement about the blocks immediately after the collisions is correct?

- A** Both blocks must travel with the same speed.
- B** The steel block must travel faster than the wooden block.
- C** The wooden block must travel faster than the steel block.
- D** The masses of the blocks and the steel ball are needed to determine which block travels faster.

238. 9702/12/F/M/23/Q7

Which expression defines force?

- A** $(\text{mass} \times \text{change in speed}) \times \text{time taken}$
- B** $\frac{\text{mass} \times \text{change in speed}}{\text{time taken}}$
- C** $(\text{change of momentum}) \times \text{time taken}$
- D** $\frac{\text{change of momentum}}{\text{time taken}}$

239. 9702/12/F/M/23/Q8

A ship of mass $8.4 \times 10^7 \text{ kg}$ is approaching a harbour with speed 16.4 ms^{-1} . By using reverse thrust it can maintain a constant total stopping force of $920\,000 \text{ N}$.

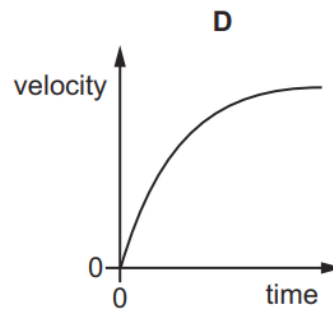
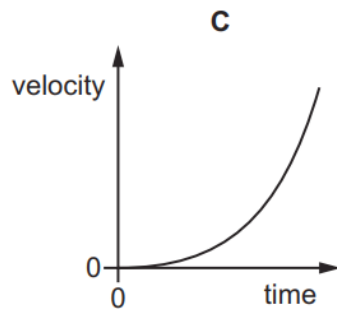
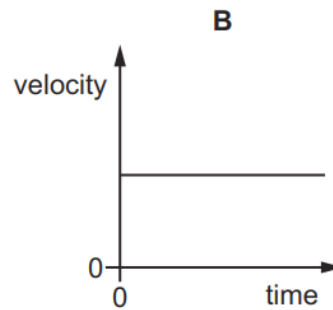
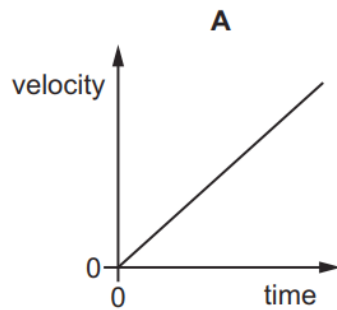
How long will it take to stop?

- A** 15 seconds
- B** 150 seconds
- C** 25 minutes
- D** 250 minutes

240. 9702/12/F/M/23/Q9

The velocity–time graphs of four different objects are shown.

Which graph represents an object falling from rest through a long distance in the Earth's atmosphere?

**241. 9702/12/F/M/23/Q10**

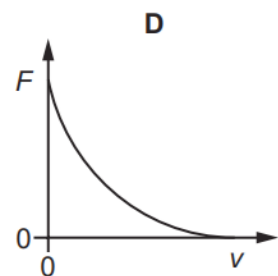
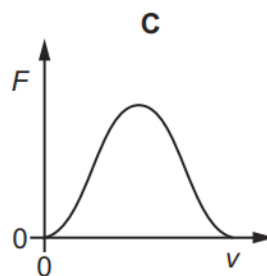
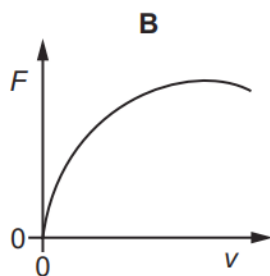
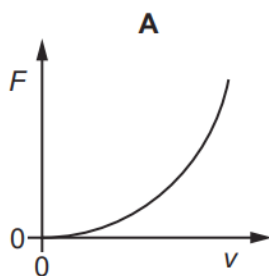
Which statement about collisions is correct?

- A** Kinetic energy is conserved in all collisions.
- B** Momentum is only conserved in perfectly elastic collisions.
- C** The relative speed of approach is equal to the relative speed of separation for perfectly elastic collisions.
- D** When two objects of different masses collide, they exert forces of different magnitudes on each other.

242. 9702/11/M/J/23/Q9

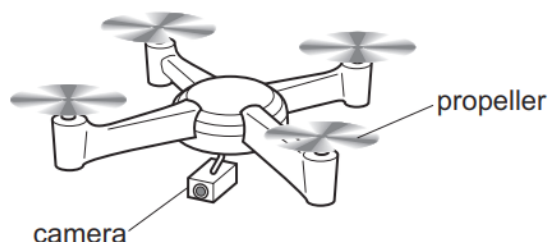
A small ball is held at the surface of liquid oil in a container. The ball is released from rest and falls through the oil. The ball has velocity v . A viscous (drag) force F acts on the ball.

Which graph could show the variation with v of F ?



243. 9702/11/M/J/23/Q7

A camera drone of mass 1.20 kg hovers at a fixed point above the ground. The drone has four propellers.



In a time of 1.00 s , each propeller pushes a mass of 0.400 kg of air vertically downwards.

Assume that the air above the propellers is stationary.

What is the speed of the air leaving each propeller?

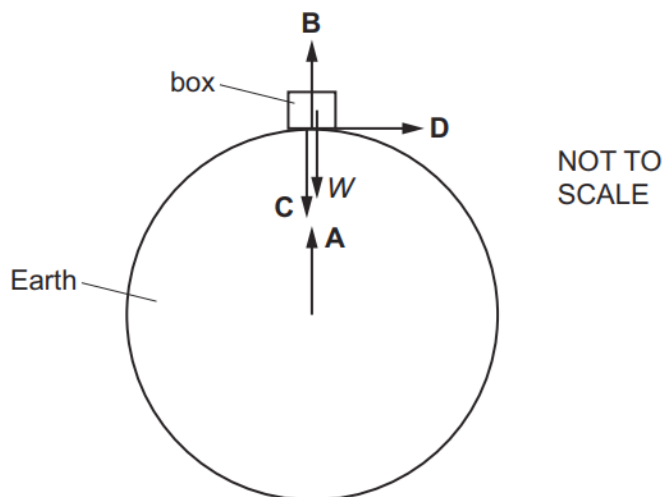
- A** 0.750 ms^{-1} **B** 3.00 ms^{-1} **C** 7.36 ms^{-1} **D** 29.4 ms^{-1}

244. 9702/11/M/J/23/Q8

A box rests on the Earth, as shown.

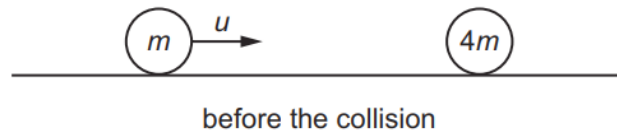
Newton's third law describes how forces of the same type act in pairs. One of the forces of a pair is the weight W of the box.

Which arrow represents the other force of this pair?



245. 9702/11/M/J/23/Q10

An object of mass m , moving at speed u along a frictionless horizontal surface, collides head-on with a stationary object of mass $4m$.



After the collision, the object of mass m rebounds along its initial path with $\frac{1}{4}$ of its kinetic energy before the collision.

What is the speed of the object of mass $4m$ after the collision?

- A** $\frac{u}{8}$ **B** $\frac{3u}{16}$ **C** $\frac{5u}{16}$ **D** $\frac{3u}{8}$

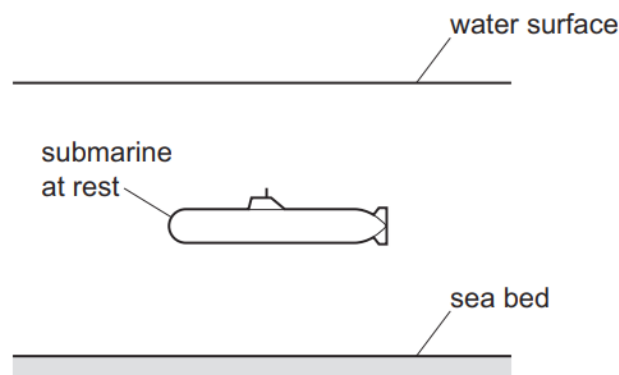
246. 9702/12/M/J/23/Q7

Which statement describes the mass of an object?

- A** the force the object experiences due to gravity
- B** the momentum of the object before a collision
- C** the resistance of the object to changes in motion
- D** the weight of the object as measured by a balance

247. 9702/12/M/J/23/Q8

A submarine of total mass 3200 kg is at rest underwater.



The total mass of the submarine is suddenly decreased by 200 kg by pumping water out of the submarine horizontally in a negligible time. The upthrust acting on the submarine is unchanged.

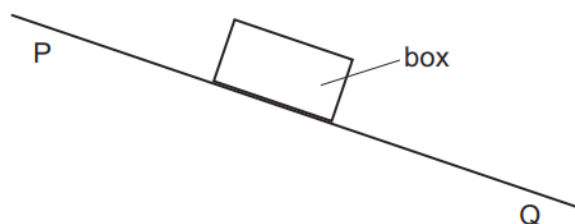
The change in the total weight of the submarine causes it to accelerate vertically upwards.

What is the initial upwards acceleration of the submarine?

- A** 0.613 ms^{-2} **B** 0.654 ms^{-2} **C** 9.81 ms^{-2} **D** 10.5 ms^{-2}

248. 9702/12/M/J/23/Q9

A box in air slides with increasing speed down a rough slope from point P to point Q.



The slope surface exerts a constant frictional force on the box.

As the box moves from P to Q, there are changes to the magnitudes of its acceleration and the total resistive force acting on it.

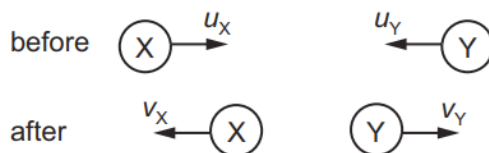
Which row describes the changes?

	magnitude of acceleration	magnitude of total resistive force
A	increases	decreases
B	decreases	decreases
C	increases	increases
D	decreases	increases

249. 9702/12/M/J/23/Q10

Two balls, X and Y, approach each other along the same straight line and collide. The collision is perfectly elastic.

Their initial speeds are u_X and u_Y respectively. After the collision they move apart with speeds v_X and v_Y respectively. Their directions are shown.



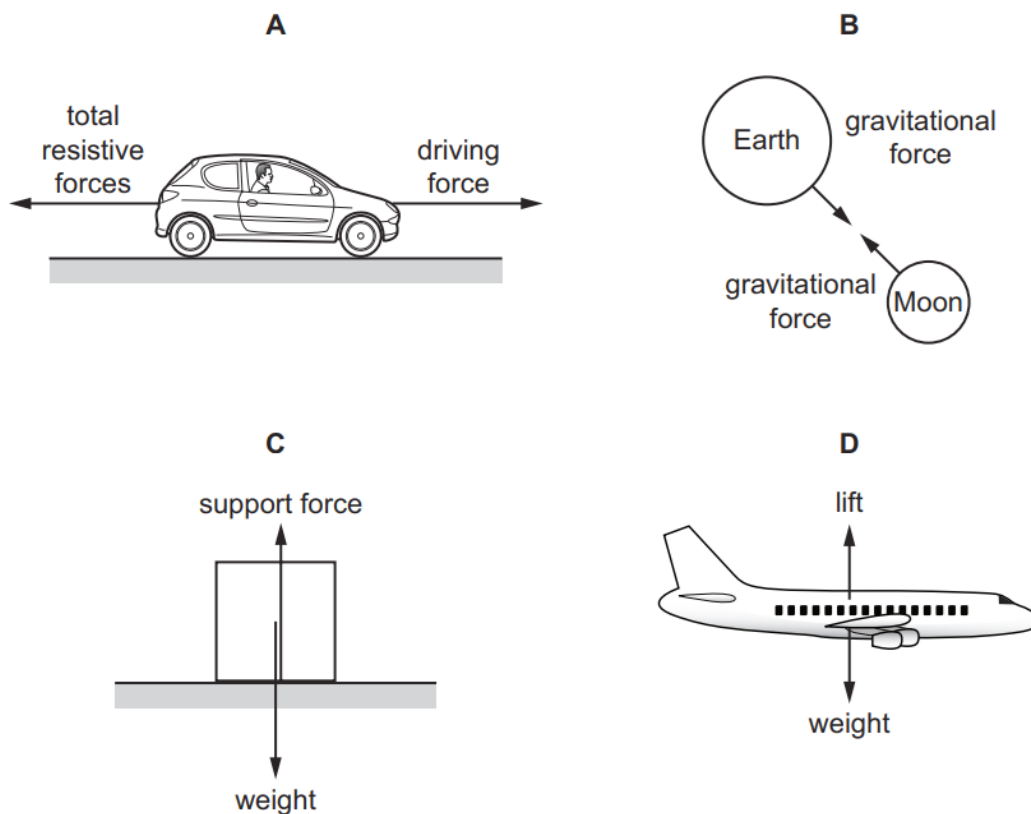
Which equation is correct?

- A** $u_X + u_Y = v_X + v_Y$
- B** $u_X + u_Y = v_X - v_Y$
- C** $u_X - u_Y = v_X + v_Y$
- D** $u_X - u_Y = v_X - v_Y$

250. 9702/12/M/J/23/Q8

Each diagram illustrates a pair of forces of equal magnitude.

Which diagram gives an example of a pair of forces that is described by Newton's third law of motion?



251. 9702/12/M/J/23/Q9

Two balls of identical shape and size but different masses are falling through the same liquid.

The sum of the drag force and upthrust acting on each ball is equal to its weight.

Which statement about the two balls is correct?

- A** The heavier ball has a larger acceleration than the lighter ball.
- B** The heavier ball has a smaller deceleration than the lighter ball.
- C** The heavier ball is falling at the same speed as the lighter ball.
- D** The heavier ball is falling at a larger speed than the lighter ball.

252. 9702/12/M/J/23/Q10

A perfectly elastic collision occurs between two objects X and Y. The mass of X is m and the mass of Y is $4m$. Object X travels at speed v before the collision and speed $\frac{3v}{5}$ in the opposite direction after the collision. Object Y is stationary before the collision.

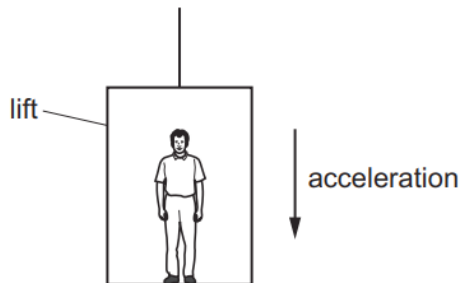


What is the kinetic energy of Y after the collision?

- A** $\frac{8}{10}mv^2$ **B** $\frac{34}{50}mv^2$ **C** $\frac{16}{50}mv^2$ **D** $\frac{1}{5}mv^2$

253. 9702/11/O/N/23/Q7

A man stands in a lift that is accelerating vertically downwards, as shown.



Which statement describes the force exerted by the man on the floor?

- A** It is equal to the weight of the man.
B It is greater than the force exerted by the floor on the man.
C It is less than the force exerted by the floor on the man.
D It is less than the weight of the man.

254. 9702/11/O/N/23/Q8

A ball of mass 200 g is thrown horizontally with a speed of 20 m s^{-1} against a vertical wall.

The ball is in contact with the wall for a time of 0.10 s before rebounding back along its original path with a speed of 10 m s^{-1} .

What is the average force exerted by the wall on the ball during the collision?

- A** 20 N **B** 60 N **C** 20 kN **D** 60 kN

255. 9702/11/O/N/23/Q9

In an experiment, a metal ball is dropped into a viscous liquid. The terminal velocity of the ball in the liquid is measured.

The experiment is repeated four times. For each repeat, a change is made to one of the following.

- 1 the density of the metal of the ball
- 2 the height from which the ball is dropped
- 3 the density of the liquid
- 4 the depth of the liquid

Which two changes separately affect the terminal velocity of the ball in the liquid?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

256. 9702/11/O/N/23/Q10

Two objects move towards each other along the same straight line.



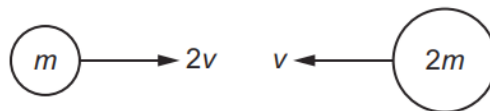
After colliding, the two objects stick together and are stationary.

Which statement **must** be correct?

- A** The total kinetic energy of the two objects does not change during the collision.
B The total momentum of the two objects before the collision is zero.
C The two objects have equal mass.
D The two objects have the same speed before the collision.

257. 9702/12/O/N/23/Q10

Two balls, of masses m and $2m$, travelling in a vacuum with initial velocities $2v$ and v respectively, collide with each other head-on, as shown.



After the collision, the ball of mass m rebounds to the left with velocity v .

What is the loss of kinetic energy in the collision?

- A** $\frac{3}{4}mv^2$ **B** $\frac{3}{2}mv^2$ **C** $\frac{9}{4}mv^2$ **D** $\frac{9}{2}mv^2$

258. 9702/12/O/N/23/Q9

An object falls from a stationary helicopter and reaches terminal velocity.

What happens to the acceleration of the object between leaving the helicopter and reaching terminal velocity?

- A It decreases to 9.81 m s^{-2} .
- B It decreases to zero.
- C It increases to 9.81 m s^{-2} .
- D It remains constant at 9.81 m s^{-2} .

259. 9702/12/O/N/23/Q8

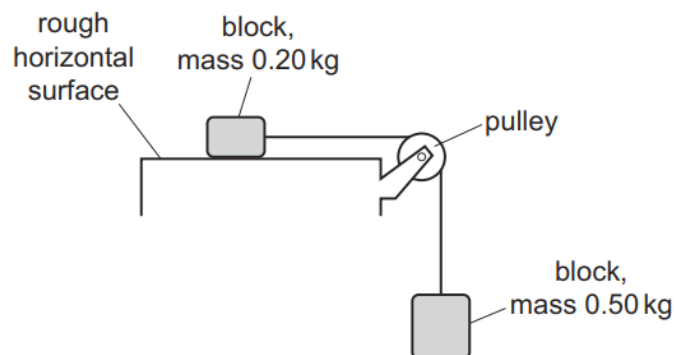
A resultant force causes an object to accelerate.

What is equal to the resultant force?

- A the acceleration of the object per unit mass
- B the change in kinetic energy of the object per unit time
- C the change in momentum of the object per unit time
- D the change in velocity of the object per unit time

260. 9702/12/O/N/23/Q7

Two blocks, of mass 0.20 kg and 0.50 kg , are connected by a light inextensible string that passes over a frictionless pulley.



The blocks are initially held stationary. The block of mass 0.20 kg rests on a rough horizontal surface.

The block of mass 0.50 kg is suspended in air. Air resistance is negligible.

When the blocks are released, they have an acceleration of magnitude 2.0 m s^{-2} .

What is the magnitude of the frictional force between the block of mass 0.20 kg and the rough surface?

- A 3.5 N
- B 3.9 N
- C 4.5 N
- D 6.3 N

261. 9702/13/O/N/23/Q7

Which statement about mass is correct?

- A** Mass has a magnitude and a direction.
- B** Mass resists changes in motion.
- C** The greater the mass of an object, the greater its acceleration when falling in a vacuum.
- D** The mass of an object depends on its location.

262. 9702/13/O/N/23/Q8

A snooker ball has a mass of 200 g. It hits the cushion of a snooker table and rebounds along its original path.

The ball arrives at the cushion with a speed of 14.0 ms^{-1} and then leaves it with a speed of 7.0 ms^{-1} . The ball and the cushion are in contact for a time of 0.60 s.

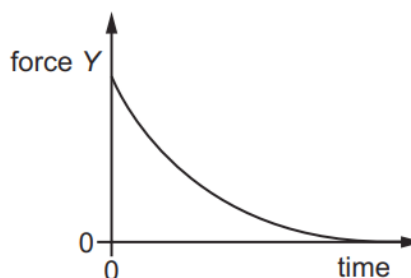
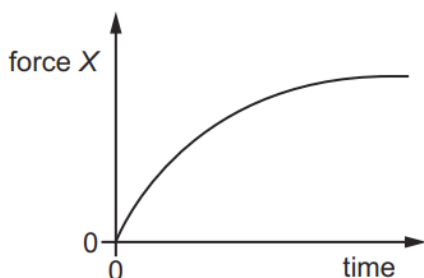
What is the average force exerted on the ball by the cushion?

- A** 1.4 N **B** 2.3 N **C** 4.2 N **D** 7.0 N

263. 9702/13/O/N/23/Q9

A ball falls from rest through air and eventually reaches a constant velocity.

For this fall, forces X and Y vary with time as shown.



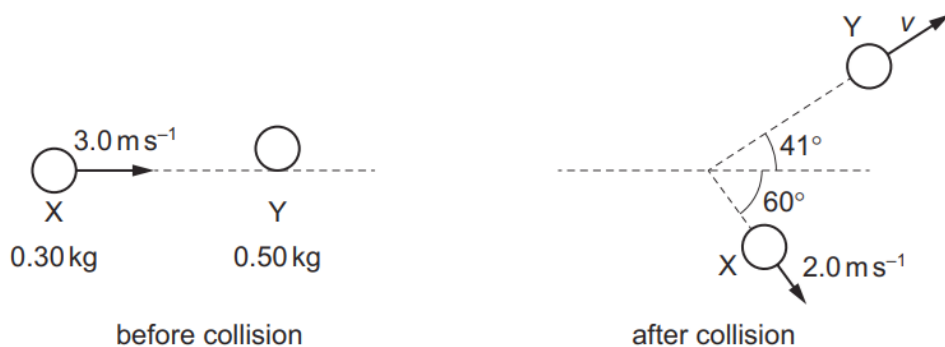
What could be forces X and Y ?

	force X	force Y
A	air resistance	resultant force
B	air resistance	weight
C	upthrust	resultant force
D	upthrust	weight

264. 9702/13/O/N/23/Q10

An object X of mass 0.30 kg is travelling in a straight line at a constant velocity of 3.0 ms^{-1} on a horizontal frictionless surface. Object X collides with a stationary object Y of mass 0.50 kg .

After the collision, X moves with a velocity of 2.0 ms^{-1} at an angle of 60° to its direction before the collision. Object Y moves with a velocity v at an angle of 41° to the direction of X before the collision, as shown.



What is the value of v ?

- A** 0.80 ms^{-1} **B** 1.2 ms^{-1} **C** 1.6 ms^{-1} **D** 1.8 ms^{-1}

ANSWERS

1. B	48. B	95. B	142. A	187. D	232. B
2. B	49. C	96. A	143. A	188. A	233. C
3. A	50. B	97. C	144. B	189. D	234. D
4. C	51. A	98. C	145. B	190. A	235. D
5. A	52. B	99. C	146. D	191. D	236. A
6. A	53. B	100. C	147. D	192. C	237. B
7. A	54. A	101. A	148. B	193. A	238. D
8. C	55. D	102. B	149. B	194. C	239. C
9. B	56. A	103. C	150. C	195. B	240. D
10. D	57. B	104. D	151. D	196. D	241. C
11. A	58. B	105. C	152. C	197. C	242. A
12. B	59. D	106. D	153. A	198. D	243. C
13. C	60. A	107. C	154. D	199. B	244. A
14. B	61. D	108. C	155. C	200. A	245. D
15. A	62. A	109. B	156. A	201. C	246. C
16. A	63. D	110. A	157. A	202. B	247. B
17. C	64. D	111. C	158. C	203. D	248. D
18. D	65. B	112. B	159. D	204. C	249. A
19. B	66. A	113. C	160. C	205. A	250. B
20. A	67. B	114. C	161. D	206. B	251. D
21. B	68. D	115. B	162. A	207. D	252. C
22. B	69. B	116. A	163. A	208. B	253. D
23. A	70. A	117. C	164. D	209. A	254. B
24. D	71. D	118. B	165. B	210. A	255. B
25. C	72. D	119. C	166. C	211. A	256. B
26. B	73. B	120. C	167. D	212. B	257. C
27. D	74. A	121. C	168. B	213. C	258. B
28. B	75. C	122. B	169. A	214. A	259. C
29. B	76. B	123. B	170. D	215. D	260. A
30. B	77. A	124. D	171. D	216. C	261. B
31. A	78. B	125. D	172. C	217. D	262. D
32. D	79. D	126. C	173. B	218. D	263. A
33. A	80. D	127. B	174. D	219. B	264. C
34. D	81. D	128. D	175. D	220. C	
35. A	82. C	129. B	176. C	221. B	
36. A	83. B	130. A	177. B	222. A	
37. C	84. D	131. B	178. B	223. B	
38. B	85. B	132. A	179. B	224. C	
39. D	86. A	133. D	180. D	225. D	
40. D	87. B	134. A	181. A	226. B	
41. B	88. B	135. B	182. B	227. A	
42. C	89. D	136. D	183. C	228. C	
43. C	90. B	137. C	184. C	229. A	
44. D	91. B	138. C	185. B	230. A	
45. A	92. A	139. C	186. D	231. C	
46. B	93. A	140. C			
47. B	94. C	141. A			

9702/22/M/J/09/Q2

- 1 An experiment is conducted on the surface of the planet Mars.
A sphere of mass 0.78 kg is projected almost vertically upwards from the surface of the planet. The variation with time t of the vertical velocity v in the upward direction is shown in Fig. 2.1.

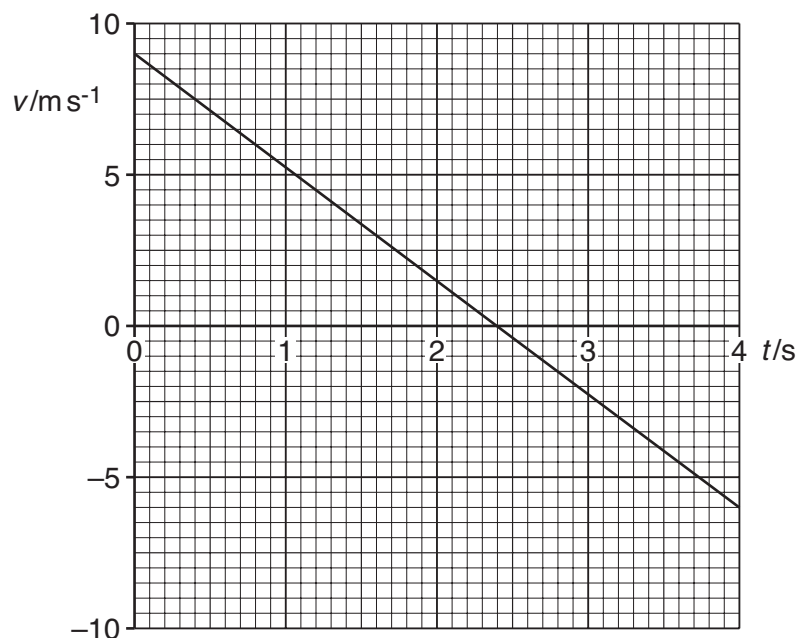


Fig. 2.1

The sphere lands on a small hill at time $t = 4.0\text{ s}$.

- (a) State the time t at which the sphere reaches its maximum height above the planet's surface.

$t = \dots\dots\dots\text{ s}$ [1]

- (b) Determine the vertical height above the point of projection at which the sphere finally comes to rest on the hill.

height = $\dots\dots\dots\text{ m}$ [3]

(c) Calculate, for the first 3.5 s of the motion of the sphere,

(i) the change in momentum of the sphere,

change in momentum =N s [2]

(ii) the force acting on the sphere.

force =N [2]

(d) Using your answer in (c)(ii),

(i) state the weight of the sphere,

weight =N [1]

(ii) determine the acceleration of free fall on the surface of Mars.

acceleration =ms⁻² [2]

9702/21/O/N/09/Q3

- 2 A stationary nucleus of mass 220u undergoes radioactive decay to produce a nucleus D of mass 216u and an α -particle of mass 4u , as illustrated in Fig. 3.1.

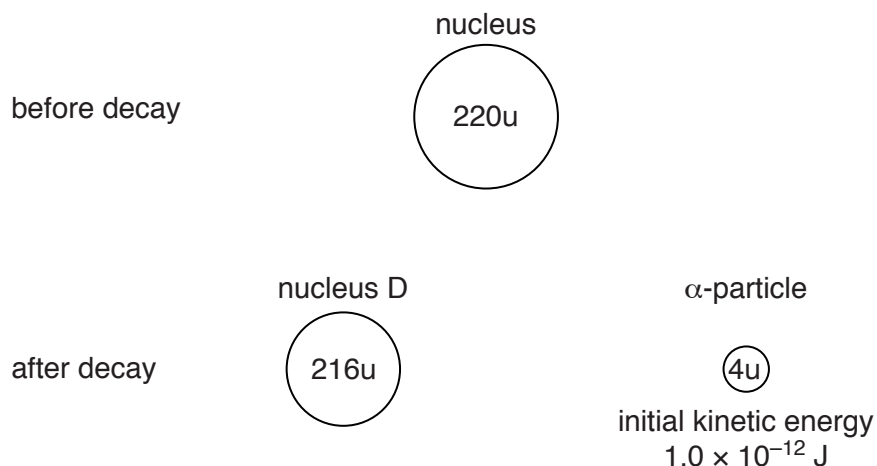


Fig. 3.1

The initial kinetic energy of the α -particle is $1.0 \times 10^{-12} \text{ J}$.

- (a) (i) State the law of conservation of linear momentum.

.....

 [2]

- (ii) Explain why the initial velocities of the nucleus D and the α -particle must be in opposite directions.

.....

 [2]

- (b) (i) Show that the initial speed of the α -particle is $1.7 \times 10^7 \text{ ms}^{-1}$.

- (ii) Calculate the initial speed of nucleus D. [2]

speed = ms^{-1} [2]

- (c) The range in air of the emitted α -particle is 4.5 cm .
 Calculate the average deceleration of the α -particle as it is stopped by the air.

deceleration = ms^{-2} [2]

9702/22/M/J/10/Q3

3 (a) (i) Define *force*.

.....
[1]

(ii) State Newton's third law of motion.

.....

[3]

(b) Two spheres approach one another along a line joining their centres, as illustrated in Fig. 3.1.

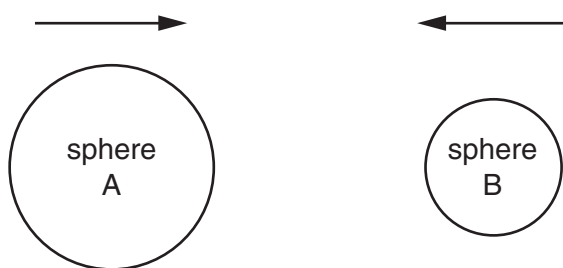


Fig. 3.1

When they collide, the average force acting on sphere A is F_A and the average force acting on sphere B is F_B .

The forces act for time t_A on sphere A and time t_B on sphere B.

(i) State the relationship between

1. F_A and F_B ,

.....[1]

2. t_A and t_B .

.....[1]

(ii) Use your answers in (i) to show that the change in momentum of sphere A is equal in magnitude and opposite in direction to the change in momentum of sphere B.

.....
[1]

9702/21/O/N/10/Q3

- 4 (a) State the relation between force and momentum.

..... [1]

- (b) A rigid bar of mass 450g is held horizontally by two supports A and B, as shown in Fig. 3.1.

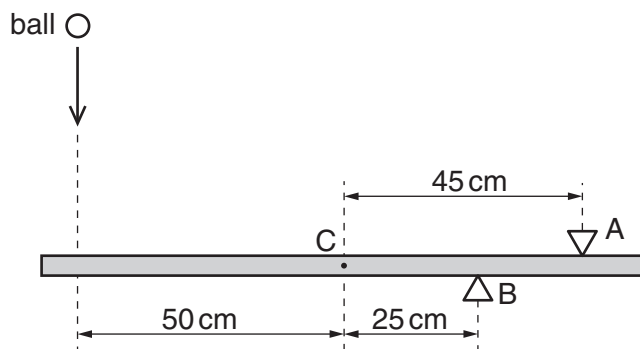


Fig. 3.1

The support A is 45 cm from the centre of gravity C of the bar and support B is 25 cm from C.

A ball of mass 140g falls vertically onto the bar such that it hits the bar at a distance of 50 cm from C, as shown in Fig. 3.1.

The variation with time t of the velocity v of the ball before, during and after hitting the bar is shown in Fig. 3.2.

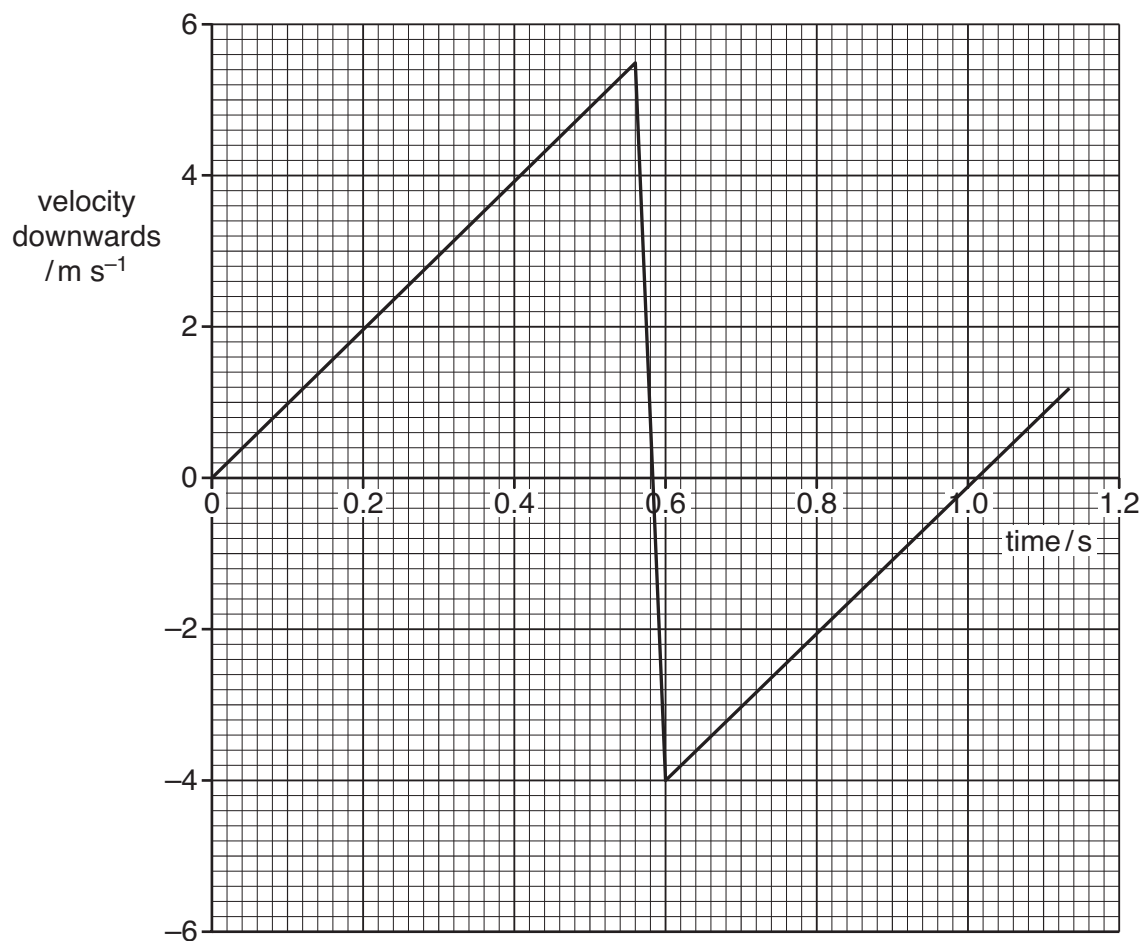


Fig. 3.2

For the time that the ball is in contact with the bar, use Fig. 3.2

(i) to determine the change in momentum of the ball,

change = kg ms^{-1} [2]

(ii) to show that the force exerted by the ball on the bar is 33 N.

[1]

(c) For the time that the ball is in contact with the bar, use data from Fig. 3.1 and (b)(ii) to calculate the force exerted on the bar by

(i) the support A,

force = N [3]

(ii) the support B.

force = N [2]

- 5 (a) State Newton's second law.

9702/22/O/N/12/Q2

.....
.....[1]

- (b) A ball of mass 65 g hits a wall with a velocity of 5.2 ms^{-1} perpendicular to the wall. The ball rebounds perpendicularly from the wall with a speed of 3.7 ms^{-1} . The contact time of the ball with the wall is 7.5 ms.

Calculate, for the ball hitting the wall,

- (i) the change in momentum,

change in momentum = N s [2]

- (ii) the magnitude of the average force.

force = N [1]

- (c) (i) For the collision in (b) between the ball and the wall, state how the following apply:

1. Newton's third law,

.....
.....
.....
.....[2]

2. the law of conservation of momentum.

.....
.....[1]

- (ii) State, with a reason, whether the collision is elastic or inelastic.

.....
.....[1]

9702/22/M/J/13/Q2

6 (a) Define *force*.

..... [1]

(b) A resultant force F acts on an object of mass 2.4 kg. The variation with time t of F is shown in Fig. 2.1.

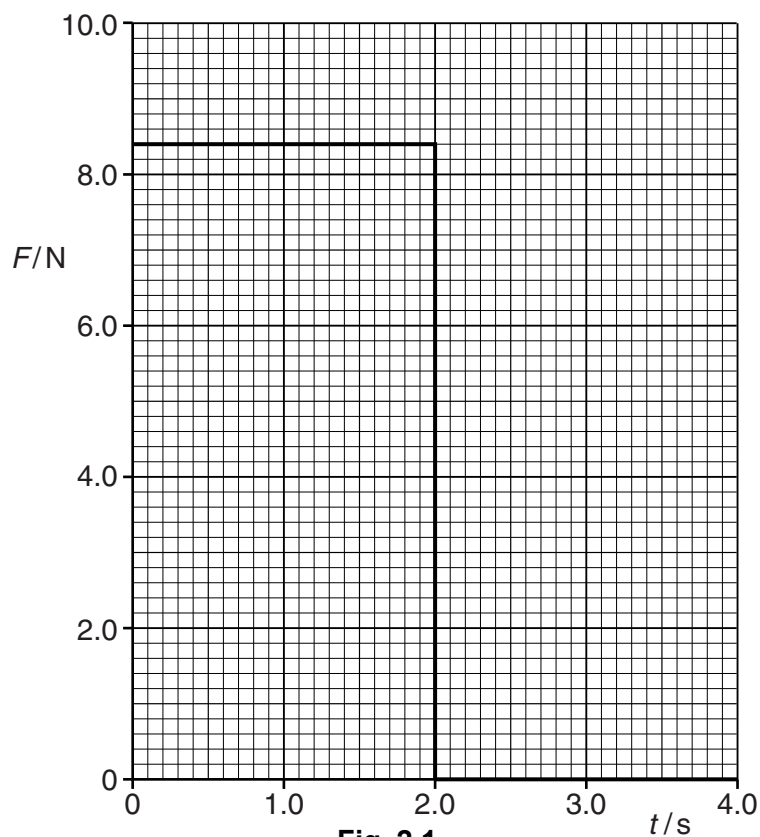
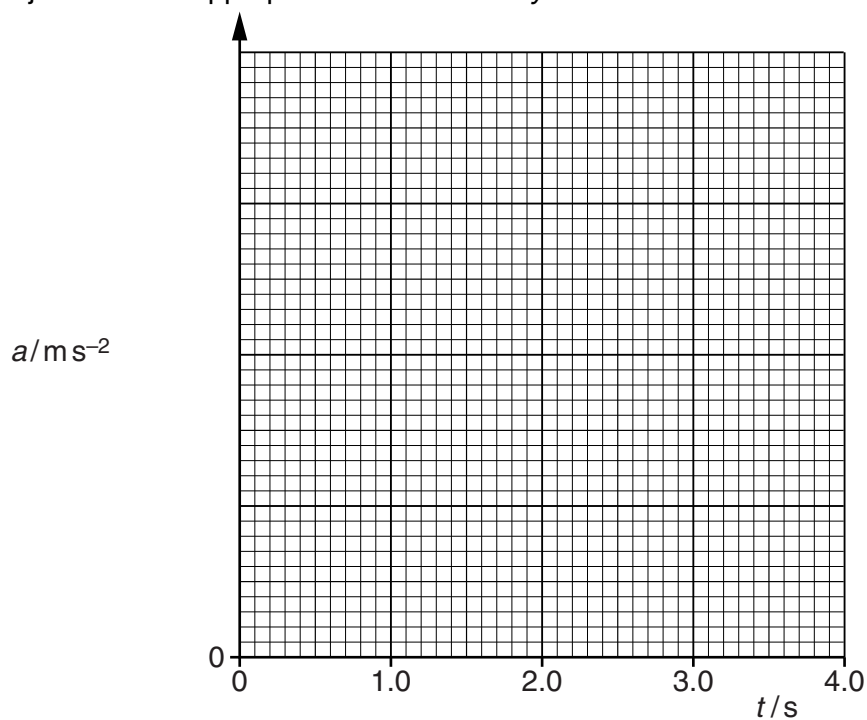


Fig. 2.1

The object starts from rest.

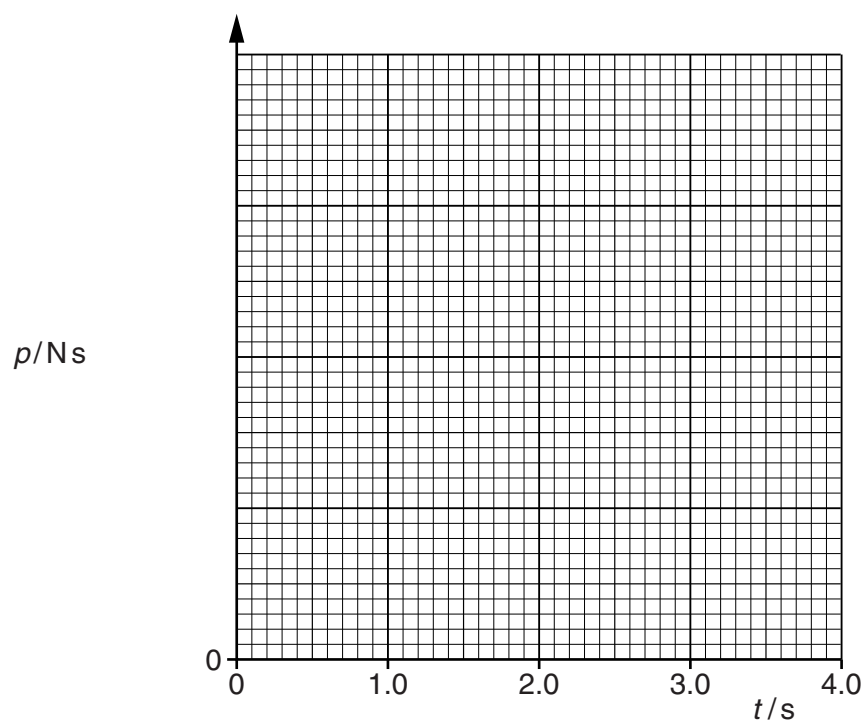
- (i) On Fig. 2.2, show quantitatively the variation with t of the acceleration a of the object. Include appropriate values on the y-axis.



[4]

Fig. 2.2

- (ii) On Fig. 2.3, show quantitatively the variation with t of the momentum p of the object. Include appropriate values on the y-axis.

**Fig. 2.3**

[5]

9702/23/M/J/13/Q3

- 7 (a) (i) State the principle of conservation of momentum.

.....

 [2]

- (ii) State the difference between an elastic and an inelastic collision.

..... [1]

- (b) An object A of mass 4.2 kg and horizontal velocity 3.6 ms^{-1} moves towards object B as shown in Fig. 3.1.

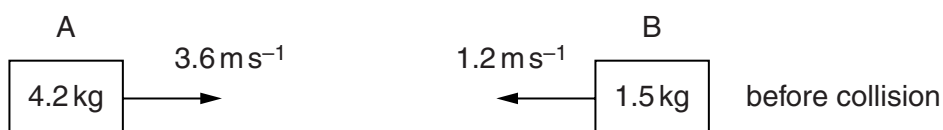


Fig. 3.1

Object B of mass 1.5 kg is moving with a horizontal velocity of 1.2 ms^{-1} towards object A.

The objects collide and then both move to the right, as shown in Fig. 3.2.

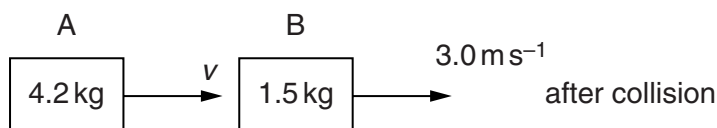


Fig. 3.2

Object A has velocity v and object B has velocity 3.0 ms^{-1} .

- (i) Calculate the velocity v of object A after the collision.

velocity = ms^{-1} [3]

- (ii) Determine whether the collision is elastic or inelastic.

[3]

- 8 (a) State Newton's first law of motion.

9702/22/M/J/14/Q3

.....
 [1]

- (b) A box slides down a slope, as shown in Fig. 3.1.

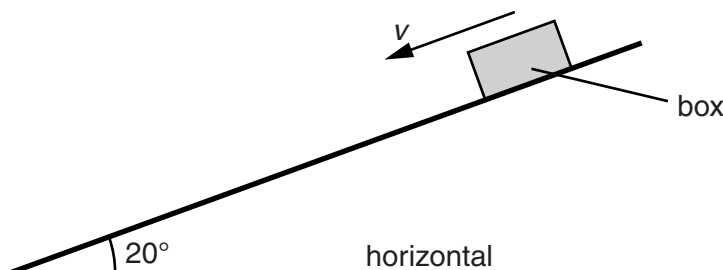


Fig. 3.1

The angle of the slope to the horizontal is 20° . The box has a mass of 65 kg. The total resistive force R acting on the box is constant as it slides down the slope.

- (i) State the names and directions of the other two forces acting on the box.

1.
 2. [2]

- (ii) The variation with time t of the velocity v of the box as it moves down the slope is shown in Fig. 3.2.

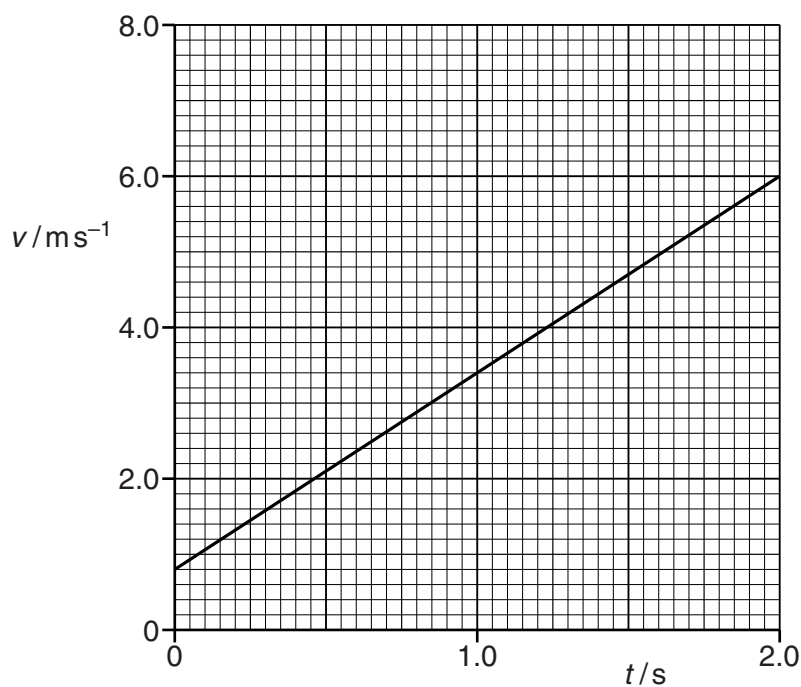


Fig. 3.2

1. Use data from Fig. 3.2 to show that the acceleration of the box is 2.6 ms^{-2} .

[2]

2. Calculate the resultant force on the box.

resultant force = N [1]

3. Determine the resistive force R on the box.

$R =$ N [3]

9702/23/O/N/14/Q4

- 9 (a) State the principle of conservation of momentum.

.....

 [2]

- (b) A ball X and a ball Y are travelling along the same straight line in the same direction, as shown in Fig. 4.1.



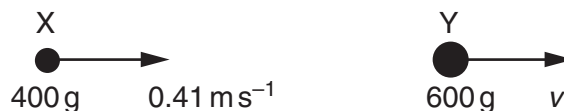
Fig. 4.1

Ball X has mass 400 g and horizontal velocity 0.65 ms⁻¹.

Ball Y has mass 600 g and horizontal velocity 0.45 ms⁻¹.

Ball X catches up and collides with ball Y. After the collision, X has horizontal velocity 0.41 ms⁻¹ and Y has horizontal velocity v , as shown in Fig. 4.2.

Fig. 4.2



Calculate

- (i) the total initial momentum of the two balls,

momentum = Ns [3]

- (ii) the velocity v ,

$v =$ ms⁻¹ [2]

- (iii) the total initial kinetic energy of the two balls.

kinetic energy = J [3]

- (c) Explain how you would check whether the collision is elastic.

.....
 [1]

- (d) Use Newton's third law to explain why, during the collision, the change in momentum of X is equal and opposite to the change in momentum of Y.

.....

 [2]

9702/21/M/J/15/Q3

- 10 Two balls X and Y are supported by long strings, as shown in Fig. 3.1.

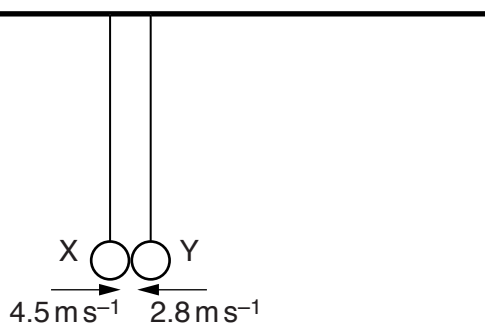


Fig. 3.1

The balls are each pulled back and pushed towards each other. When the balls collide at the position shown in Fig. 3.1, the strings are vertical. The balls rebound in opposite directions.

Fig. 3.2 shows data for X and Y during this collision.

ball	mass	velocity just before collision / ms^{-1}	velocity just after collision / ms^{-1}
X	50g	+4.5	-1.8
Y	M	-2.8	+1.4

Fig. 3.2

The positive direction is horizontal and to the right.

- (a) Use the conservation of linear momentum to determine the mass M of Y.

$$M = \dots\dots\dots \text{g} \quad [3]$$

- (b) State and explain whether the collision is elastic.

.....

[1]

- (c) Use Newton's second and third laws to explain why the magnitude of the change in momentum of each ball is the same.

.....

[3]

9702/22/M/J/15/Q4

- 11 (a)** A gas molecule has a mass of $6.64 \times 10^{-27} \text{ kg}$ and a speed of 1250 ms^{-1} . The molecule collides normally with a flat surface and rebounds with the same speed, as shown in Fig. 4.1.

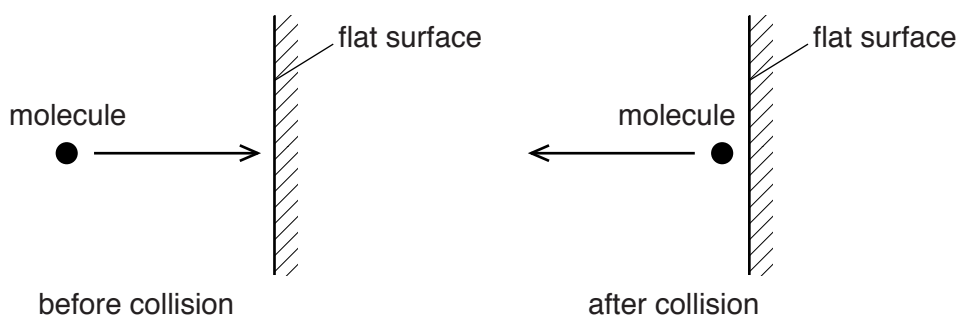


Fig. 4.1

Calculate the change in momentum of the molecule.

change in momentum = N s [2]

- (b) (i)** Use the kinetic model to explain the pressure exerted by gases.

.....

[3]

- (ii)** Explain the effect of an increase in density, at constant temperature, on the pressure of a gas.

.....
[1]

- (c) For the spheres in (b), the variation with time of the momentum of sphere A before during and after the collision with sphere B is shown in Fig. 3.2.

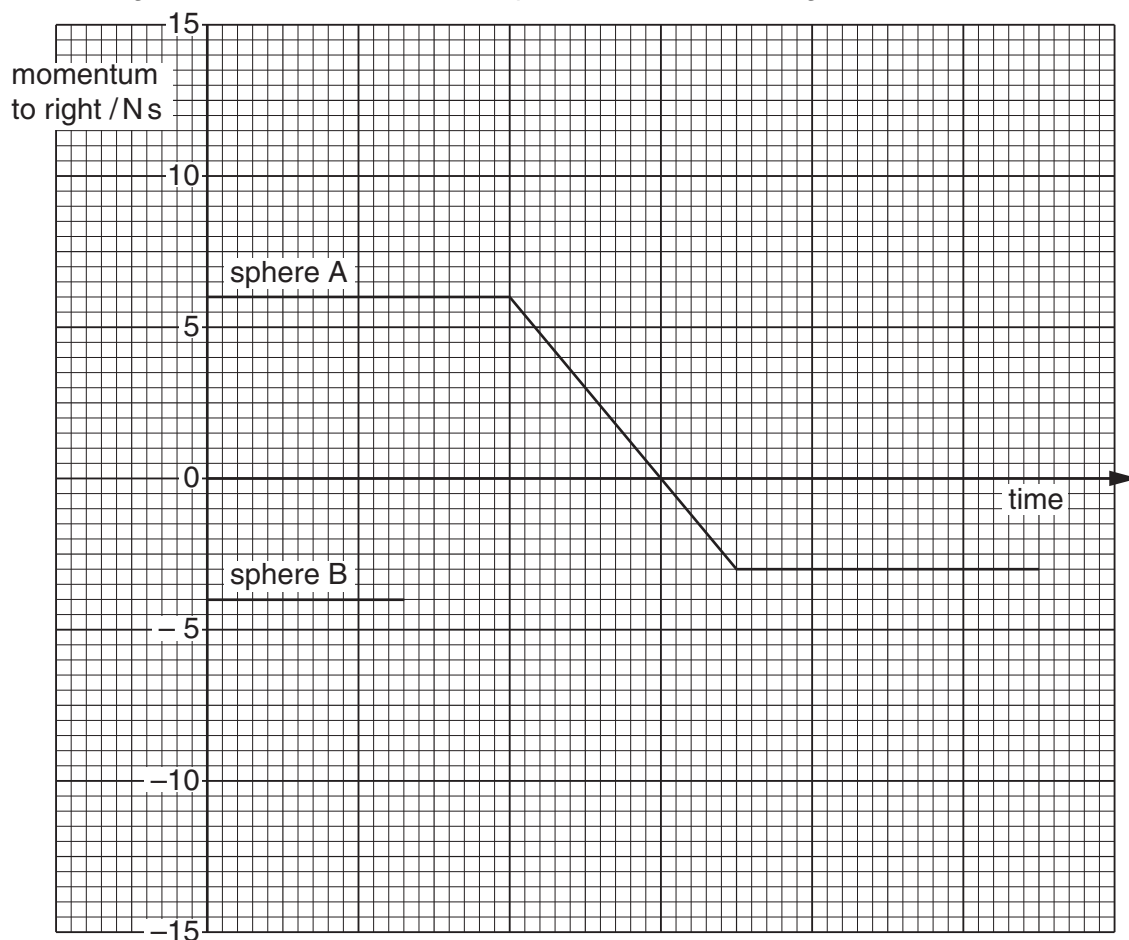


Fig. 3.2

The momentum of sphere B before the collision is also shown on Fig. 3.2.

Complete Fig. 3.2 to show the variation with time of the momentum of sphere B during and after the collision with sphere A.

[3]

12 A ball of mass 150 g is at rest on a horizontal floor, as shown in Fig. 3.1.

9702/21/M/J/16/Q3

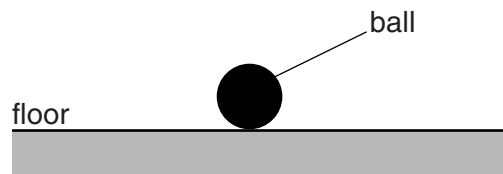


Fig. 3.1

- (a) (i)** Calculate the magnitude of the normal contact force from the floor acting on the ball.

force = N [1]

- (ii)** Explain your working in **(i)**.

.....
.....
.....[1]

- (b) The ball is now lifted above the floor and dropped so that it falls vertically, as illustrated in Fig. 3.2.

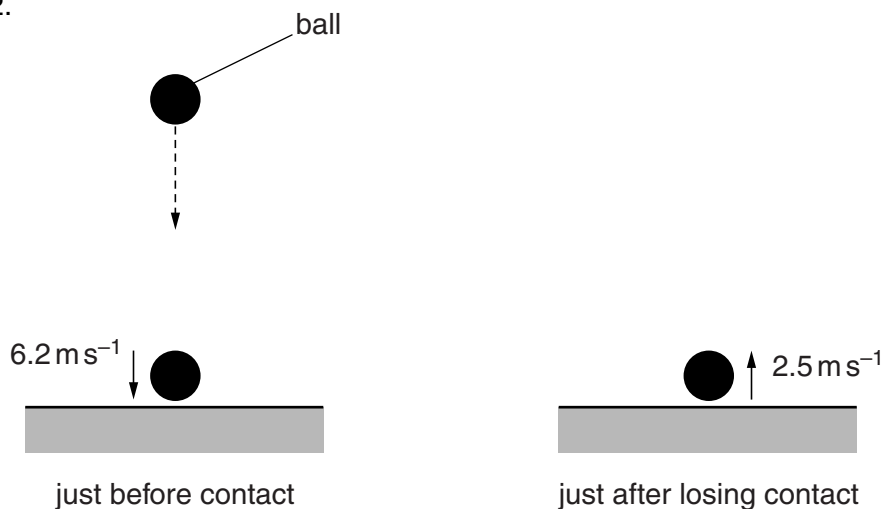


Fig. 3.2

Just before contact with the floor, the ball has velocity 6.2 m s^{-1} downwards. The ball bounces from the floor and its velocity just after losing contact with the floor is 2.5 m s^{-1} upwards. The ball is in contact with the floor for 0.12 s .

- (i) State Newton's second law of motion.

.....
[1]

- (ii) Calculate the average resultant force on the ball when it is in contact with the floor.

magnitude of force = N

direction of force
[3]

- (iii) State and explain whether linear momentum is conserved during the collision of the ball with the floor.

.....

[2]

13 (a) State the law of conservation of momentum.

9702/23/M/J/16/Q5

.....

 [2]

(b) Two particles A and B collide elastically, as illustrated in Fig. 5.1.

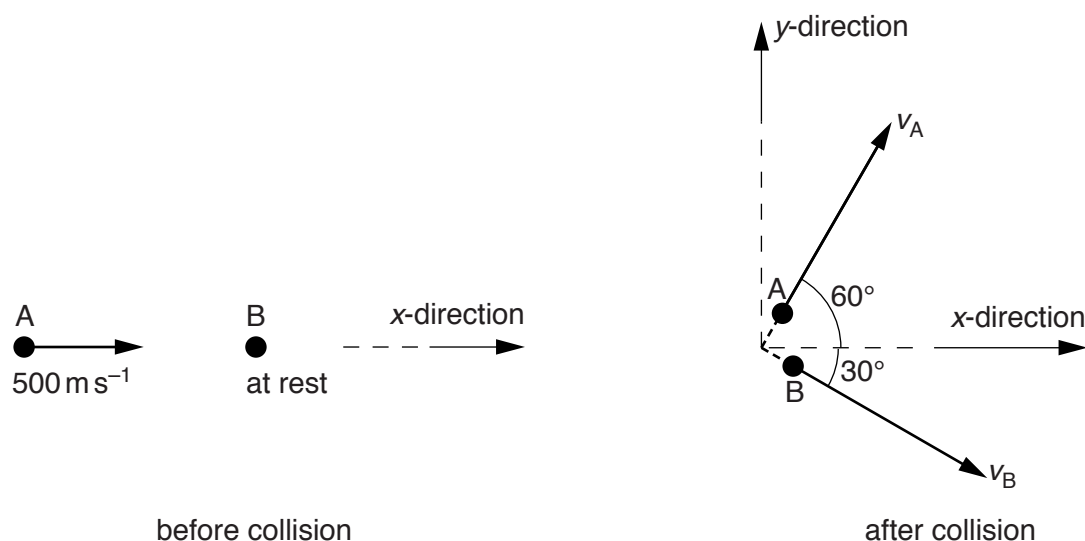


Fig. 5.1

The initial velocity of A is 500 m s^{-1} in the x-direction and B is at rest.

The velocity of A after the collision is v_A at 60° to the x-direction. The velocity of B after the collision is v_B at 30° to the x-direction.

The mass m of each particle is $1.67 \times 10^{-27} \text{ kg}$.

(i) Explain what is meant by the particles colliding *elastically*.

..... [1]

(ii) Calculate the total initial momentum of A and B.

momentum =Ns [1]

(iii) State an expression in terms of m , v_A and v_B for the total momentum of A and B after the collision

1. in the x -direction,

.....

2. in the y -direction.

.....

[2]

(iv) Calculate the magnitudes of the velocities v_A and v_B after the collision.

$v_A = \dots\dots\dots \text{ms}^{-1}$

$v_B = \dots\dots\dots \text{ms}^{-1}$
[3]

9702/21/M/J/17/Q2 (b)

- 14 A paraglider P of mass 95 kg is pulled by a wire attached to a boat, as shown in Fig. 2.1.

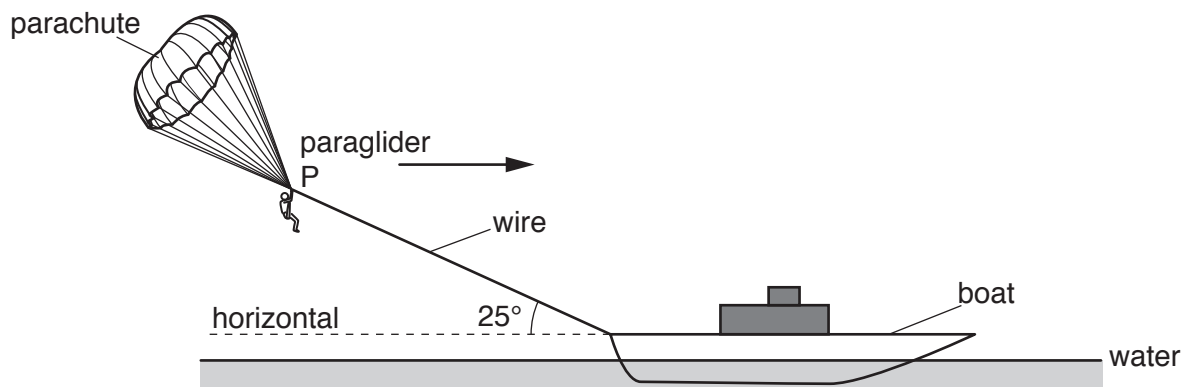


Fig. 2.1

The wire makes an angle of 25° with the horizontal water surface. P moves in a straight line parallel to the surface of the water.

The variation with time t of the velocity v of P is shown in Fig. 2.2.

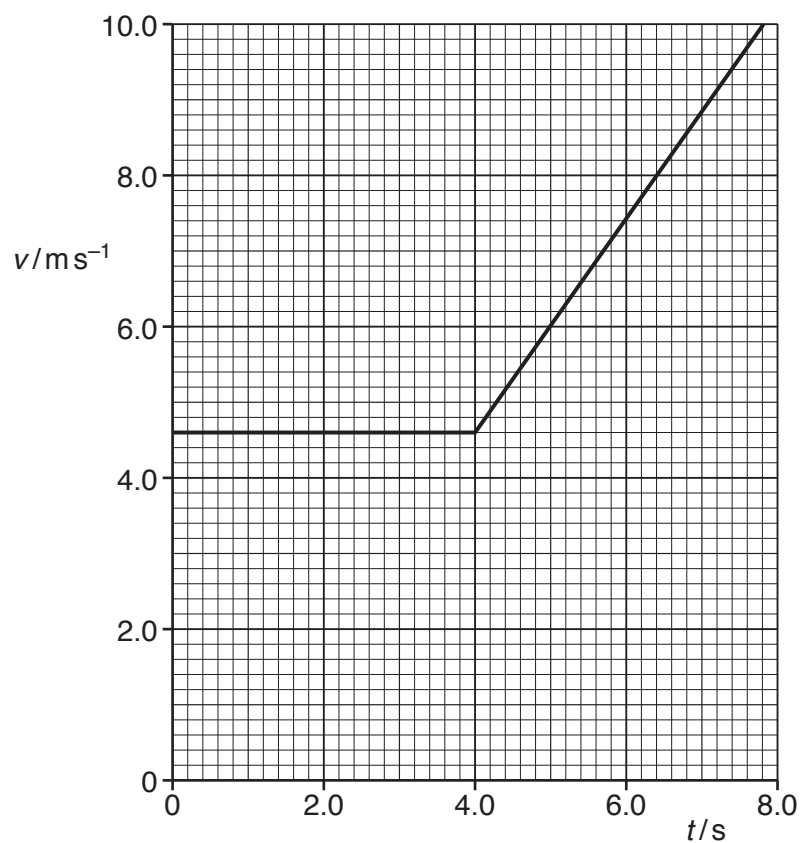


Fig. 2.2

- (i) Show that the acceleration of P is 1.4 ms^{-2} at time $t = 5.0 \text{ s}$.

[2]

- (ii) Calculate the total distance moved by P from time $t = 0$ to $t = 7.0 \text{ s}$.

distance =m [2]

- (iii) Calculate the change in kinetic energy of P from time $t = 0$ to $t = 7.0 \text{ s}$.

change in kinetic energy =J [2]

- (iv) The tension in the wire at time $t = 5.0 \text{ s}$ is 280 N .

Calculate, for the horizontal motion,

1. the vertical lift force F supporting P,

$F = \dots\dots\dots \text{ N}$ [3]

2. the force R due to air resistance acting on P in the horizontal direction.

$R = \dots\dots\dots \text{ N}$ [3]

9702/22/M/J/17/Q4

- 15 (a) State Newton's first law of motion.

.....
[1]

- (b) An object A of mass 100 g is moving in a straight line with a velocity of 0.60 m s^{-1} to the right. An object B of mass 200 g is moving in the same straight line as object A with a velocity of 0.80 m s^{-1} to the left, as shown in Fig. 4.1.



Fig. 4.1

Objects A and B collide. Object A then moves with a velocity of 0.40 m s^{-1} to the left.

- (i) Calculate the magnitude of the velocity of B after the collision.

magnitude of velocity = m s^{-1} [2]

- (ii) The collision between A and B is inelastic.

Explain how the collision is inelastic and still obeys the law of conservation of energy.

.....

[1]

9702/21/O/N/17/Q1 (b)

- 16 (a)** A ball of weight 1.5 N falls vertically from rest in air. The drag force F_D acting on the ball is given by the expression in **(a)**. The ball reaches a constant (terminal) speed of 33 m s^{-1} .

Assume that the upthrust acting on the ball is negligible and that the density of the air is uniform.

For the instant when the ball is travelling at a speed of 25 m s^{-1} , determine

- (i)** the drag force F_D on the ball,

$$F_D = \dots\dots\dots \text{ N [2]}$$

- (ii)** the acceleration of the ball.

$$\text{acceleration} = \dots\dots\dots \text{ m s}^{-2} \text{ [2]}$$

- (b)** Describe the acceleration of the ball in **(b)** as its speed changes from zero to 33 m s^{-1} .

.....
.....
.....
.....[3]

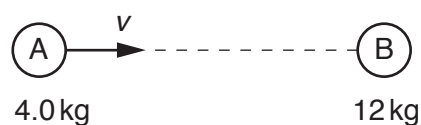
- 17 (a) State the principle of conservation of momentum.

9702/23/O/N/17/Q3

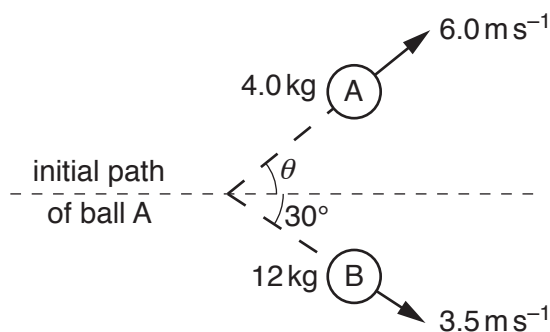
.....

[2]

- (b) Ball A moves with speed v along a horizontal frictionless surface towards a stationary ball B, as shown in Fig. 3.1.



before collision

Fig. 3.1

after collision

Fig. 3.2 (not to scale)

Ball A has mass 4.0 kg and ball B has mass 12 kg .

The balls collide and then move apart as shown in Fig. 3.2.

Ball A has velocity 6.0 m s^{-1} at an angle of θ to the direction of its initial path.

Ball B has velocity 3.5 m s^{-1} at an angle of 30° to the direction of the initial path of ball A.

- (i) By considering the components of momentum at right-angles to the direction of the initial path of ball A, calculate θ .

$$\theta = \dots\dots\dots^\circ \quad [3]$$

- (ii) Use your answer in (i) to show that the initial speed v of ball A is 12 m s^{-1} .
 Explain your working.

[2]

- (iii) By calculation of kinetic energies, state and explain whether the collision is elastic or inelastic.

.....
[3]

9702/21/M/J/18/Q3

- 18 (a) State what is meant by the *mass* of a body.

.....
[1]

- (b) Two blocks travel directly towards each other along a horizontal, frictionless surface. The blocks collide, as illustrated in Fig. 3.1.

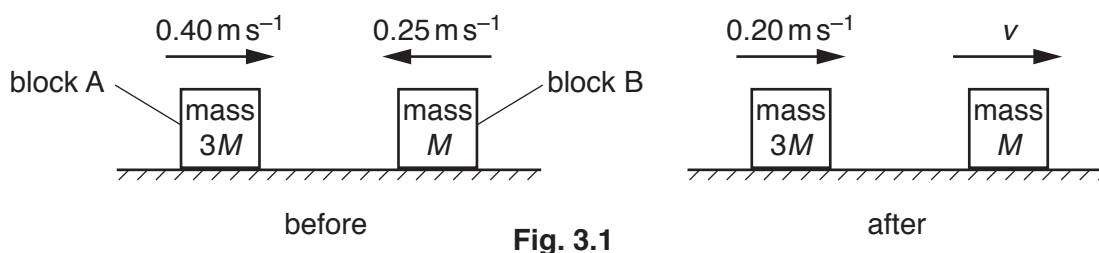


Fig. 3.1

Block A has mass $3M$ and block B has mass M .

Before the collision, block A moves to the right with speed 0.40 ms^{-1} and block B moves to the left with speed 0.25 ms^{-1} .

After the collision, block A moves to the right with speed 0.20 ms^{-1} and block B moves to the right with speed v .

- (i) Use Newton's third law to explain why, during the collision, the change in momentum of block A is equal and opposite to the change in momentum of block B.

.....

[2]

(ii) Determine speed v .

$$v = \dots\dots\dots \text{ms}^{-1} \text{ [3]}$$

(iii) Calculate, for the blocks,

1. the relative speed of approach,

$$\text{relative speed of approach} = \dots\dots\dots \text{ms}^{-1}$$

2. the relative speed of separation.

$$\text{relative speed of separation} = \dots\dots\dots \text{ms}^{-1} \text{ [2]}$$

(iv) Use your answers in (b)(iii) to state and explain whether the collision is elastic or inelastic.

.....
.....[1]

- 19 (a) State the principle of conservation of momentum.

9702/22/M/J/18/Q2

.....

[2]

- (b) A stationary firework explodes into three different fragments that move in a horizontal plane, as illustrated in Fig. 2.1.

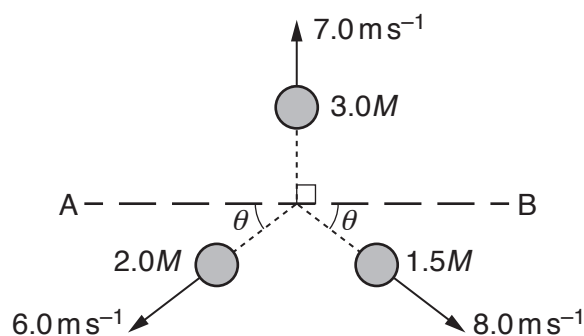


Fig. 2.1

The fragment of mass $3.0M$ has a velocity of 7.0 ms^{-1} perpendicular to line AB.
 The fragment of mass $2.0M$ has a velocity of 6.0 ms^{-1} at angle θ to line AB.
 The fragment of mass $1.5M$ has a velocity of 8.0 ms^{-1} at angle θ to line AB.

- (i) Use the principle of conservation of momentum to determine θ .

$\theta = \dots\dots\dots^\circ$ [3]

- (ii) Calculate the ratio

$$\frac{\text{kinetic energy of fragment of mass } 2.0M}{\text{kinetic energy of fragment of mass } 1.5M}$$

ratio =[2]

9702/21/O/N/18/Q2

- 20 A wooden block moves along a horizontal frictionless surface, as shown in Fig. 2.1.

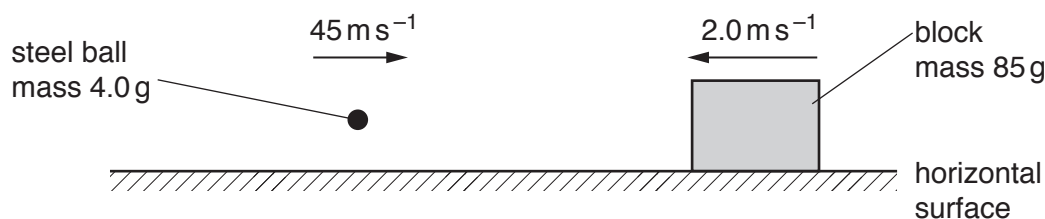


Fig. 2.1

The block has mass 85 g and moves to the left with a velocity of 2.0 m s^{-1} . A steel ball of mass 4.0 g is fired to the right. The steel ball, moving horizontally with a speed of 45 m s^{-1} , collides with the block and remains embedded in it. After the collision the block and steel ball both have speed v .

- (a) Calculate v .

$$v = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (b) (i) For the block and ball, state

1. the relative speed of approach before collision,

$$\text{relative speed of approach} = \dots\dots\dots \text{ m s}^{-1}$$

2. the relative speed of separation after collision.

$$\text{relative speed of separation} = \dots\dots\dots \text{ m s}^{-1} \quad [1]$$

- (ii) Use your answers in (i) to state and explain whether the collision is elastic or inelastic.

.....
 [1]

- (c) Use Newton's third law to explain the relationship between the rate of change of momentum of the ball and the rate of change of momentum of the block during the collision.

.....

 [2]

9702/22/O/N/18/Q3

- 21 (a) State the principle of conservation of momentum.

.....

.....

..... [2]

- (b) The propulsion system of a toy car consists of a propeller attached to an electric motor, as illustrated in Fig. 3.1.

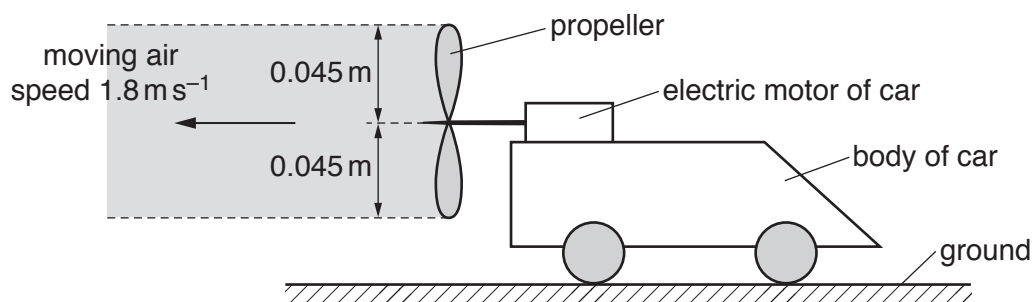


Fig. 3.1

The car is on horizontal ground and is initially held at rest by its brakes. When the motor is switched on, it rotates the propeller so that air is propelled horizontally to the left. The density of the air is 1.3 kg m^{-3} .

Assume that the air moves with a speed of 1.8 ms^{-1} in a uniform cylinder of radius 0.045 m . Also assume that the air to the right of the propeller is stationary.

- (i) Show that, in a time interval of 2.0 s , the mass of air propelled to the left is 0.030 kg .

[2]

(ii) Calculate

1. the increase in the momentum of the mass of air in **(b)(i)**,

increase in momentum = N s

2. the force exerted on this mass of air by the propeller.

force = N
[3]

(iii) Explain how Newton's third law applies to the movement of the air by the propeller.

.....
.....
..... [2]

(iv) The total mass of the car is 0.20 kg. The brakes of the car are released and the car begins to move with an initial acceleration of 0.075 m s^{-2} .

Determine the initial frictional force acting on the car.

frictional force = N [2]

9702/23/O/N/18/Q3

- 22 (a) State Newton's second law of motion.

.....
.....[1]

- (b) A toy rocket consists of a container of water and compressed air, as shown in Fig. 3.1.

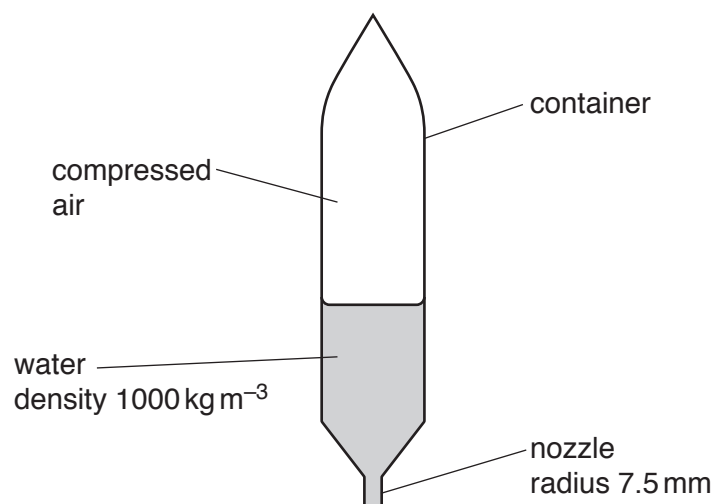


Fig. 3.1

Water is pushed vertically downwards through a nozzle by the compressed air. The rocket moves vertically upwards.

The nozzle has a circular cross-section of radius 7.5 mm. The density of the water is 1000 kg m^{-3} . Assume that the water leaving the nozzle has the shape of a cylinder of radius 7.5 mm and has a constant speed of 13 m s^{-1} relative to the rocket.

- (i) Show that the mass of water leaving the nozzle in the first 0.20 s after the rocket launch is 0.46 kg.

[2]

(ii) Calculate

1. the change in the momentum of the mass of water in **(b)(i)** due to leaving the nozzle,

change in momentum =Ns

2. the force exerted on this mass of water by the rocket.

force =N
[3]

(iii) State and explain how Newton's third law applies to the movement of the rocket by the water.

.....
.....
.....[2]

(iv) The container has a mass of 0.40 kg. The initial mass of water before the rocket is launched is 0.70 kg. The mass of the compressed air in the rocket is negligible. Assume that the resistive force on the rocket due to its motion is negligible.

For the rocket at a time of 0.20 s after launching,

1. show that its total mass is 0.64 kg,

2. calculate its acceleration.

acceleration = ms⁻²
[3]

9702/21/M/J/19/Q2

- 23 A block X slides along a horizontal frictionless surface towards a stationary block Y, as illustrated in Fig. 2.1.

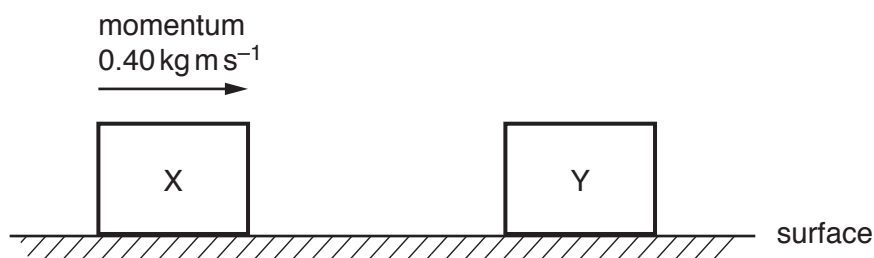


Fig. 2.1

There are no resistive forces acting on block X as it moves towards block Y. At time $t = 0$, block X has momentum 0.40 kg m s^{-1} . A short time later, the blocks collide and then separate.

The variation with time t of the momentum of block Y is shown in Fig. 2.2.

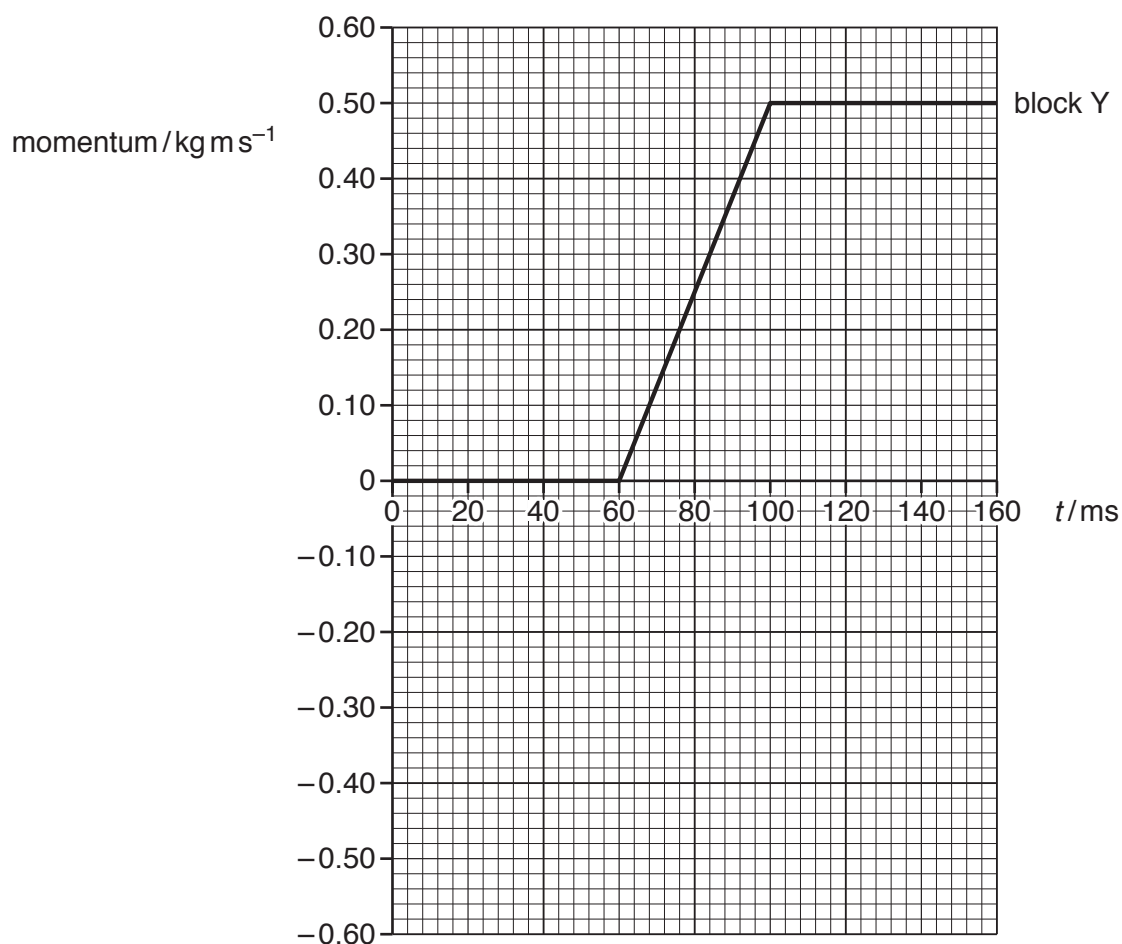


Fig. 2.2

(a) Define *linear momentum*.

.....[1]

(b) Use Fig. 2.2 to:

(i) determine the time interval over which the blocks are in contact with each other

time interval = ms [1]

(ii) describe, without calculation, the magnitude of the acceleration of block Y from:

1. time $t = 80$ ms to $t = 100$ ms

.....

2. time $t = 100$ ms to $t = 120$ ms.

.....

[2]

(c) Use Fig. 2.2 to determine the magnitude of the force exerted by block X on block Y.

force = N [2]

(d) On Fig. 2.2, sketch the variation of the momentum of block X with time t from $t = 0$ to $t = 160$ ms. [3]

9702/21/M/J/20/Q2

- 24 (a) State Newton's second law of motion.

.....
 [1]

- (b) A delivery company suggests using a remote-controlled aircraft to drop a parcel into the garden of a customer. When the aircraft is vertically above point P on the ground, it releases the parcel with a velocity that is horizontal and of magnitude 5.4 m s^{-1} . The path of the parcel is shown in Fig. 2.1.

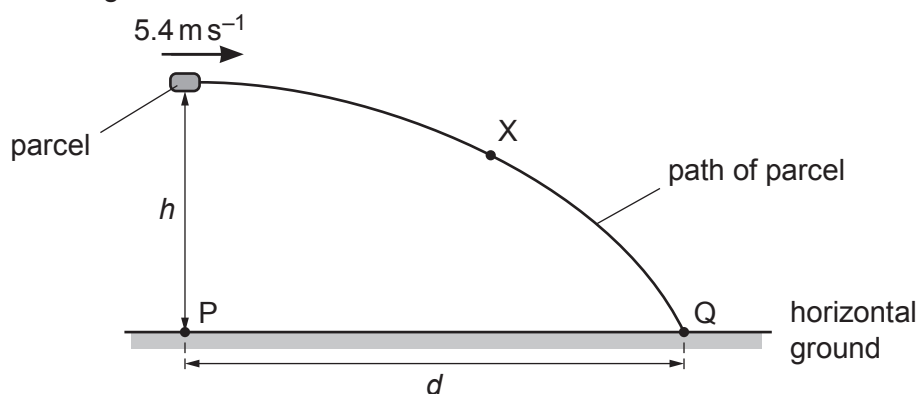


Fig. 2.1 (not to scale)

The parcel takes a time of 0.81 s after its release to reach point Q on the horizontal ground. Assume air resistance is negligible.

- (i) On Fig. 2.1, draw an arrow from point X to show the direction of the acceleration of the parcel when it is at that point. [1]
 (ii) Determine the height h of the parcel above the ground when it is released.

$h = \dots\dots\dots \text{ m}$ [2]

- (iii) Calculate the horizontal distance d between points P and Q.

$d = \dots\dots\dots \text{ m}$ [1]

- (c) Another parcel is accidentally released from rest by a different aircraft when it is hovering at a great height above the ground. Air resistance is now significant.
- (i) On Fig. 2.2, draw arrows to show the directions of the forces acting on the parcel as it falls vertically downwards. Label each arrow with the name of the force.

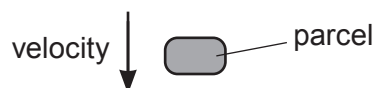


Fig. 2.2

[2]

- (ii) By considering the forces acting on the parcel, state and explain the variation, if any, of the acceleration of the parcel as it moves downwards before it reaches constant (terminal) speed.

.....

.....

.....

.....

.....

..... [3]

- (iii) Describe the energy conversion that occurs when the parcel is falling through the air at constant (terminal) speed.

.....

..... [1]

9702/22/M/J/20/Q2

- 25 (a) Fig. 2.1 shows the velocity–time graph for an object moving in a straight line.

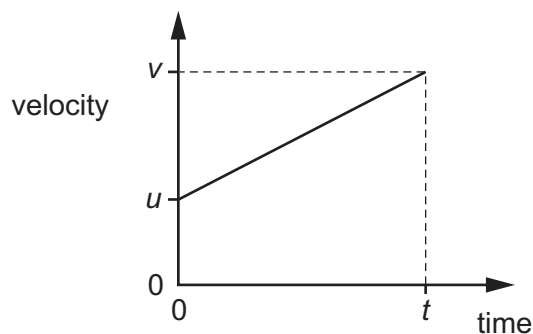


Fig. 2.1

- (i) Determine an expression, in terms of u , v and t , for the area under the graph.

area = [1]

- (ii) State the name of the quantity represented by the area under the graph.

..... [1]

- (b) A ball is kicked with a velocity of 15 m s^{-1} at an angle of 60° to horizontal ground. The ball then strikes a vertical wall at the instant when the path of the ball becomes horizontal, as shown in Fig. 2.2.

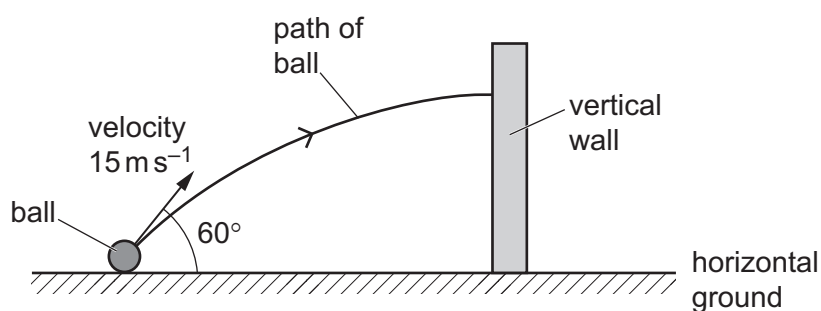


Fig. 2.2 (not to scale)

Assume that air resistance is negligible.

- (i) By considering the vertical motion of the ball, calculate the time it takes to reach the wall.

time = s [3]

- (ii) Explain why the horizontal component of the velocity of the ball remains constant as it moves to the wall.

.....

..... [1]

- (iii) Show that the ball strikes the wall with a horizontal velocity of 7.5 m s^{-1} .

[1]

- (c) The mass of the ball in (b) is 0.40 kg . It is in contact with the wall for a time of 0.12 s and rebounds horizontally with a speed of 4.3 m s^{-1} .

- (i) Use the information from (b)(iii) to calculate the change in momentum of the ball due to the collision.

change in momentum = kg m s^{-1} [2]

- (ii) Calculate the magnitude of the average force exerted on the ball by the wall.

average force = N [1]

9702/23/M/J/20/Q2

- 26 (a) State Newton's first law of motion.

.....
 [1]

- (b) A skier is pulled in a straight line along horizontal ground by a wire attached to a kite, as shown in Fig. 2.1.

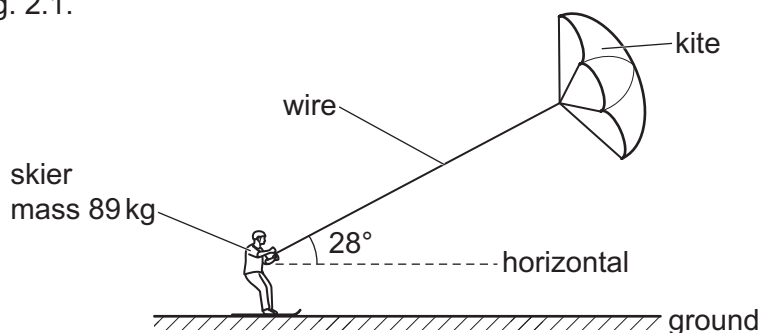


Fig. 2.1 (not to scale)

The mass of the skier is 89 kg. The wire is at an angle of 28° to the horizontal. The variation with time t of the velocity v of the skier is shown in Fig. 2.2.

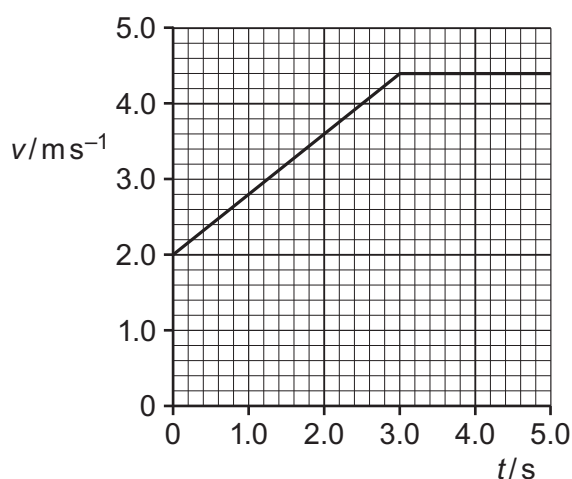


Fig. 2.2

- (i) Use Fig. 2.2 to determine the distance moved by the skier from time $t = 0$ to $t = 5.0$ s.

distance = m [2]

- (ii) Use Fig. 2.2 to show that the acceleration a of the skier is 0.80 ms^{-2} at time $t = 2.0 \text{ s}$.

[2]

- (iii) The tension in the wire at time $t = 2.0 \text{ s}$ is 240 N .

Calculate:

1. the horizontal component of the tension force acting on the skier

horizontal component of force = N [1]

2. the total resistive force R acting on the skier in the horizontal direction.

$R =$ N [2]

- (iv) The skier is now lifted upwards by a gust of wind. For a few seconds the skier moves horizontally through the air with the wire at an angle of 45° to the horizontal, as shown in Fig. 2.3.

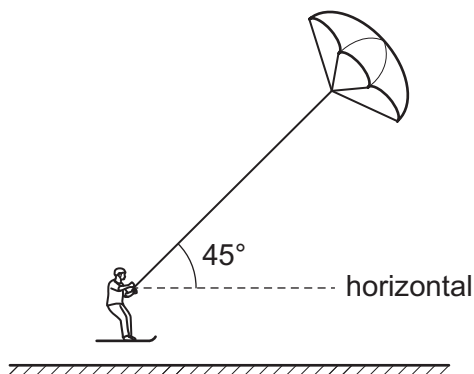


Fig. 2.3 (not to scale)

By considering the vertical components of the forces acting on the skier, determine the new tension in the wire when the skier is moving horizontally through the air.

tension = N [2]

27. 9702/21/O/N/20/Q3

(a) Define *force*.

.....
 [1]

(b) A ball falls vertically downwards towards a horizontal floor and then rebounds along its original path, as illustrated in Fig. 3.1.

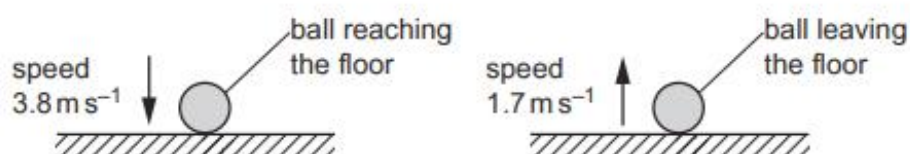


Fig. 3.1

The ball reaches the floor with speed 3.8 m s^{-1} . The ball is then in contact with the floor for a time of 0.081 s before leaving it with speed 1.7 m s^{-1} . The mass of the ball is 0.062 kg .

(i) Calculate the loss of kinetic energy of the ball during the collision.

loss of kinetic energy = J [2]

(ii) Determine the magnitude of the change in momentum of the ball during the collision.

change in momentum = N s [2]

(iii) Show that the magnitude of the average resultant force acting on the ball during the collision is 4.2 N .

[1]

(iv) Use the information in (iii) to calculate the magnitude of:

1. the average force of the floor on the ball during the collision

average force = N

2. the average force of the ball on the floor during the collision.

average force = N
[2]

[Total: 8]

28. 9702/22/O/N/20/Q2

- (a) A cylinder is suspended from the end of a string. The cylinder is stationary in water with the axis of the cylinder vertical, as shown in Fig. 2.1.

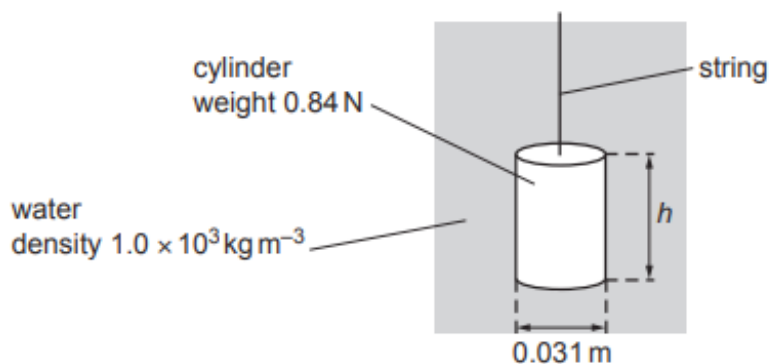


Fig. 2.1 (not to scale)

The cylinder has weight 0.84 N , height h and a circular cross-section of diameter 0.031 m . The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$. The difference between the pressures on the top and bottom faces of the cylinder is 520 Pa .

- (i) Calculate the height h of the cylinder.

$h = \dots\dots\dots \text{ m}$ [2]

- (ii) Show that the upthrust acting on the cylinder is 0.39 N .

[2]

- (iii) Calculate the tension T in the string.

$T = \dots\dots\dots \text{ N}$ [1]

- (b) The string is now used to move the cylinder in (a) vertically upwards through the water. The variation with time t of the velocity v of the cylinder is shown in Fig. 2.2.

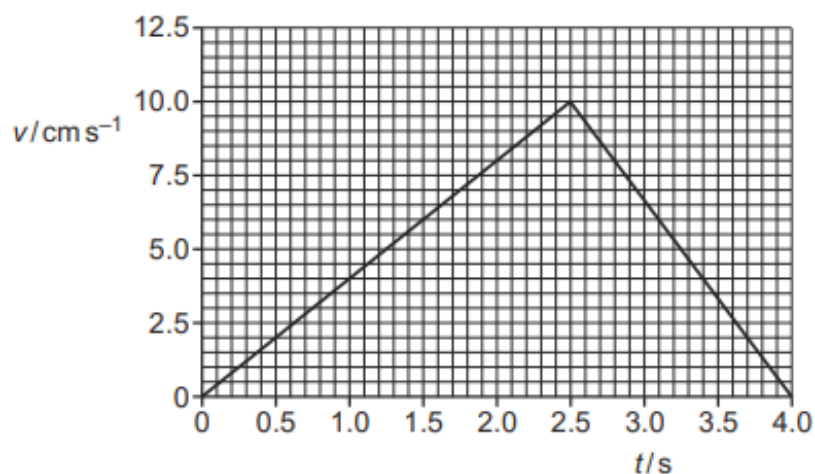


Fig. 2.2

- (i) Use Fig. 2.2 to determine the acceleration of the cylinder at time $t = 2.0$ s.

acceleration = m s^{-2} [2]

- (ii) The top face of the cylinder is at a depth of 0.32 m below the surface of the water at time $t = 0$.

Use Fig. 2.2 to determine the depth of the top face below the surface of the water at time $t = 4.0$ s.

depth = m [2]

(c) The cylinder in (b) is released from the string at time $t = 4.0\text{ s}$. The cylinder falls, from rest, vertically downwards through the water. Assume that the upthrust acting on the cylinder remains constant as it falls.

(i) State the name of the force that acts on the cylinder when it is moving and does not act on the cylinder when it is stationary.

..... [1]

(ii) State and explain the variation, if any, of the acceleration of the cylinder as it falls downwards through the water.

.....

.....

..... [2]

[Total: 12]

29. 9702/22/M/J/21/Q1

- (a) Complete Table 1.1 by stating whether each of the quantities is a vector or a scalar.

Table 1.1

quantity	vector or scalar
acceleration	
electrical resistance	
momentum	

[2]

- (b) State the conditions for an object to be in equilibrium.

.....

.....

.....

..... [2]

- (c) A floating solid cylinder is attached by a wire to the sea bed, as shown in Fig. 1.1.

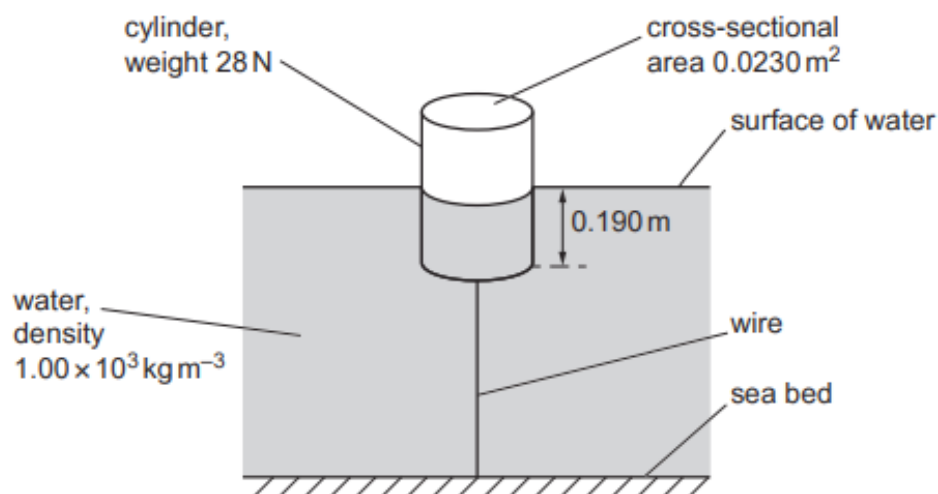


Fig. 1.1 (not to scale)

The density of the water is $1.00 \times 10^3 \text{ kg m}^{-3}$. The base of the cylinder is at a depth of 0.190 m below the surface of the water. The cylinder has a weight of 28 N and a cross-sectional area of 0.0230 m^2 .

The wire and the central axis of the cylinder are both vertical. The cylinder is in equilibrium.

- (i) Calculate, to three significant figures, the upthrust acting on the cylinder due to the water.

upthrust = N [2]

- (ii) Show that the tension T in the wire is 15 N.

[1]

- (iii) The wire has a cross-sectional area of 3.2 mm^2 .

Calculate the stress in the wire.

stress = Pa [2]

- (iv) The surface of the water gradually rises until it is level with the top face of the cylinder.

State and explain, qualitatively, the variation of the strain energy stored in the wire as the water surface rises.

.....

.....

.....

..... [2]

30. 9702/22/M/J/21/Q2

A ball is thrown vertically downwards to the ground, as illustrated in Fig. 2.1.

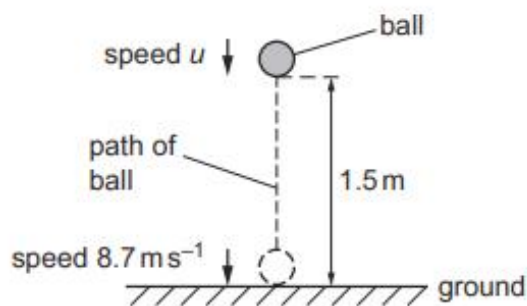


Fig. 2.1

The ball is thrown with speed u from a height of 1.5 m. The ball then hits the ground with speed 8.7 m s^{-1} . Assume that air resistance is negligible.

- (a) Calculate speed u .

$u = \dots\dots\dots \text{ m s}^{-1}$ [2]

- (b) State how Newton's third law applies to the collision between the ball and the ground.

.....

 [2]

- (c) The ball is in contact with the ground for a time of 0.091 s. The ball rebounds vertically and leaves the ground with speed 5.4 m s^{-1} . The mass of the ball is 0.059 kg.

- (i) Calculate the magnitude of the change in momentum of the ball during the collision.

change in momentum = N s [2]

- (ii) Determine the magnitude of the average resultant force that acts on the ball during the collision.

average resultant force = N [1]

- (iii) Use your answer in (c)(ii) to calculate the magnitude of the average force exerted by the ground on the ball during the collision.

average force = N [2]

- (d) The ball was thrown downwards at time $t = 0$ and hits the ground at time $t = T$.

On Fig. 2.2, sketch a graph to show the variation of the speed of the ball with time t from $t = 0$ to $t = T$. Numerical values are not required.

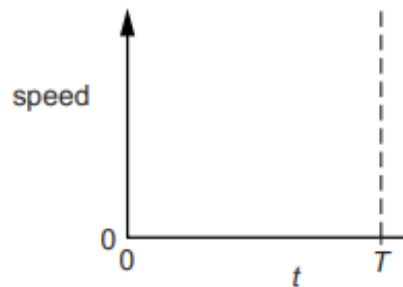


Fig. 2.2

[1]

- (e) In practice, air resistance is not negligible.

State and explain the variation, if any, with time t of the gradient of the graph in (d) when air resistance is not negligible.

.....

 [2]

31. 9702/21/O/N/21/Q2

(a) Define *momentum*.

.....
 [1]

(b) Two balls X and Y, of equal diameter but different masses 0.24 kg and 0.12 kg respectively, slide towards each other on a frictionless horizontal surface, as shown in Fig. 2.1.

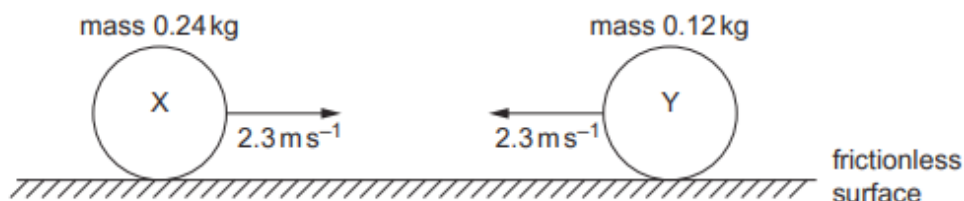


Fig. 2.1

Both balls have initial speed 2.3 ms^{-1} before they collide with each other. Fig. 2.2 shows the variation with time t of the force F_Y exerted on ball Y by ball X during the collision.

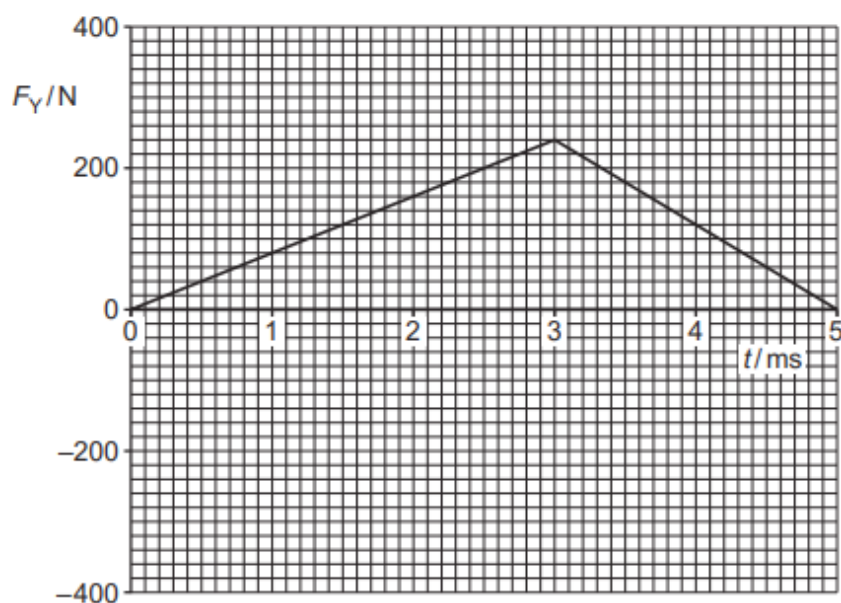


Fig. 2.2

(i) Calculate the kinetic energy of ball X before the collision.

kinetic energy = J [3]

- (ii) The area enclosed by the lines and the time axis in Fig. 2.2 represents the change in momentum of ball Y during the collision.

Determine the magnitude of the change in momentum of ball Y.

change in momentum = N s [2]

- (iii) Calculate the magnitude of the velocity of ball Y after the collision.

velocity = ms^{-1} [2]

- (c) On Fig. 2.3, sketch the variation with time t of the force F_X exerted on ball X by ball Y during the collision in (b).

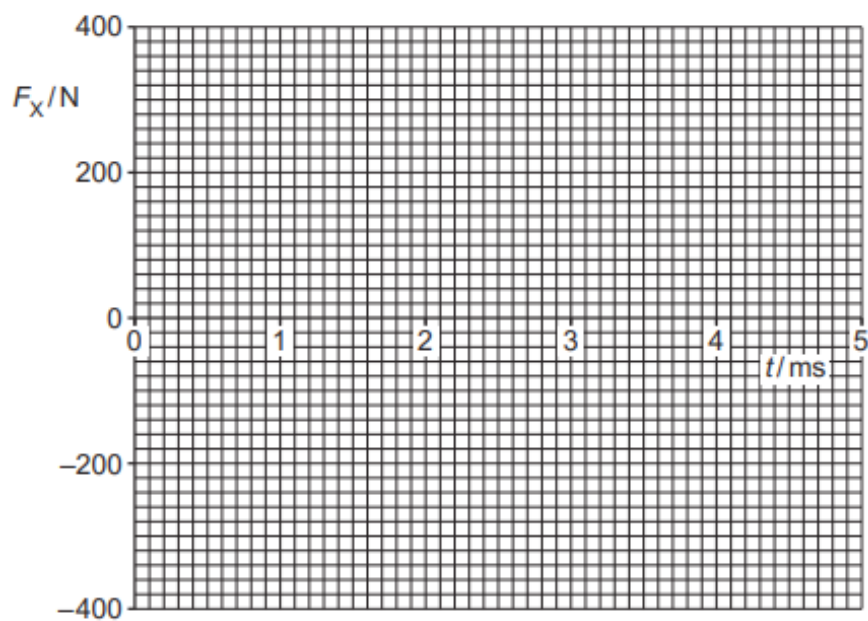


Fig. 2.3

32. 9702/22/F/M/22/Q1

A sphere of radius 2.1 mm falls with terminal (constant) velocity through a liquid, as shown in Fig. 1.1.

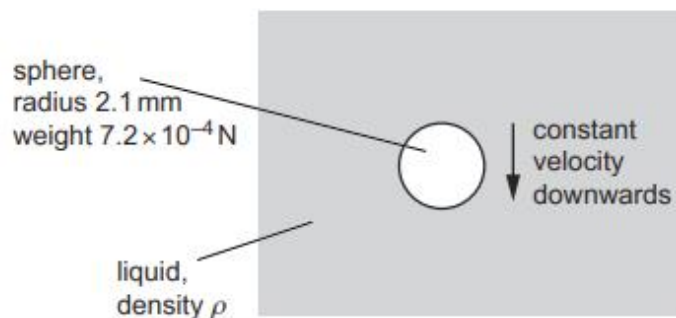


Fig. 1.1

Three forces act on the moving sphere. The weight of the sphere is $7.2 \times 10^{-4} \text{ N}$ and the upthrust acting on it is $4.8 \times 10^{-4} \text{ N}$. The viscous force F_V acting on the sphere is given by

$$F_V = krv$$

where r is the radius of the sphere, v is its velocity and k is a constant. The value of k in SI units is 17.

(a) Determine the SI base units of k .

SI base units [2]

(b) Use the value of the upthrust acting on the sphere to calculate the density ρ of the liquid.

$\rho = \dots \text{ kg m}^{-3}$ [3]

- (c) (i) On the sphere in Fig. 1.1, draw three arrows to show the directions of the weight W , the upthrust U and the viscous force F_v . Label these arrows W , U and F_v respectively. [1]
- (ii) Determine the magnitude of the terminal (constant) velocity of the sphere.

velocity = ms^{-1} [2]

[Total: 8]

33. 9702/22/F/M/22/Q3

A jet of water hits a vertical wall at right angles, as shown in Fig. 3.1.

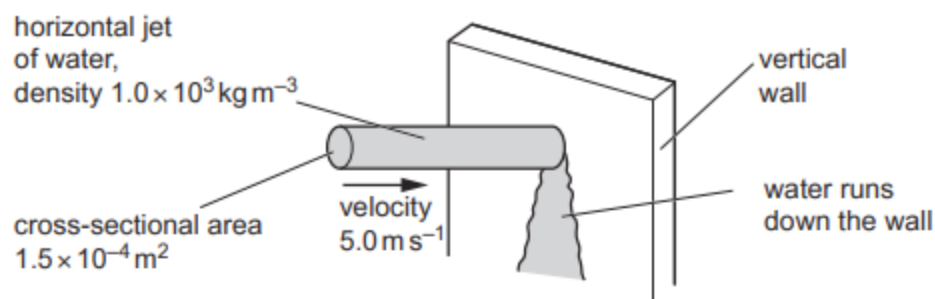


Fig. 3.1 (not to scale)

The water hits the vertical wall with a velocity of 5.0 m s^{-1} in a horizontal direction. The cross-sectional area of the jet is $1.5 \times 10^{-4} \text{ m}^2$. The density of the water is $1.0 \times 10^3 \text{ kg m}^{-3}$.

The water runs down the wall after hitting it.

(a) Show that, over a time of 1.6 s, the mass of water hitting the wall is 1.2 kg.

[2]

(b) Calculate:

- (i) the decrease in the horizontal momentum of the mass of water in (a) due to hitting the wall

decrease in momentum = N s [1]

- (ii) the magnitude of the horizontal force exerted on the water by the wall.

force = N [1]

(c) State and explain the magnitude of the horizontal force exerted on the wall by the water.

.....
..... [1]

(d) Calculate the pressure exerted on the wall by the water.

pressure = Pa [2]

[Total: 7]

34. 9702/21/M/J/22/Q1

(a) Define velocity.

.....
..... [1]

- (b) A rock of mass 7.5 kg is projected vertically upwards from the surface of a planet. The rock leaves the surface of the planet with a speed of 4.0 ms^{-1} at time $t = 0$. The variation with time t of the velocity v of the rock is shown in Fig. 1.1.

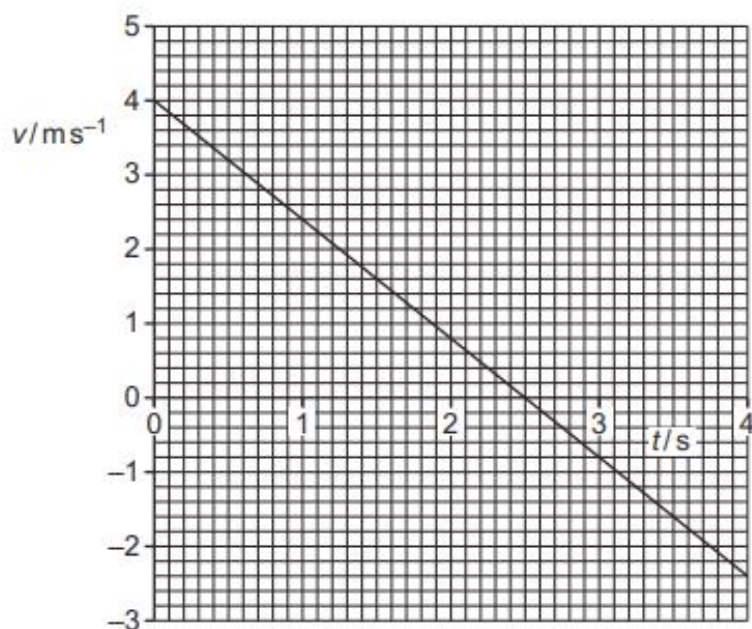


Fig. 1.1

Assume that the planet does not have an atmosphere and that the viscous force acting on the rock is always zero.

- (i) Determine the height of the rock above the surface of the planet at time $t = 4.0\text{ s}$.

height = m [3]

- (ii) Determine the change in the momentum of the rock from time $t = 0$ to time $t = 4.0\text{ s}$.

change in momentum = N s [2]

- (iii) Determine the weight W of the rock on this planet.

$W =$ N [2]

- (c) In practice, the planet in (b) does have an atmosphere that causes a viscous force to act on the moving rock.

State and explain the variation, if any, in the resultant force acting on the rock as it moves vertically upwards.

.....
.....
.....
..... [2]

[Total: 10]

35. 9702/22/M/J/22/Q2

A sphere is attached by a metal wire to the horizontal surface at the bottom of a river, as shown in Fig. 2.1.

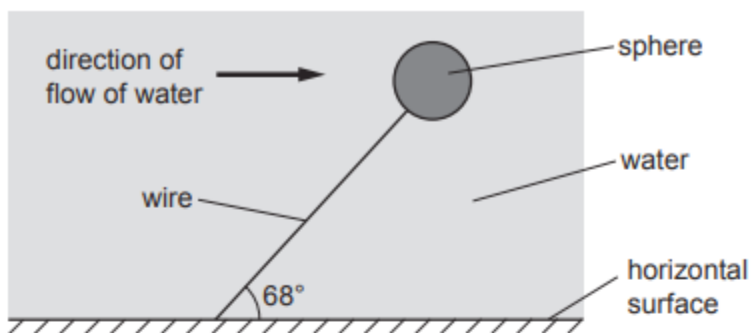


Fig. 2.1 (not to scale)

The sphere is fully submerged and in equilibrium, with the wire at an angle of 68° to the horizontal surface. The weight of the sphere is 32 N . The upthrust acting on the sphere is 280 N . The density of the water is $1.0 \times 10^3\text{ kg m}^{-3}$.

Assume that the force on the sphere due to the water flow is in a horizontal direction.

- (a) By considering the components of force in the vertical direction, determine the tension in the wire.

tension = N [2]

- (b) For the sphere, calculate:

- (i) the volume

volume = m^3 [1]

- (ii) the density.

density = kg m^{-3} [2]

- (c) The centre of the sphere is initially at a height of 6.2m above the horizontal surface. The speed of the water then increases, causing the sphere to move to a different position. This movement of the sphere causes its gravitational potential energy to decrease by 77 J.

Calculate the final height of the centre of the sphere above the horizontal surface.

height = m [3]

- (d) The extension of the wire increases when the sphere changes position as described in (c). The wire obeys Hooke's law.

- (i) State a symbol equation that gives the relationship between the tension T in the wire and its extension x . Identify any other symbol that you use.

.....
..... [1]

- (ii) Before the sphere changed position, the initial elastic potential energy of the wire was 0.65 J. The change in position of the sphere causes the extension of the wire to double.

Calculate the final elastic potential energy of the wire after the sphere has changed position.

final elastic potential energy = J [2]

[Total: 11]

36. 9702/22/M/J/22/Q4

(a) State the principle of conservation of momentum.

.....

 [2]

(b) Two balls, X and Y, move along a horizontal frictionless surface, as shown from above in Fig. 4.1.

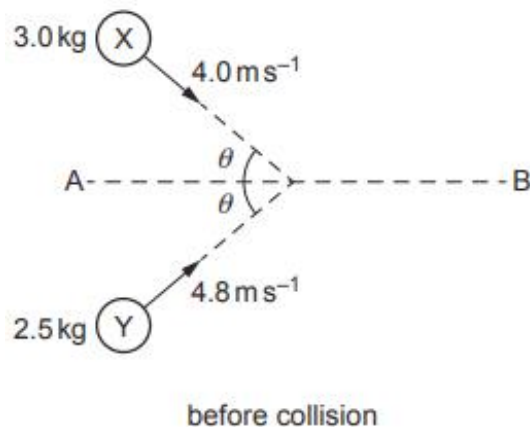


Fig. 4.1 (not to scale)

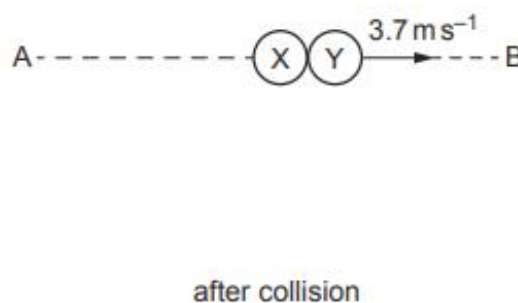


Fig. 4.2 (not to scale)

Ball X has a mass of 3.0 kg and a velocity of 4.0 m s^{-1} in a direction at angle θ to a line AB. Ball Y has a mass of 2.5 kg and a velocity of 4.8 m s^{-1} in a direction at angle θ to the line AB.

The balls collide and stick together. After colliding, the balls have a velocity of 3.7 m s^{-1} along the line AB on the horizontal surface, as shown in Fig. 4.2.

(i) By considering the components of the momenta along the line AB, calculate θ .

$\theta = \dots\dots\dots^\circ$ [3]

- (ii) By calculation of kinetic energies, state and explain whether the collision of the balls is inelastic or perfectly elastic.

.....
..... [2]

[Total: 7]

37. 9702/23/M/J/22/Q3

(a) Define velocity.

.....
..... [1]

(b) A constant driving force of 2400N acts on a car of mass 1200kg. The car accelerates from rest in a straight line along a horizontal road.

Assume that the resistive forces acting on the car are negligible.

(i) Calculate the acceleration of the car.

acceleration = m s^{-2} [1]

(ii) On Fig. 3.1, sketch a graph showing the variation with time t of the velocity v of the car for the first 20 seconds of its motion.

Label this line A.

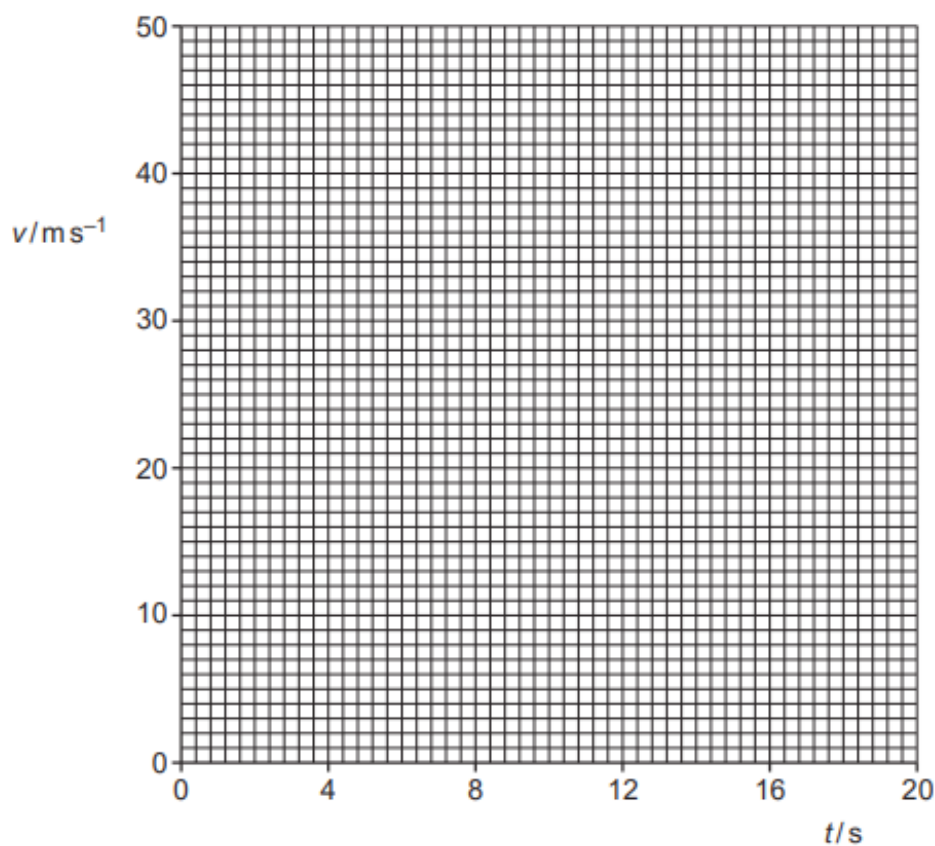


Fig. 3.1

[2]

(c) In reality, a resistive force due to air resistance acts on the car in (b). This resistive force increases with speed until it becomes equal in magnitude to the driving force at time $t = 12$ s.

(i) On Fig. 3.1, sketch a second line to show the variation with time t of the velocity v of the car for the first 20 seconds of its motion. Label this line B. [3]

(ii) At time $t = 20$ s, the driving force is increased to 3000 N and remains constant at this value.

Describe how the velocity of the car changes due to this increase in the driving force.

.....
.....
.....
..... [2]

[Total: 9]

38. 9702/22/O/N/22/Q2

A spherical balloon is filled with a fixed mass of gas. A small block is connected by a string to the balloon, as shown in Fig. 2.1.

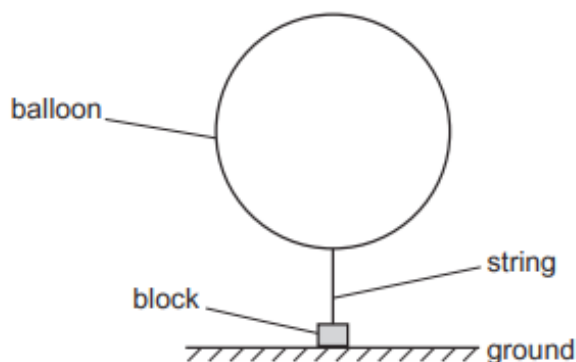


Fig. 2.1 (not to scale)

The block is held on the ground by an external force so that the string is vertical. The density of the air surrounding the balloon is 1.2 kg m^{-3} . The upthrust acting on the balloon is 0.071 N . The upthrust acting on the string and block is negligible.

- (a) By using Archimedes' principle, calculate the radius r of the balloon.

$r = \dots\dots\dots \text{ m}$ [2]

- (b) The total weight of the balloon, string and block is 0.053 N .

The external force holding the block on the ground is removed so that the released block is lifted vertically upwards by the balloon.

Calculate the acceleration of the block immediately after it is released.

acceleration = $\dots\dots\dots \text{ ms}^{-2}$ [3]

- (c) The balloon continues to lift the block. The string breaks as the block is moving vertically upwards with a speed of 1.4 m s^{-1} . After the string breaks, the detached block briefly continues moving upwards before falling vertically downwards to the ground. The block hits the ground with a speed of 3.6 m s^{-1} .

Assume that the air resistance on the block is negligible.

- (i) By considering the motion of the block after the string breaks, calculate the height of the block above the ground when the string breaks.

height = m [2]

- (ii) The string breaks at time $t = 0$ and the block hits the ground at time $t = T$.

On Fig. 2.2, sketch a graph to show the variation of the velocity v of the block with time t from $t = 0$ to $t = T$.

Numerical values of t are not required. Assume that v is positive in the upward direction.

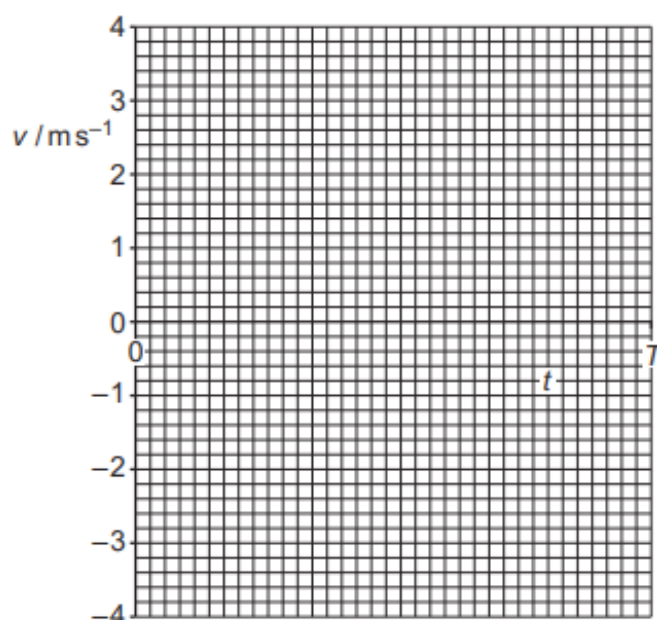


Fig. 2.2

[2]

39. 9702/23/O/N/22/Q2

The engine of a toy rocket pushes gases vertically downwards and this results in the rocket accelerating vertically upwards from the ground.

The rocket starts to move from rest at time $t = 0$. The variation with time t of the vertical velocity v of the rocket for the first 0.30 s of the flight is shown in Fig. 2.1.

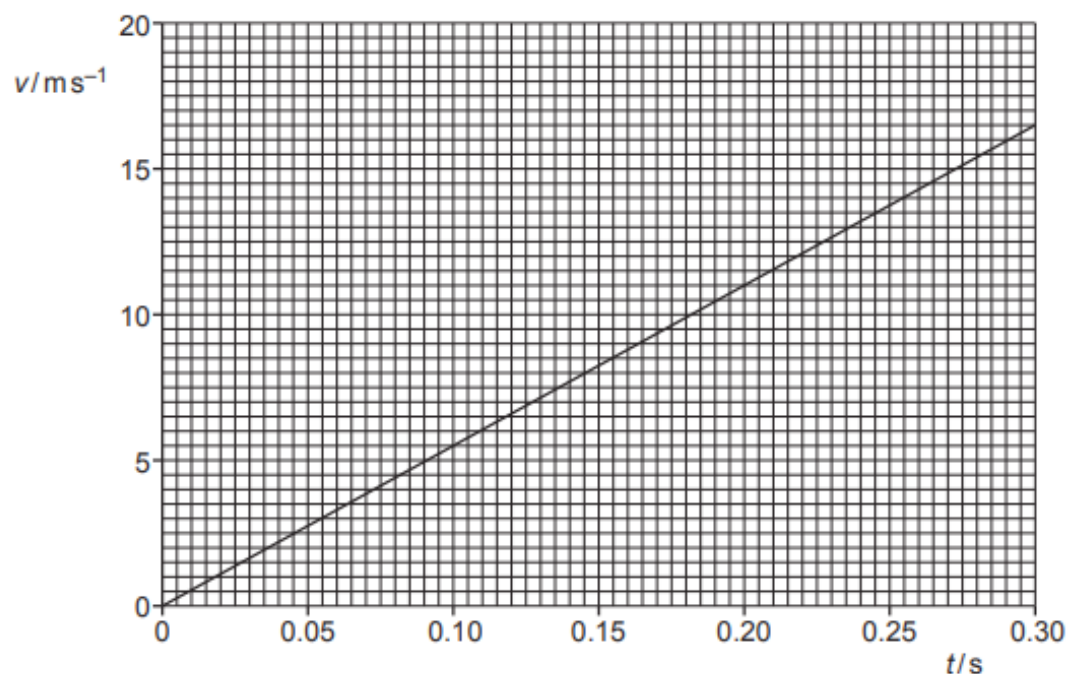


Fig. 2.1

As the rocket moves, the thrust force T provided by the rocket engine is 16 N.
Assume that the mass of the rocket is constant for this part of its flight.

Assume that air resistance is negligible.

(a) For this part of the rocket's flight:

- (i)** show that the acceleration of the rocket is 55 ms^{-2}

[1]

- (ii)** state an expression for the resultant force F experienced by the rocket in terms of the thrust force T and the weight W of the rocket

[1]

(iii) calculate the mass of the rocket.

mass = kg [2]

(b) At time $t = 0.30\text{ s}$, a small piece of metal separates from the rocket.

Calculate:

(i) the height of the rocket above the ground at $t = 0.30\text{ s}$

height = m [2]

(ii) the speed at which the piece of metal strikes the ground.

speed = m s^{-1} [3]

[Total: 9]

40. 9702/22/F/M/23/Q4

Two blocks slide directly towards each other along a frictionless horizontal surface, as shown in Fig. 4.1. The blocks collide and then move as shown in Fig. 4.2.

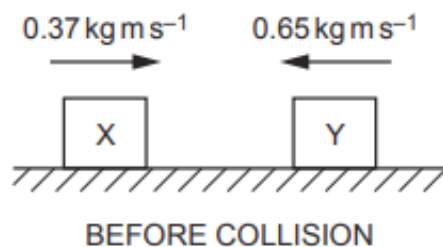


Fig. 4.1

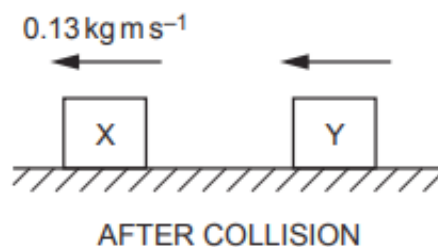


Fig. 4.2

Block X initially moves to the right with a momentum of 0.37 kg m s^{-1} . Block Y initially moves to the left with a momentum of 0.65 kg m s^{-1} . After the blocks collide, block X moves to the left back along its original path with a momentum of 0.13 kg m s^{-1} . Block Y also moves to the left after the collision.

- (a)** Block X has an initial kinetic energy of 0.30 J .

Calculate the mass of block X.

mass = kg [3]

- (b)** Determine the magnitude of the momentum of block Y after the collision.

momentum = kg m s^{-1} [1]

(c) Block X exerts an average force of 7.7 N on block Y during the collision.

Calculate the time that the blocks are in contact with each other.

time = s [2]

[Total: 6]

41. 9702/21/M/J/23/Q3

A block is pulled in a straight line along a rough horizontal surface by a varying force X , as shown in Fig. 3.1.

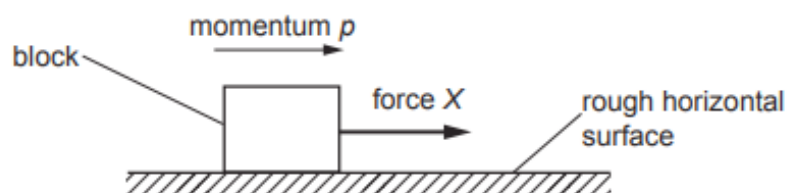


Fig. 3.1

Air resistance is negligible. Assume that the frictional force exerted on the block by the surface is constant and has magnitude 2.0 N .

The variation with time t of the momentum p of the block is shown in Fig. 3.2.

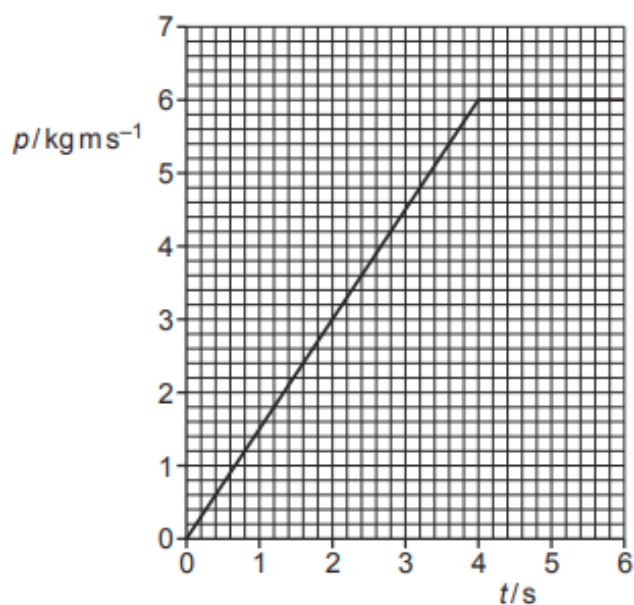


Fig. 3.2

(a) State Newton's second law of motion.

.....
..... [1]

(b) Use Fig. 3.2 to determine, for the block at time $t = 2.0\text{ s}$, the magnitude of:

(i) the resultant force on the block

resultant force = N [1]

(ii) the force X .

$X =$ N [1]

(c) On Fig. 3.3, sketch a graph to show the variation of force X with time t from $t = 0$ to $t = 6.0\text{ s}$.

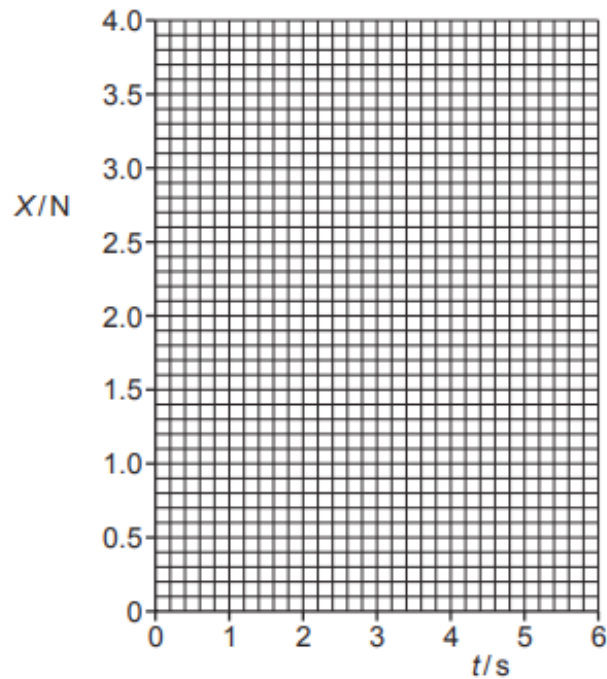


Fig. 3.3

[3]

[Total: 6]

42. 9702/22/M/J/23/Q3

A block is pulled by a force X in a straight line along a rough horizontal surface, as shown in Fig. 3.1.

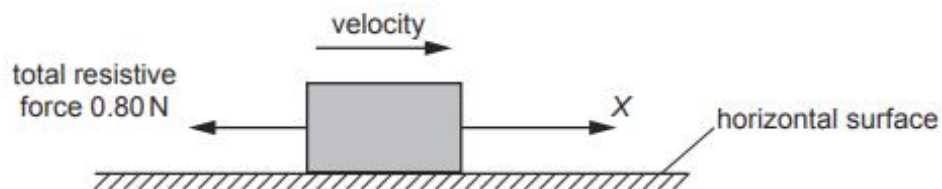


Fig. 3.1

Assume that the total resistive force opposing the motion of the block is 0.80 N at all speeds of the block.

The variation with time t of the magnitude of the force X is shown in Fig. 3.2.

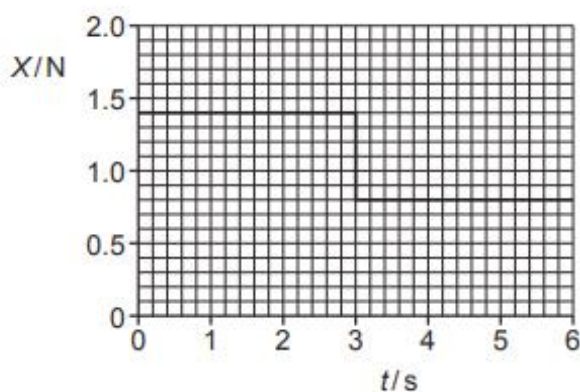


Fig. 3.2

(a) (i) Define force.

.....
 [1]

(ii) Determine the change in momentum of the block from time $t = 0$ to time $t = 3.0$ s.

change in momentum = kg m s^{-1} [2]

- (b) (i) Describe and explain the motion of the block between time $t = 3.0\text{ s}$ and time $t = 6.0\text{ s}$.

.....
.....
.....
..... [2]

- (ii) Force X produces a total power of 2.0 W when moving the block between time $t = 3.0\text{ s}$ and time $t = 6.0\text{ s}$.

Calculate the distance moved by the block during this time interval.

distance = m [3]

- (c) The block is at rest at time $t = 0$.

On Fig. 3.3, sketch a graph to show the variation of the momentum of the block with time t from $t = 0$ to $t = 6.0\text{ s}$.

Numerical values of momentum are not required.

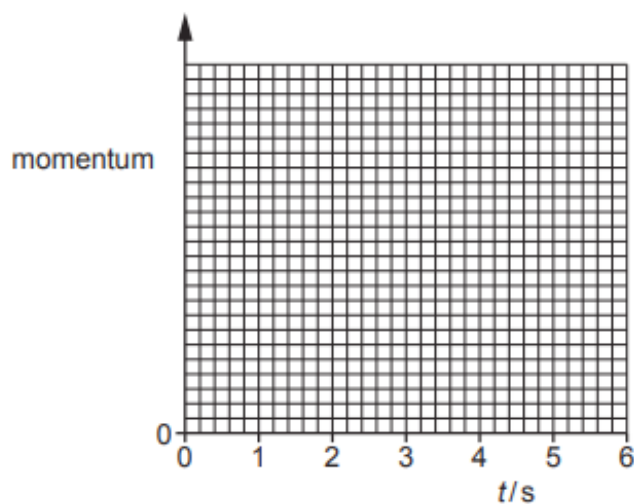


Fig. 3.3

[2]

43. 9702/23/M/J/23/Q2

A sphere floats in equilibrium on the surface of sea water of density 1050 kg m^{-3} , as shown in Fig. 2.1.

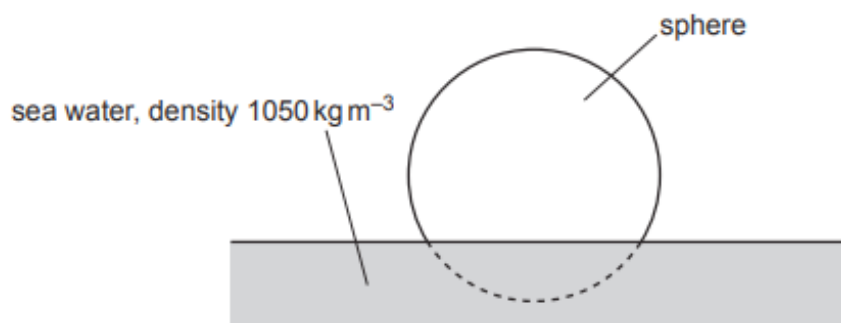


Fig. 2.1

- (a)** 21% of the volume of the sphere is below the surface of the water.

Calculate the density of the sphere.

density = kg m^{-3} [2]

- (b)** The sphere is now held so that its entire volume is below the surface of the water. The sphere is then released.

- (i)** Calculate the initial acceleration of the sphere.

acceleration = ms^{-2} [3]

- (ii) The sphere accelerates upwards but remains entirely below the surface of the water.

State and explain what happens to the acceleration of the sphere as its velocity begins to increase.

.....

.....

.....

..... [3]

[Total: 8]

44. 9702/23/M/J/23/Q3

(a) State the principle of conservation of momentum.

.....

.....

..... [2]

(b) A firework is initially stationary. It explodes into three fragments A, B and C that move in a horizontal plane, as shown in the view from above in Fig. 3.1.

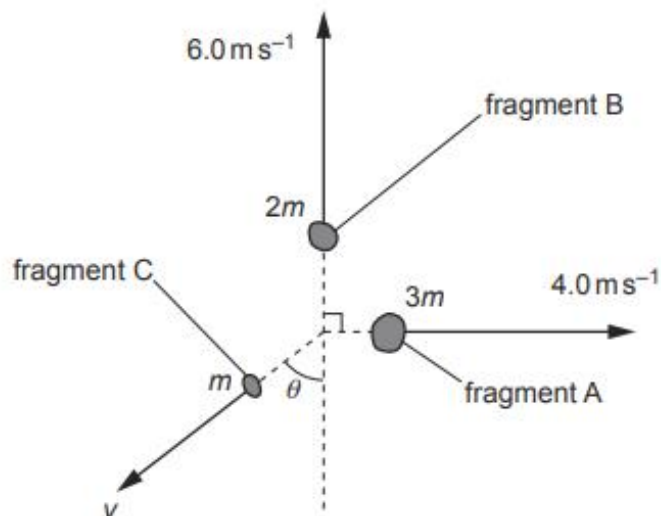


Fig. 3.1

Fragment A has a mass of $3m$ and moves away from the explosion at a speed of 4.0 ms^{-1} .

Fragment B has a mass of $2m$ and moves away from the explosion at a speed of 6.0 ms^{-1} at right angles to the direction of A.

Fragment C has a mass of m and moves away from the explosion at a speed v and at an angle θ as shown in Fig. 3.1.

Calculate:

(i) the angle θ

$\theta = \dots\dots\dots^\circ$ [3]

(ii) the speed v .

$v = \dots\dots\dots \text{ms}^{-1}$ [2]

- (c) The firework in (b) contains a chemical that has mass 5.0 g and has chemical energy per unit mass 700 J kg^{-1} . When the firework explodes, all of the chemical energy is transferred to the kinetic energy of fragments A, B and C.

(i) Show that the total chemical energy in the firework is 3.5 J.

[1]

(ii) Calculate the mass m .

$m = \dots\dots\dots \text{kg}$ [3]

[Total: 11]

45. 9702/21/O/N/23/Q2

A hot-air balloon floats just above the ground. The balloon is stationary and is held in place by a vertical rope, as shown in Fig. 2.1.

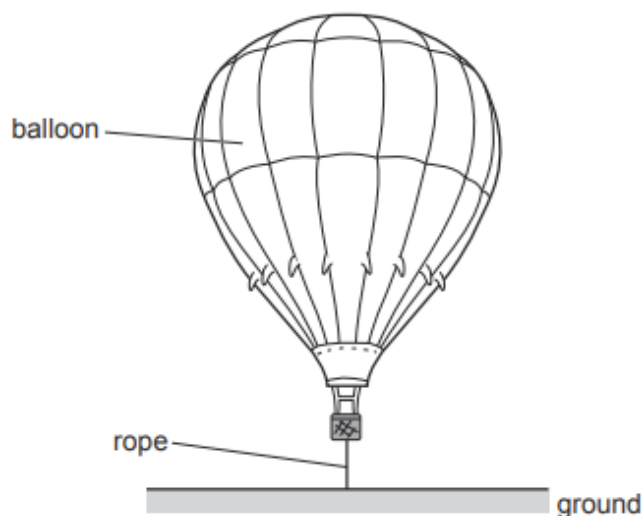


Fig. 2.1

The balloon has a weight W of $3.39 \times 10^4 \text{ N}$. The tension T in the rope is $4.00 \times 10^2 \text{ N}$.

Upthrust U acts on the balloon.

The density of the surrounding air is 1.23 kg m^{-3} .

- (a) (i) On Fig. 2.1, draw labelled arrows to show the directions of the three forces acting on the balloon. [2]
- (ii) Calculate the volume, to three significant figures, of the balloon.

volume = m^3 [3]

- (iii) The balloon is released from the rope.

Calculate the initial acceleration of the balloon.

acceleration = ms^{-2} [3]

- (b) The balloon is stationary at a height of 500 m above the ground. A tennis ball is released from rest and falls vertically from the balloon.

A passenger in the balloon uses the equation $v^2 = u^2 + 2as$ to calculate that the ball will be travelling at a speed of approximately 100 m s^{-1} when it hits the ground.

Explain why the actual speed of the ball will be much lower than 100 m s^{-1} when it hits the ground.

.....

 [3]

- (c) Before the balloon is released, the rope holding the balloon has a strain of 2.4×10^{-5} . The rope has an unstretched length of 2.5 m. The rope obeys Hooke's law.

- (i) Show that the extension of the rope is $6.0 \times 10^{-5} \text{ m}$.

[1]

- (ii) Calculate the elastic potential energy E_p of the rope.

$E_p = \dots\dots\dots \text{ J}$ [2]

- (iii) The rope holding the balloon is replaced with a new one of the same original length and cross-sectional area. The tension is unchanged and the new rope also obeys Hooke's law.

The new rope is made from a material of a lower Young modulus.

State and explain the effect of the lower Young modulus on the elastic potential energy of the rope.

.....

 [2]

46. 9702/21/O/N/23/Q3

A trolley A moves along a horizontal surface at a constant velocity towards another trolley B which is moving at a lower constant speed in the same direction. Fig. 3.1 shows the trolleys at time $t = 0$.



Fig. 3.1

Table 3.1 shows data for the trolleys.

Table 3.1

trolley	mass/kg	initial speed/ m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 3.2 shows the variation with time t of the velocity v for trolley B.

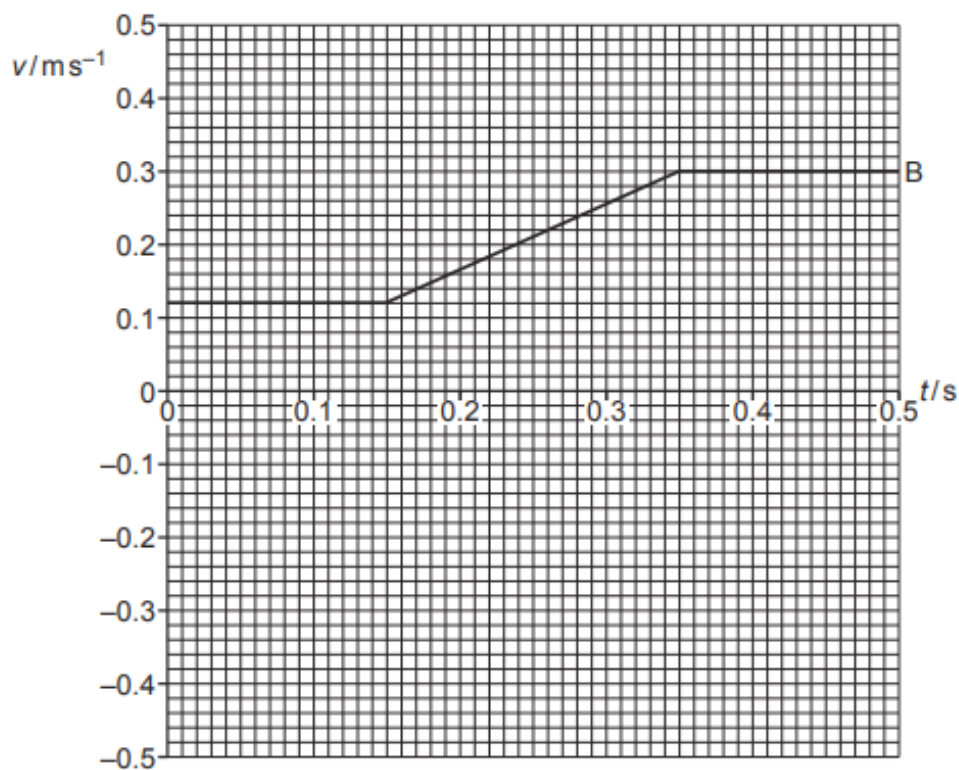


Fig. 3.2

(a) State what is represented by the area under a velocity–time graph.

..... [1]

(b) Use Table 3.1 and Fig. 3.2 to determine:

(i) the acceleration of trolley B during the collision

acceleration of B = ms^{-2} [2]

(ii) the magnitude and direction of the final velocity of trolley A.

magnitude = ms^{-1}

direction [3]

(c) On Fig. 3.2, sketch the variation of the velocity of trolley A with time t from $t = 0$ to $t = 0.50$ s. [3]

[Total: 9]

47. 9702/22/O/N/23/Q2

A high-altitude balloon is stationary in still air. A solid sphere is suspended from the balloon by a string, as shown in Fig. 2.1.

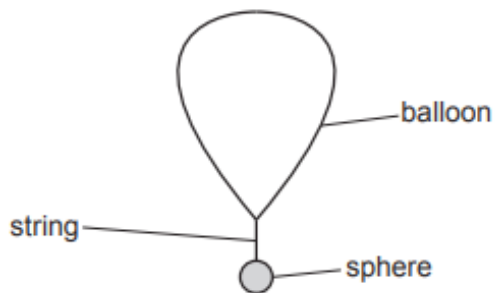


Fig. 2.1 (not to scale)

The volume of the balloon is 7.5 m^3 . The total weight of the balloon, string and sphere is 65 N . The upthrust acting on the string and sphere is negligible.

- (a) Calculate the density of the air surrounding the balloon.

density = kg m^{-3} [2]

- (b)** The string breaks, releasing the sphere.

- (i) State the magnitude of the acceleration of the sphere immediately after the string breaks.

acceleration = ms^{-2} [1]

- (ii) State and explain the variation, if any, in the magnitude of the acceleration of the sphere when it is moving downwards **before** it reaches terminal (constant) velocity.

..... [3]

(c) The sphere has a mass of 4.0 kg.

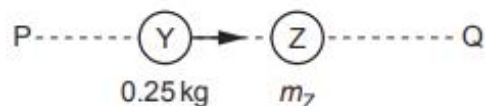
Calculate the total resistive force acting on the sphere at the instant when its acceleration is 1.9ms^{-2} .

resistive force = N [2]

[Total: 8]

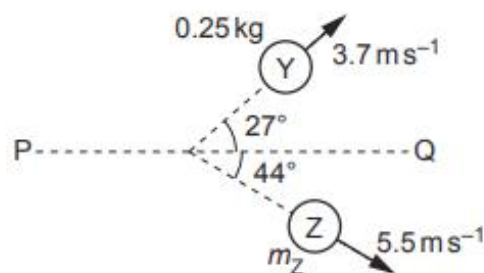
48. 9702/22/O/N/23/Q4

- (a) A ball Y moves along a horizontal frictionless surface and collides with a ball Z, as illustrated in the views from above in Fig. 4.1 and Fig. 4.2.



BEFORE COLLISION

Fig. 4.1 (not to scale)



AFTER COLLISION

Fig. 4.2 (not to scale)

Ball Y has a mass of 0.25 kg and initially moves along a line PQ.
Ball Z has a mass m_Z and is initially stationary.

After the collision, ball Y has a final velocity of 3.7 ms^{-1} at an angle of 27° to line PQ and ball Z has a final velocity of 5.5 ms^{-1} at an angle of 44° to line PQ.

- (i) Calculate the component of the final momentum of ball Y in the direction perpendicular to line PQ.

component of momentum = N s [2]

- (ii) By considering the component of the final momentum of each ball in the direction perpendicular to line PQ, calculate m_Z .

$m_Z = \dots \text{ kg}$ [1]

- (iii) During the collision, the average force exerted on Y by Z is F_Y and the average force exerted on Z by Y is F_Z .

Compare the magnitudes and directions of F_Y and F_Z . Numerical values are not required.

magnitudes:

directions:

[2]

- (b) Two blocks, A and B, move directly towards each other along a horizontal frictionless surface, as shown in the view from above in Fig. 4.3.



Fig. 4.3

The blocks collide perfectly elastically. Before the collision, block A has a speed of 4 m s^{-1} and block B has a speed of 6 m s^{-1} . After the collision, block B moves back along its original path with a speed of 2 m s^{-1} .

Calculate the speed of block A after the collision.

speed = m s^{-1} [1]

[Total: 6]

49. 9702/23/O/N/23/Q3

- (a)** State the conditions for a system to be in equilibrium.

.....

.....

..... [2]

- (b)** Fig. 3.1 shows an airship in flight. The airship is propelled by identical fans that can be angled to control the motion of the airship.

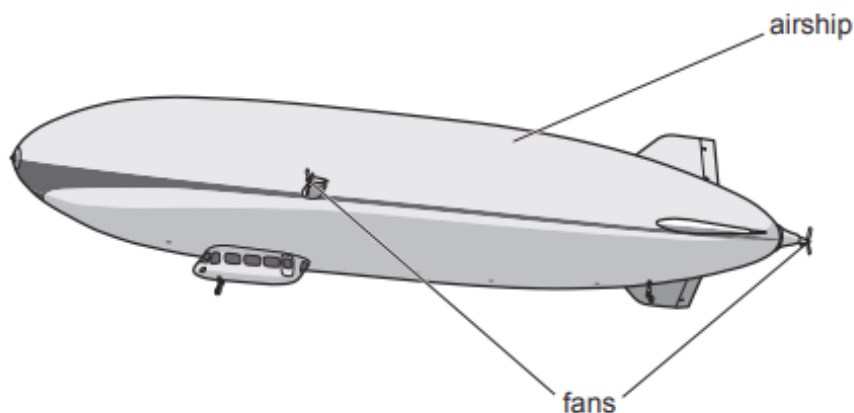


Fig. 3.1

The upthrust on the airship is 93 000 N.
The density of the surrounding air is 1.2 kg m^{-3} .

- (i)** Calculate the volume of air displaced by the airship.

volume = m^3 [1]

- (ii)** When fully loaded, the weight of the airship is greater than the upthrust.
To maintain horizontal flight, the fans provide a total vertical force of $3.0 \times 10^3 \text{ N}$ upwards on the airship.

Calculate the mass of the airship.

mass = kg [2]

- (c) At a certain time, the airship in (b) is stationary. The thrust force exerted by a fan on the airship is 2800 N.

To produce this force, a mass of 64 kg of air is propelled through the blades of the fan in a time of 0.50 s. Assume that this air is initially stationary at the entrance to the fan.

Calculate:

- (i) the change in momentum Δp of the air propelled through the fan blades in this time

$$\Delta p = \dots\dots\dots \text{ kg m s}^{-1} \quad [2]$$

- (ii) the speed of the air as it leaves the fan

$$\text{speed} = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (iii) the total kinetic energy of this air due to its movement through the fan.

$$\text{kinetic energy} = \dots\dots\dots \text{ J} \quad [2]$$

[Total: 11]

ANSWERS

- 1 (a) 2.4 s A1 [1]
 (b) in (b) and (c), allow answers as (+) or (–)
 recognises distance travelled as area under graph line C1
 $\text{height} = (\frac{1}{2} \times 2.4 \times 9.0) - (\frac{1}{2} \times 1.6 \times 6.0)$ C1
 $= 6.0 \text{ m (allow 6 m)}$ A1 [3]
 (answer 15.6 scores 2 marks
 answer 10.8 or 4.8 scores 1 mark)
 alternative solution: $s = ut - \frac{1}{2}at^2$
 $= (9 \times 4) - \frac{1}{2} \times (9 / 2.4) \times 4^2$
 $= 6.0 \text{ m}$
 (answer 66 scores 2 marks
 answer 36 or 30 scores 1 mark)
 (c) (i) change in momentum = $0.78 (9.0 + 4.2)$ (allow 4.2 ± 0.2) C1
 $= 10.3 \text{ N s}$ (allow 10 N s) A1 [2]
 (ii) force = $\Delta p / \Delta t$ or $m\Delta v / \Delta t$ C1
 $= 10.3 / 3.5 / 0.08$
 $= 2.9 \text{ N}$ A1 [2]
 (d) (i) 2.9 N A1 [1]
 (ii) $g = \text{weight} / \text{mass}$ C1
 $= 2.9 / 0.78$
 $= 3.7 \text{ m s}^{-2}$ A1 [2]

- 2 (a) (i) either sum / total momentum (of system of bodies) is constant
 or total momentum before = total momentum after M1
 for an isolated system / no (external) force acts on system A1 [2]
 (ii) zero momentum before / after decay M1
 so α -particle and nucleus D must have momenta in opposite directions A1 [2]
 (b) (i) kinetic energy = $\frac{1}{2}mv^2$ C1
 $1.0 \times 10^{-12} = \frac{1}{2} \times 4 \times \underline{1.66} \times 10^{-27} \times v^2$ M1
 $v = 1.7 \times 10^7 \text{ m s}^{-1}$ A0 [2]
 (ii) $1.7 \times 10^7 \times 4u = 216u \times V$ C1
 $V = 3.1 \times 10^5 \text{ m s}^{-1}$ A1 [2]
 (accept $3.2 \times 10^5 \text{ m s}^{-1}$, do not accept 220 rather than 216)
 (c) $(1.7 \times 10^7)^2 = 2 \times \text{deceleration} \times 4.5 \times 10^{-2}$ C1
 $\text{deceleration} / a = 3.2 \times 10^{15} \text{ m s}^{-2}$ A1 [2]
 (accept calculation based on calculating $F = 2.22 \times 10^{11} \text{ N}$
 and then use of $F = ma$)

3	(a)	(i) force is rate of change of momentum	B1	[1]	
		(ii) force on body A is equal in magnitude to force on body B (from A)	M1		
		forces are in opposite directions	A1		
		forces are of the same kind	A1	[3]	
	(b)	(i) 1 $F_A = -F_B$	B1	[1]	
		2 $t_A = t_B$	B1	[1]	
		(ii) $\Delta p = F_A t_A = -F_B t_B$	B1	[1]	
	(c)	graph: momentum change occurs at same times for both spheres	B1		
		final momentum of sphere B is to the right	M1		
		and of magnitude 5 N s	A1	[3]	
4	(a)	force = rate of change of momentum (allow symbols if defined)	B1	[1]	
	(b)	(i) $\Delta p = 140 \times 10^{-3} \times (5.5 + 4.0)$ $= 1.33 \text{ kg m s}^{-1}$	C1		
			A1	[2]	
		(ii) force = $1.33 / 0.04$ $= 33.3 \text{ N}$	M1		
		A0	[1]		
	(c)	(i) taking moments about B $(33 \times 75) + (0.45 \times g \times 25) = F_A \times 20$ $F_A = 129 \text{ N}$	C1		
			C1		
			A1	[3]	
		(ii) $F_B = 33 + 129 + 0.45g$ $= 166 \text{ N}$	C1		
		A1	[2]		
5	(a)	(resultant) force = rate of change of momentum / allow proportional to or change in momentum / time (taken)	B1	[1]	
			C1		
	(b)	(i) $\Delta p = (-) 65 \times 10^{-3} (5.2 + 3.7)$ $= (-) 0.58 \text{ N s}$	A1	[2]	
		(ii) $F = 0.58 / 7.5 \times 10^{-3}$ $= 77(.3) \text{ N}$	A1	[1]	
	(c)	(i) 1. force on the wall from the ball is equal to the force on ball from the wall but in the opposite direction (statement of Newton's third law can score one mark)	M1		
			A1	[2]	
		2. momentum change of ball is equal and opposite to momentum change of the wall / change of momentum of ball and wall is zero	B1	[1]	
		(ii) kinetic energy (of ball and wall) is reduced / not conserved so inelastic (Allow relative speed of approach does not equal relative speed of separation.)	B1	[1]	
6	(a)	force = rate of change of momentum	A1	[1]	
	(b)	(i) horizontal line on graph from $t = 0$ to t about $2.0 \text{ s} \pm \frac{1}{2}$ square, $a > 0$	M1		
			horizontal line at 3.5 on graph from 0 to 2 s	A1	
			vertical line at $t = 2.0 \text{ s}$ to $a = 0$ or sharp step without a line	B1	
			horizontal line from $t = 2 \text{ s}$ to $t = 4 \text{ s}$ with $a = 0$	B1	[4]
	(ii)	straight line and positive gradient	M1		
			starting at (0,0)	A1	
			finishing at (2,16.8)	A1	
			horizontal line from 16.8	M1	
			from 2.0 to 4.0	A1	[5]

7	(i)	the total momentum of a system (of interacting bodies) remains constant provided there are no resultant external forces / isolated system	M1 A1 [2]
	(ii)	elastic: total kinetic energy is conserved, inelastic: loss of kinetic energy [allow elastic: relative speed of approach equals relative speed of separation]	B1 [1]
	(b) (i)	initial mom: $4.2 \times 3.6 - 1.2 \times 1.5$ ($= 15.12 - 1.8 = 13.3$)	C1
		final mom: $4.2 \times v + 1.5 \times 3$	C1
		$v = (13.3 - 4.5) / 4.2 = 2.1 \text{ ms}^{-1}$	A1 [3]
	(ii)	initial kinetic energy $= \frac{1}{2} m_A(v_A)^2 + \frac{1}{2} m_B(v_B)^2$ $= 27.21 + 1.08 = 28(.28)$	M1
		final kinetic energy $= 9.26 + 6.75 = 16$	M1
		initial KE is not the same as final KE hence inelastic <i>provided final KE less than initial KE</i>	A1 [3]
		[allow in terms of relative speeds of approach and separation]	
8	(a)	a body/mass/object continues (at rest or) at constant/uniform velocity unless acted on by a <u>resultant</u> force	B1 [1]
	(b) (i)	weight <u>vertically</u> down normal/reaction/contact (force) perpendicular/normal <u>to the slope</u>	B1 [2]
	(ii) 1.	acceleration = gradient or $(v - u)/t$ or $\Delta v/t$	C1
		$= (6.0 - 0.8)/(2.0 - 0.0) = 2.6 \text{ ms}^{-2}$	M1 [2]
	2.	$F = ma$ $= 65 \times 2.6$ $= 169 \text{ N}$ (allow to 2 or 3 s.f.)	A1 [1]
	3.	weight component seen: $mg \sin \theta$ (218 N) $218 - R = 169$	C1
		$R = 49 \text{ N}$ (require 2 s.f.)	C1
			A1 [3]
9	(a)	for a system (of interacting bodies) the <u>total</u> momentum remains constant provided there is no <u>resultant</u> force acting (on the system)	M1 A1 [2]
	(b) (i)	total momentum $= m_1 v_1 + m_2 v_2$ $= 0.4 \times 0.65 + 0.6 \times 0.45$ $= 0.26 + 0.27 = 0.53 \text{ N s}$	C1 C1 A1 [3]
	(ii)	$0.53 = 0.4 \times 0.41 + 0.6 \times v$ $v = 0.366 / 0.6 = 0.61 \text{ ms}^{-1}$	C1 A1 [2]
	(iii)	$KE = \frac{1}{2} m v^2$ total initial KE $= \frac{1}{2} \times 0.4 \times (0.65)^2 + \frac{1}{2} \times 0.6 \times (0.45)^2$ $= 0.0845 + 0.06075 = 0.15(0.145) \text{ J}$	C1 C1 A1 [3]
	(c)	check relative speed of approach equals relative speed of separation or: total final kinetic energy equals the total initial kinetic energy	B1 [1]
	(d)	the forces on the two bodies (or on X and Y) are equal and opposite time same for both forces <u>and</u> force is change in momentum/time	B1 B1 [2]

10 (a) $4.5 \times 50 - 2.8 \times M (= \dots)$	C1	
$(\dots) = -1.8 \times 50 + 1.4 \times M$	C1	
$(M =) 75 \text{ g}$	A1	[3]
(b) <u>total</u> initial kinetic energy/KE not equal to the total final kinetic energy/KE or <u>relative</u> speed of approach is not equal to relative speed of separation so not elastic or is inelastic	B1	[1]
(c) force on X is equal and opposite to force on Y (Newton III)	M1	
force equals/is proportional to rate of change of momentum (Newton II)	M1	
time of collision same for both balls hence change in momentum is the same	A1	[3]
11 (a) $(p =) mv$	C1	
$\Delta p (= -6.64 \times 10^{-27} \times 1250 - 6.64 \times 10^{-27} \times 1250) = 1.66 \times 10^{-23} \text{ N s}$	A1	[2]
(b) (i) molecule collides with wall/container and there is a change in momentum change in momentum / time is force or $\Delta p = Ft$	B1	
<u>many/all/sum of</u> molecular collisions over surface/area of container produces pressure	B1	[3]
(ii) more collisions per unit time so greater pressure	B1	[1]
12 (a) (i) force $(= mg = 0.15 \times 9.81) = 1.5 \text{ (1.47) N}$	A1	[1]
(ii) resultant force (on ball) is zero so normal contact force = weight or the forces are in opposite directions so normal contact force = weight or normal contact force up = weight down	A1	[1]
(b) (i) (resultant) force proportional/equal to rate of change of momentum	B1	[1]
(ii) change in momentum $= 0.15 \times (6.2 + 2.5) (= 1.305 \text{ N s})$	C1	
magnitude of force $= 1.305 / 0.12$ $= 11 \text{ (10.9) N}$	A1	
or		
(average) acceleration $= (6.2 + 2.5) / 0.12 (= 72.5 \text{ m s}^{-2})$	(C1)	
magnitude of force $= 0.15 \times 72.5$ $= 11 \text{ (10.9) N}$	(A1)	
(direction of force is) upwards/up	B1	[3]
(iii) there is a change/gain in momentum of the floor	M1	
this is equal (and opposite) to the change/loss in momentum of the ball so momentum is conserved	A1	[2]
or		
change of (total) momentum of <u>ball and floor</u> is zero so momentum is conserved	(M1) (A1)	
or		
(total) momentum of <u>ball and floor</u> before is equal to the (total) momentum of <u>ball and floor</u> after so momentum is conserved	(M1) (A1)	

13	(a) the <u>total</u> momentum of a system (of colliding particles) remains constant provided there is no resultant external force acting on the system/isolated or closed system	M1 A1 [2]
(b) (i)	the <u>total</u> kinetic energy before (the collision) is equal to the total kinetic energy after (the collision)	B1 [1]
(ii)	$p (= mv = 1.67 \times 10^{-27} \times 500) = 8.4 (8.35) \times 10^{-25} \text{ N s}$	A1 [1]
(iii)	1. $mv_A \cos 60^\circ + mv_B \cos 30^\circ$ or $m(v_A^2 + v_B^2)^{1/2}$	B1
	2. $mv_A \sin 60^\circ + mv_B \sin 30^\circ$	B1 [2]
(iv)	8.35×10^{-25} or $500m = mv_A \cos 60^\circ + mv_B \cos 30^\circ$ and $0 = mv_A \sin 60^\circ + mv_B \sin 30^\circ$ or using a vector triangle	C1
	$v_A = 250 \text{ m s}^{-1}$	A1
	$v_B = 430 (433) \text{ m s}^{-1}$	A1 [3]
14	(i) $a = (v - u) / t$ or gradient or $\Delta v / (\Delta) t$ e.g. $a = (8.8 - 4.6) / (7.0 - 4.0) = 1.4 \text{ m s}^{-2}$	C1 A1
(ii)	$s = 4.6 \times 4 + [(8.8 + 4.6) / 2] \times 3$ $= 18.4 + 20.1$ $= 39 (38.5) \text{ m}$	C1 A1
(iii)	$\Delta E = \frac{1}{2} \times 95 [(8.8)^2 - (4.6)^2]$ $= 3678 - 1005$ $= 2700 (2673) \text{ J}$	C1 A1
(iv) 1	weight = $95 \times 9.81 (= 932 \text{ N})$ vertical tension force = $280 \sin 25^\circ$ or $280 \cos 65^\circ (= 118.3 \text{ N})$ $F = 932 + 118$ $= 1100 (1050) \text{ N}$	C1 C1 A1
(iv) 2	horizontal tension force = $280 \cos 25^\circ$ or $280 \sin 65^\circ (= 253.8 \text{ N})$ resultant force = $95 \times 1.4 (= 133 \text{ N})$ $133 = 253.8 - R$ $R = 120 (120.8) \text{ N}$	C1 C1 A1

15	a body/mass/object continues (at rest or) at constant/uniform velocity unless acted on by a resultant force	
(b) (i)	initial momentum = final momentum $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ $0.60 \times 100 - 0.80 \times 200 = -0.40 \times 100 + v \times 200$ $v = (-) 0.3(0) \text{ m s}^{-1}$	C1 A1
(ii)	<u>kinetic</u> energy is not conserved/is lost (but) <u>total</u> energy is conserved/constant or some of the (initial) <u>kinetic</u> energy is transformed into other forms of energy	B1 C1
16	(a) (i) $K\rho = 1.5 / 33$ $= 1.38 \times 10^{-3}$ $F_D = 1.38 \times 10^{-3} \times 25^2$ or $F_D / 1.5 = 25^2 / 33^2$ $F_D = 0.86 \text{ N}$ (ii) $a = (1.5 - 0.86) / (1.5 / 9.81)$ or $a = 9.81 - [0.86 / (1.5 / 9.81)]$ $a = 4.2 \text{ ms}^{-2}$ (b) initial acceleration is $g/9.81 \text{ (ms}^{-2}\text{)}/\text{acceleration of free fall}$ acceleration decreases final acceleration is zero	A1 C1 A1 B1 B1 B1
17	(a) <u>sum/total</u> momentum (of system of bodies) is constant or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after for an isolated system/no (resultant) <u>external</u> force (b) (i) $p = mv$ $(4.0 \times 6.0 \times \sin \theta) - (12 \times 3.5 \times \sin 30^\circ) = 0$ or $(m_A v_A \times \sin \theta) - (m_B v_B \times \sin 30^\circ) = 0$ $\theta = 61^\circ$ (ii) shows the horizontal <u>momentum</u> component of ball A or of ball B as $(4.0 \times 6.0 \times \cos \theta)$ or $(12 \times 3.5 \times \cos 30^\circ)$ $(4.0 \times 6.0 \times \cos 61^\circ) + (12 \times 3.5 \times \cos 30^\circ) = 4.0v$ so $v = 12 \text{ (ms}^{-1}\text{)}$ (iii) initial $E_K (= \frac{1}{2} \times 4.0 \times 12^2) = 290 \text{ (288) (J)}$ final $E_K (= \frac{1}{2} \times 4.0 \times 6.0^2 + \frac{1}{2} \times 12 \times 3.5^2) = 150 \text{ (145.5) (J)}$ (initial $E > \text{final } E$) so inelastic [both M1 marks required to award this mark]	M1 A1 C1 M1 A1 C1 A1 M1 M1 A1

18 (a) mass is the property (of a body/object) resisting changes in motion or mass is the quantity of matter (in a body)	B1
(b) (i) force on A (by B) equal <u>and opposite</u> to force on B (by A) or both A and B exert equal and opposite forces on each other	B1
force is rate of change of momentum <u>and</u> time (of contact) is same	B1
(ii) $p = mv$ or $3M \times 0.40$ or $M \times 0.25$ or $3M \times 0.2$ or Mv	C1
$(3M \times 0.40) - (M \times 0.25) = (3M \times 0.2) + Mv$	C1
$v = (3 \times 0.40) - 0.25 - (3 \times 0.2)$	A1
$= 0.35 \text{ ms}^{-1}$	
(iii) 1. relative speed of approach $= 0.40 + 0.25$	A1
$= 0.65 \text{ ms}^{-1}$	
2. relative speed of separation $= 0.35 - 0.20$	A1
$= 0.15 \text{ ms}^{-1}$	
(iv) (relative) speed of separation not equal to/less than (relative) speed of approach or answers (to (b)(iii)) are not equal and so inelastic collision	B1
19 (a) <u>sum/total</u> momentum (of a system of bodies) is constant	M1
or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after	
for an isolated system or no (resultant) external force	A1
(b) (i) $(p =) mv$ or $(3.0M \times 7.0)$ or $(2.0M \times 6.0)$ or $(1.5M \times 8.0)$	C1
$3.0M \times 7.0 = 2.0M \times 6.0 \sin \theta + 1.5M \times 8.0 \sin \theta$	C1
$\theta = 61^\circ$	A1
or (vector triangle method)	
momentum vector triangle drawn	(C1)
$\theta = 61^\circ$ (2 marks for $\pm 1^\circ$, 1 mark for $\pm 2^\circ$)	(A2)
or (use of cosine rule)	
$p = mv$ or $(3.0M \times 7.0)$ or $(2.0M \times 6.0)$ or $(1.5M \times 8.0)$	(C1)
$(21M)^2 = (12M)^2 + (12M)^2 - (2 \times 12M \times 12M \times \cos 2\theta)$	(C1)
$\theta = 61^\circ$	(A1)
(ii) $(E =) \frac{1}{2}mv^2$	C1
ratio $= (\frac{1}{2} \times 2.0M \times 6.0^2) / (\frac{1}{2} \times 1.5M \times 8.0^2)$	A1
$= 0.75$	

20 (a)	$(p =) mv$ or 4.0×45 or 2.0×85 or $89v$	C1
	$(4.0 \times 45) - (2.0 \times 85) = 89v$	A1
	$v = 0.11 \text{ ms}^{-1}$	
(b) (i)	1. speed of approach = 47 ms^{-1} and 2. speed of separation = 0	A1
(ii)	speed of separation less than/not equal to speed of approach and so inelastic collision	A1
(c)	force is equal to rate of change of momentum	B1
	force on ball (by block) <u>equal and opposite</u> to force on block (by ball) so rates of change of momentum are <u>equal and opposite</u>	B1
	or force on ball (by block) <u>equal and opposite</u> to force on block (by ball)	(B1)
	force is equal to rate of change of momentum so rates of change of momentum are <u>equal and opposite</u>	(B1)
21 (a)	<u>sum/total</u> momentum (of a system of bodies) is constant or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after for an isolated system or no (resultant) external force	M1
(b) (i)	$m = \rho V$	C1
	$= 1.3 \times \pi \times 0.045^2 \times 1.8 \times 2.0 = 0.030 \text{ (kg)}$	A1
(ii)	1. $(\Delta)p = (\Delta)mv$	C1
	$= 0.030 \times 1.8$	A1
	$= 0.054 \text{ N s}$	
	2. $F = 0.054/2.0$ or $0.030 \times 1.8/2.0$	A1
	$= 0.027 \text{ N}$	
(iii)	force on air (by propeller) equal to force on propeller (by air) and opposite (in direction)	M1 A1
(iv)	resultant force = 0.20×0.075 (= 0.015 N) frictional force = $0.027 - 0.015$ $= 0.012 \text{ N}$	C1 A1

22 (a)	(resultant) force proportional/equal to rate of change of momentum	B1
(b)(i)	$\rho = m/V$	C1
	$V = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 (= 4.59 \times 10^{-4} \text{ m}^3)$	A1
	$m = \pi \times (7.5 \times 10^{-3})^2 \times 13 \times 0.2 \times 1000 = 0.46 \text{ kg}$	
(ii)	1. $(\Delta)p = (\Delta m)v$	C1
	$(\Delta)p = 0.46 \times 13$	A1
	$= 6.0 \text{ N s}$	
	2. $F = 6.0/0.20$	A1
	$= 30 \text{ N}$	
(iii)	force on water (by rocket/nozzle) equal to force on rocket/nozzle (by water) in the opposite direction	M1 A1
(iv)	1. mass = $0.40 + 0.70 - 0.46 = 0.64 \text{ kg}$	A1
	2. acceleration = $[30 - (0.64 \times 9.81)]/0.64$ or $30/0.64 - 9.81$	C1
	$= 37 \text{ m s}^{-2}$	A1

23 (a)	(momentum =) mass $\times$ velocity	B1
(b)(i)	time = 40 ms	A1
(ii)	1. (the magnitude of the acceleration is) constant	B1
	2. (the magnitude of the acceleration is) zero	B1
(c)	$F = \Delta p / (\Delta)t$ or $F = \text{gradient}$	C1
	e.g. $F = 0.50 / 40 \times 10^{-3}$	A1
	$= 13 \text{ N}$	
(d)	horizontal line from (0, 0.40) to (60, 0.40)	B1
	straight line from (60, 0.40) to (100, -0.10)	B1
	horizontal line from (100, -0.10) to (160, -0.10)	B1

24 (a)	(resultant) force proportional to rate of change of momentum	B1
(b)(i)	arrow drawn vertically downwards from point X	B1
(ii)	$s = ut + \frac{1}{2}at^2$	C1
	$h = \frac{1}{2} \times 9.81 \times 0.81^2$	
	$= 3.2 \text{ m}$	A1
(iii)	$d = 5.4 \times 0.81$	A1
	$= 4.4 \text{ m}$	
(c)(i)	downward pointing arrow labelled weight	B1
	upward pointing arrow labelled air resistance	B1
(ii)	air resistance increases	B1
	weight constant or <u>resultant</u> force decreases	B1
	(so) acceleration decreases	B1
(iii)	gravitational potential energy to thermal/internal energy	B1

- 25 (a) (i)** area = $ut + \frac{1}{2}(v - u)t$ **A1**
or
 area = $vt - \frac{1}{2}(v - u)t$
or
 area = $\frac{1}{2}(u + v)t$
- (ii)** displacement **A1**
- (b) (i)** $u = 15 \sin 60^\circ (= 13 \text{ m s}^{-1})$ **C1**
 $t = 15 \sin 60^\circ / 9.81$ **C1**
 $= 1.3 \text{ s}$ **A1**
- (ii)** the force in the horizontal direction is zero **B1**
- (iii)** velocity = $15 \cos 60^\circ = 7.5 \text{ (m s}^{-1}\text{)}$ **A1**
or
 (velocity =) $15 \sin 30^\circ = 7.5 \text{ (m s}^{-1}\text{)}$
- (c) (i)** $p = mv$ **or** 0.40×7.5 **or** 0.40×4.3 **C1**
 $\Delta p = 0.40 (7.5 + 4.3)$ **A1**
 $= 4.7 \text{ kg m s}^{-1}$
- (ii)** force = $4.7 / 0.12$ **or** $0.40 \times [(7.5 + 4.3) / 0.12]$ **A1**
 $= 39 \text{ N}$
-

- 26 (a)** a body continues at (rest or) constant velocity unless acted upon by a resultant force **B1**
- (b) (i)** distance = $[\frac{1}{2} \times (2.0 + 4.4) \times 3.0] + [4.4 \times 2.0]$ **C1**
 $= 9.6 + 8.8$ **A1**
 $= 18 \text{ m}$
- (ii)** $a = (v - u) / t$ **or** gradient **or** $\Delta v / (\Delta)t$ **C1**
 e.g. $a = (4.4 - 2.0) / 3.0 = 0.80 \text{ m s}^{-2}$ **A1**
- (iii) 1.** force = $240 \cos 28^\circ$ **or** $240 \sin 62^\circ$ **A1**
 $= 210 \text{ N}$
- 2.** resultant force = $89 \times 0.80 (= 71.2 \text{ N})$ **C1**
 $R = 210 - 71$ **A1**
 $= 140 \text{ N}$
- (iv)** $T \sin 45^\circ = mg$ **C1**
 $T = (89 \times 9.81) / \sin 45^\circ$ **A1**
 $= 1200 \text{ N}$
-

27.

(a)	(force =) rate of change of momentum
(b)(i)	$E = \frac{1}{2}mv^2$ or $\frac{1}{2} \times 0.062 \times 3.8^2$ or $\frac{1}{2} \times 0.062 \times 1.7^2$ loss of KE = $\frac{1}{2} \times 0.062 \times (3.8^2 - 1.7^2)$ = 0.36 J
(b)(ii)	$p = mv$ or 0.062×3.8 or 0.062×1.7 change in momentum = $0.062 \times (1.7 + 3.8)$ = 0.34 N s
b)(iii)	(average resultant force =) $0.34 / 0.081 = 4.2$ (N) or (average resultant force =) $0.062 \times (1.7 + 3.8) / 0.081 = 4.2$ (N)
b)(iv)	1. average force = $4.2 + (0.062 \times 9.81)$ = 4.8 N 2. average force = 4.8 N

28.

(a)(i)	$(\Delta)p = \rho g(\Delta)h$ $520 = 1000 \times 9.81 \times h$ $h = 0.053$ m
(a)(ii)	(upthrust =) $(\Delta)p \times A$ $= (\Delta)p \times \pi(d/2)^2$ or $(\Delta)p \times \pi r^2$ $= 520 \times \pi(0.031/2)^2 = 0.39$ (N)
(a)(iii)	$T = 0.84 - 0.39$ $= 0.45$ N
(b)(i)	$a = (v - u) / t$ or $(\Delta)v / (\Delta)t$ or gradient $= \text{e.g. } 8.0 \times 10^{-2} / 2.0$ $= 4.0 \times 10^{-2} \text{ ms}^{-2}$
(b)(ii)	distance = $(\frac{1}{2} \times 2.5 \times 0.10) + (\frac{1}{2} \times 1.5 \times 0.10)$ or $(\frac{1}{2} \times 4.0 \times 0.10)$ (= 0.20 (m)) depth = $0.32 - 0.20$ = 0.12 m
(c)(i)	viscous (force)
(c)(ii)	viscous force increases (with speed/time/depth) (so) acceleration decreases

29.

(a)	acceleration: vector electrical resistance: scalar momentum: vector <i>1 mark for two correct, 2 marks for all three correct</i>
(b)	resultant force (in any direction) is zero resultant torque/moment (about any point) is zero
(c)(i)	upthrust = $\rho g(\Delta)h \times A$ $= (1.00 \times 10^3 \times 9.81 \times 0.190) \times 0.0230$ $= 42.9 \text{ N}$
(c)(ii)	$(T =) 43 - 28 = 15 \text{ (N)}$ or $(T =) 42.9 - 28 = 14.9 \text{ or } 15 \text{ (N)}$
(c)(iii)	$\sigma = F/A \text{ or } T/A$ $= 15 / (3.2 \times 10^{-6})$ $= 4.7 \times 10^6 \text{ Pa}$
(c)(iv)	upthrust (on cylinder) increases (and weight constant) tension/stress increases and (so) strain energy increases

30.

(a)	$v^2 = u^2 + 2as$ $u^2 = 8.7^2 - (2 \times 9.81 \times 1.5)$ $u = 6.8 \text{ m s}^{-1}$
(b)	(magnitude of) force on ball (by ground) equal to force on ground (by ball) (direction of) force on ball (by ground) opposite to force on ground (by ball)
(c)(i)	$(p =) 0.059 \times 8.7 \text{ or } 0.059 \times 5.4$ change in momentum = $0.059 (8.7 + 5.4)$ $= 0.83 \text{ N s}$
(c)(ii)	resultant force = $0.83 / 0.091 \text{ or } 0.059 [(8.7 + 5.4) / 0.091]$ $= 9.1 \text{ N}$
(c)(iii)	$(W =) 0.059 \times 9.81$ $(W =) 0.58 \text{ (N)}$ force = $9.1 + 0.58$ $= 9.7 \text{ N}$

(d)	straight line with a positive gradient and starting from a non-zero value of speed at $t = 0$ and ending when $t = T$
(e)	air resistance increases
	resultant force/acceleration decreases so gradient (of curve) decreases

31.

(a)	mass $\times$ velocity
(b)(i)	kinetic energy $= \frac{1}{2}mv^2$
	$= \frac{1}{2} \times 0.24 \times 2.3^2$
	$= 0.63 \text{ J}$
(b)(ii)	change in momentum $= \frac{1}{2} \times 240 \times 5.0 \times 10^{-3}$
	$= 0.60 \text{ N s}$
(b)(iii)	(change in velocity of Y) $= 0.60 / 0.12$
	($= 5.0 \text{ m s}^{-1}$)
	final velocity of Y $= 5.0 - 2.3$
	$= 2.7 \text{ m s}^{-1}$
	or
	(final momentum of Y) $= 0.60 - 0.12 \times 2.3$
	($= 0.324 \text{ N s}$)
	final velocity of Y $= 0.324 / 0.12$
	$= 2.7 \text{ m s}^{-1}$
(c)	sloping straight line from (0, 0) to $t = 3.0 \text{ ms}$ and another straight line continuous with the first from $t = 3.0 \text{ ms}$ to (5.0, 0)
	lines showing maximum force of magnitude 240 N
	lines wholly in the negative F region of the graph

32.

(a)	Fv : kg m s^{-2}
	k : $\text{kg m s}^{-2} / \text{m} \times \text{m s}^{-1}$
	$= \text{kg m}^{-1} \text{ s}^{-1}$
(b)	$F = \rho g V$
	$V = 4/3 \times \pi \times (2.1 \times 10^{-3})^3$ ($= 3.88 \times 10^{-8} \text{ m}^3$)
	$\rho = 4.8 \times 10^{-4} / 9.81 \times V$
	$= 1300 \text{ kg m}^{-3}$
(c)(i)	W downwards, U upwards, Fv upwards
(c)(ii)	$F_v = 7.2 \times 10^{-4} - 4.8 \times 10^{-4}$
	$= 2.4 \times 10^{-4} \text{ (N)}$
	velocity $= 2.4 \times 10^{-4} / (17 \times 2.1 \times 10^{-3})$
	$= 6.7 \times 10^{-3} \text{ m s}^{-1}$

33.

(a)	$(m =) \rho V$
	$= 1.0 \times 10^3 \times 1.5 \times 10^{-4} \times 5.0 \times 1.6 = 1.2 \text{ (kg)}$
(b)(i)	$(\Delta)p = 1.2 \times 5.0$ $= 6.0 \text{ N s}$
(b)(ii)	$F = 6.0 / 1.6$ or $1.2 \times 5.0 / 1.6$ $= 3.8 \text{ N}$
(c)	Newton's third law applies (so) 3.8 N.
(d)	$p = F / A$ $= 3.8 / 1.5 \times 10^{-4}$ $= 2.5 \times 10^4 \text{ Pa}$

34.

(a)	change in displacement / time (taken)
(b)(i)	(displacement =) area under graph
	(at $t = 4.0 \text{ s}$) $v = (-) 2.4$
	height $= \frac{1}{2} \times 2.5 \times 4.0 - \frac{1}{2} \times 1.5 \times 2.4$ $= 3.2 \text{ m}$
(b)(ii)	change in momentum $= 7.5 (-4.0 - 2.4)$ $= (-) 48 \text{ N s}$
(b)(iii)	$W = \Delta p / (\Delta)t$ or $\Delta mv / (\Delta)t$ $= 48 / 4.0$ $= 12 \text{ N}$ or $W = ma$ or mg or $m(v - u) / t$ $= 7.5 \times 1.6$ or $7.5 \times (4 + 2.4) / 4.0$ $= 12 \text{ N}$
(c)	speed/velocity decreases so viscous force decreases viscous force decreases (and weight constant) so resultant force decreases

35.

(a)	$T \sin 68^\circ + 32 = 280$
	$T = 270 \text{ N}$
(b)(i)	$F = \rho g V$
	$V = 280 / (1.0 \times 10^3 \times 9.81)$ $= 0.029 \text{ m}^3$
(b)(ii)	$\rho = (32 / 9.81) / 0.029$
	$= 110 \text{ kg m}^{-3}$
(c)	$(\Delta)E = mg(\Delta)h \text{ or } (\Delta)E = W(\Delta)h$
	$(\Delta)h = (-) 77 / 32$
	$(\Delta)h = (-) 2.4$
	final height = $6.2 - 2.4$ $= 3.8 \text{ m}$
(d)(i)	$T = kx$ where k is a constant
	or $T = (EA / L)x$ where A is (cross-sectional) area, E is Young modulus, L is (original/unstretched) length
(d)(ii)	$E = \frac{1}{2} kx^2$ or $E = \frac{1}{2} Fx$ and $F = kx$
	$E = 0.65 \times 2^2$ $= 2.6 \text{ J}$
	or
	$E = \frac{1}{2} Fx$ $0.65 = \frac{1}{2} \times 270 \times x$ and so $x = 4.8 \times 10^{-3} \text{ m}$ $k = F / x = 270 / 4.8 \times 10^{-3}$ $= 5.6 \times 10^4$

(d)(ii)	final $E = \frac{1}{2} \times 540 \times 9.6 \times 10^{-3}$ or $E = \frac{1}{2} \times (270 \times 2) \times (4.8 \times 10^{-3} \times 2)$ or $E = \frac{1}{2} \times 5.6 \times 10^4 \times (9.6 \times 10^{-3})^2$ $= 2.6 \text{ J}$
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36.

(a)	<u>sum/total</u> momentum before = <u>sum/total</u> momentum after or <u>sum/total</u> momentum (of a system of objects) is constant if no (resultant) external force/for a closed system
(b)(i)	$(3.0 \times 4.0 \times \cos \theta) \text{ or } (2.5 \times 4.8 \times \cos \theta) \text{ or } (5.5 \times 3.7)$ $(3.0 \times 4.0 \times \cos \theta) + (2.5 \times 4.8 \times \cos \theta) = (5.5 \times 3.7)$ $\theta = 32^\circ$
(b)(ii)	(initial $E_k = \frac{1}{2} \times 3.0 \times 4.0^2 + \frac{1}{2} \times 2.5 \times 4.8^2 = 53 \text{ (J)}$ or (final $E_k = \frac{1}{2} \times 5.5 \times 3.7^2 = 38 \text{ (J)}$) values of initial E_k and final E_k both correct and inelastic stated

37.

(a)	(velocity =) change in displacement / time
(b)(i)	$a = 2400 / 1200$ $= 2.0 \text{ m s}^{-2}$
(b)(ii)	straight line from the origin with positive gradient (labelled A) ending at (20, 40)
(c)(i)	line starting at origin (with the same gradient as A) and beneath A at all points gradient decreasing to zero straight horizontal line from $t = 12 \text{ s}$ and ending at $t = 20 \text{ s}$ (and labelled B)
(c)(ii)	the velocity/speed will increase to a new terminal/constant/maximum velocity/speed or the car has an acceleration to a new (higher) terminal/constant/maximum velocity/speed

38.

(a)	$F = \rho g V$
	$0.071 = 1.2 \times 9.81 \times V$
	$(V = 6.03 \times 10^{-3} \text{ m}^3)$
	$4/3 \times \pi \times r^3 = 6.03 \times 10^{-3}$
	$r = 0.11 \text{ m}$
(b)	$m = 0.053 / 9.81$
	$(= 5.4 \times 10^{-3} \text{ kg})$
	$F = 0.071 - 0.053$
	$(= 0.018 \text{ N})$
	$a = (0.071 - 0.053) / (0.053 / 9.81)$
	$= 3.3 \text{ m s}^{-2}$
(c)(i)	$v^2 = u^2 + 2as$
	$3.6^2 = (-1.4)^2 + 2 \times 9.81 \times s \text{ or } 3.6^2 = 1.4^2 + 2 \times 9.81 \times s$
	$s = 0.56 \text{ m}$
(c)(ii)	single straight line from any positive non-zero value of v at $t = 0$ to any negative non-zero value of v at $t = T$
	line starting at $(0, 1.4)$ and ending at $(T, -3.6)$

39.

(a)(i)	(e.g. $a = [16.5 - 0] / [0.30 - 0] = 55 \text{ (m s}^{-2}\text{)}$)
(a)(ii)	$(F =) T - W$
(a)(iii)	$ma = T - mg$
	$m = 16 / (55 + 9.81)$
	$m = 0.25 \text{ kg}$
(b)(i)	$s = ut + \frac{1}{2}at^2 \text{ or } v^2 = u^2 + 2as \text{ or } s = vt - \frac{1}{2}at^2 \text{ or } s = \frac{1}{2}(u + v)t \text{ or } s = \text{area under graph}$
	$s = \frac{1}{2} \times 55 \times 0.30^2$
	or
	$s = 16.5^2 / (2 \times 55)$
	or
	$s = 16.5 \times 0.30 - \frac{1}{2} \times 55 \times 0.30^2$
	or
(b)(ii)	$s = \frac{1}{2} \times 16.5 \times 0.30$
	$s = 2.5 \text{ m}$
	$u = 16.5 \text{ (m s}^{-1}\text{)}$
	$v^2 = u^2 + 2as$
	$v^2 = 16.5^2 + 2 \times 9.81 \times 2.5$
	$v = 18 \text{ m s}^{-1}$

40.

(a)	$E = \frac{1}{2}mv^2$
	$p = mv$
	$m = 0.37^2 / (2 \times 0.30) \text{ or } 0.37 / 1.6 \text{ or } (0.30 \times 2) / 1.6^2$ $= 0.23 \text{ kg}$
(b)	$0.37 - 0.65 = -0.13 - p$ $p = 0.15 \text{ kg m s}^{-1}$
(c)	$7.7 = (0.13 + 0.37) / (\Delta)t$ or $7.7 = (0.65 - 0.15) / (\Delta)t$ time = 0.065 s

41.

(a)	(resultant) force (on an object) is proportional to / equal to the rate of change of momentum
(b)(i)	resultant force = e.g. $6.0 / 4.0$ $= 1.5 \text{ N}$
b)(ii)	force X = $1.5 + 2.0$ $= 3.5 \text{ N}$
(c)	from $t = 0$ to $t = 4.0 \text{ s}$: horizontal line at any non-zero value of X from $t = 0$ to $t = 4.0 \text{ s}$: horizontal line at $X = 3.5 \text{ N}$ from $t = 4.0 \text{ s}$ to $t = 6.0 \text{ s}$: horizontal line at $X = 2.0 \text{ N}$

42.

(a)(i)	rate of change of momentum
(a)(ii)	change in momentum = $(1.4 - 0.80) \times 3.0$ $= 1.8 \text{ kg m s}^{-1}$
(b)(i)	resultant force (on block) is zero (so) velocity is constant
(b)(ii)	$P = Fv \text{ or } P = Fs / t$ $v = 2.0 / 0.80 (= 2.5 \text{ m s}^{-1})$ distance = 2.5×3.0 $= 7.5 \text{ m}$ or

	$P = W/t$ or $P = Fs/t$
	$W = 2.0 \times 3.0 (= 6.0 \text{ J})$
	distance = $6.0 / 0.80$ $= 7.5 \text{ m}$
(c)	0 to 3.0 s: upward sloping straight line from the origin. 3.0 to 6.0 s: horizontal line at non-zero value of momentum with no 'step change' in momentum at 3.0 s

43.

(a)	$F = 1050 \times 9.81 \times 0.21V$ or $W = \rho \times 9.81 \times V$ $0.21V \times 1050 (\times 9.81) = V (\times 9.81) \times \rho$ $\rho = 220 \text{ kg m}^{-3}$
(b)(i)	$F = 1050 \times 9.81 \times V$ or $W = 220 \times 9.81 \times V$ $(V \times 1050 \times 9.81) - (V \times 220 \times 9.81) = (V \times 220) \times a$ $a = 37 \text{ m s}^{-2}$
(b)(ii)	the (downward) drag / viscous force increases (with speed) resultant force decreases (as upthrust and weight remain the same) acceleration decreases (as its velocity increases)

44.

(a)	<u>sum / total</u> momentum before = <u>sum / total</u> momentum after or <u>sum / total</u> momentum (of a system of objects) is constant if no (resultant) external force / for an isolated system
(b)(i)	$3m \times 4 = m \times v \sin \theta$ $(v \sin \theta = 12)$ $2m \times 6 = m \times v \cos \theta$ $(v \cos \theta = 12)$ therefore $\sin \theta = \cos \theta$ or $\tan \theta = 1$ $\theta = 45^\circ$
(b)(ii)	$mv \times \cos 45^\circ = 12m$ or $mv \times \sin 45^\circ = 12m$ or $(mv)^2 = (3m \times 4)^2 + (2m \times 6)^2$ $v = 17 \text{ m s}^{-1}$
(c)(i)	(chemical energy) = $0.0050 \times 700 = 3.5 \text{ (J)}$ or (chemical energy) = $5.0 \times 0.700 = 3.5 \text{ (J)}$

(c)(ii)	$E = \frac{1}{2}mv^2$
	total $E = (0.5 \times 3m \times 4^2) + (0.5 \times 2m \times 6^2) + (0.5 \times m \times 17^2)$
	$3.5 = 204m$
	$m = 0.017 \text{ kg}$

45.

(a)(i)	arrow upwards ($\uparrow$) and labelled upthrust / U arrow downwards ($\downarrow$) and labelled weight / W / mg arrow downwards ($\downarrow$) and labelled tension / T <i>1 mark: One or two correctly labelled arrows</i> <i>2 marks: Three correctly labelled arrows</i>
(a)(ii)	$U = T + W$ or upthrust = tension + weight $\rho Vg = T + W$ $V = [(4.00 \times 10^2) + (3.39 \times 10^4)] / (1.23 \times 9.81)$ $V = 2.84 \times 10^3 \text{ m}^3$
(a)(iii)	$m = W/g$ or $a = F/m$ $a = (4.00 \times 10^2) / [(3.39 \times 10^4) / 9.81]$ $a = 0.12 \text{ m s}^{-2}$
(b)	there is air resistance (which increases with speed) (average) resultant force is less (than weight) (average) acceleration is less (than $g/9.81$, so speed is less than 100 m s^{-1})
(c)(i)	(extension =) $2.5 \times 2.4 \times 10^{-5} = 6.0 \times 10^{-5} \text{ (m)}$
(c)(ii)	$E_{(P)} = \frac{1}{2}Fx$ or $E_{(P)} = \frac{1}{2}kx^2$ and $F = kx$ $E_{(P)} = \frac{1}{2} \times 4.00 \times 10^2 \times 6.0 \times 10^{-5}$ or $E_{(P)} = \frac{1}{2} \times 6.7 \times 10^6 \times (6.0 \times 10^{-5})^2$ $E_{(P)} = 0.012 \text{ J}$
(c)(iii)	longer extension or smaller spring constant elastic potential energy is greater

46.

(a)	displacement
(b)(i)	$a = \text{gradient}$ or $a = \Delta v / (\Delta)t$ or $a = (v - u) / t$ e.g. $a = (0.30 - 0.12) / (0.35 - 0.15)$ $a = 0.90 \text{ m s}^{-2}$
(b)(ii)	$(0.25 \times 0.48) + (0.75 \times 0.12) = (0.25 v) + (0.75 \times 0.30)$ or $(0.48 - 0.12) = (0.30 - v)$ or $(\frac{1}{2} \times 0.25 \times 0.48^2) + (\frac{1}{2} \times 0.75 \times 0.12^2) = (\frac{1}{2} \times 0.25 \times v^2) + (\frac{1}{2} \times 0.75 \times 0.30^2)$ $v = (-)0.060 \text{ m s}^{-1}$ direction: to the left / from the right / opposite to (its) initial velocity / opposite to (initial / final) velocity of B
(c)	sketch: horizontal line from (0, 0.48) to (0.15, 0.48) horizontal line from (0.35, -0.06) to (0.5, -0.06) straight line between (0.15, 0.48) and (0.35, -0.06)

47.

(a)	$65 = \rho g V$ or $65 = mg$ and $m = \rho V$ $\rho = 65 / (9.81 \times 7.5)$ $= 0.88 \text{ kg m}^{-3}$
(b)(i)	acceleration = 9.8 m s^{-2}
(b)(ii)	air resistance (acting on sphere) increases resultant force (on sphere) decreases (magnitude of) acceleration decreases
(c)	$F = ma$ $= 4.0 \times 1.9$ (= 7.6 N) resistive force = $(4.0 \times 9.81) - (4.0 \times 1.9)$ = 32 N

48.

(a)(i)	component of momentum = $0.25 \times 3.7 \times \sin 27^\circ$
	= 0.42 N s
(a)(ii)	$m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.42$
	or
	$m_{(Z)} \times 5.5 \times \sin 44^\circ = 0.25 \times 3.7 \times \sin 27^\circ$
(a)(iii)	magnitudes: equal
	directions: opposite
(b)	$4 + 6 = 2 + v$
	$v = 8 \text{ m s}^{-1}$

49.

(a)	resultant force (in any direction) is zero
	resultant moment / torque (about any point) is zero
(b)(i)	$F = \rho Vg$
	$V = 93\,000 / (1.2 \times 9.81)$
	= 7900 m ³
(b)(ii)	weight = $93\,000 + 3(.0) \times 10^3$
	$m = (93\,000 + 3.0 \times 10^3) / 9.81$
	= 9800 kg
(c)(i)	$(\Delta p) = F(\Delta t)$ or $F = \Delta p / (\Delta t)$ or $F = \text{rate of change of momentum}$ or $F = m(v - u) / t$
	$\Delta p = 2800 \times 0.50$
	= 1400 kg m s ⁻¹
(c)(ii)	$(\Delta p) = mv$
	$v = 1400 / 64$
	= 22 m s ⁻¹
(c)(iii)	$E_K = \frac{1}{2}mv^2$
	= $\frac{1}{2} \times 64 \times 22^2$
	= 15 000 J
	or
	$E_K = p^2 / 2m$
	= $1400^2 / (2 \times 64)$
	= 15 000 J